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(54) DEPLOYABLE HANDLE DEVICES

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(73) Assignee: **NuWhirl Systems Corp**, Corona, CA (US)

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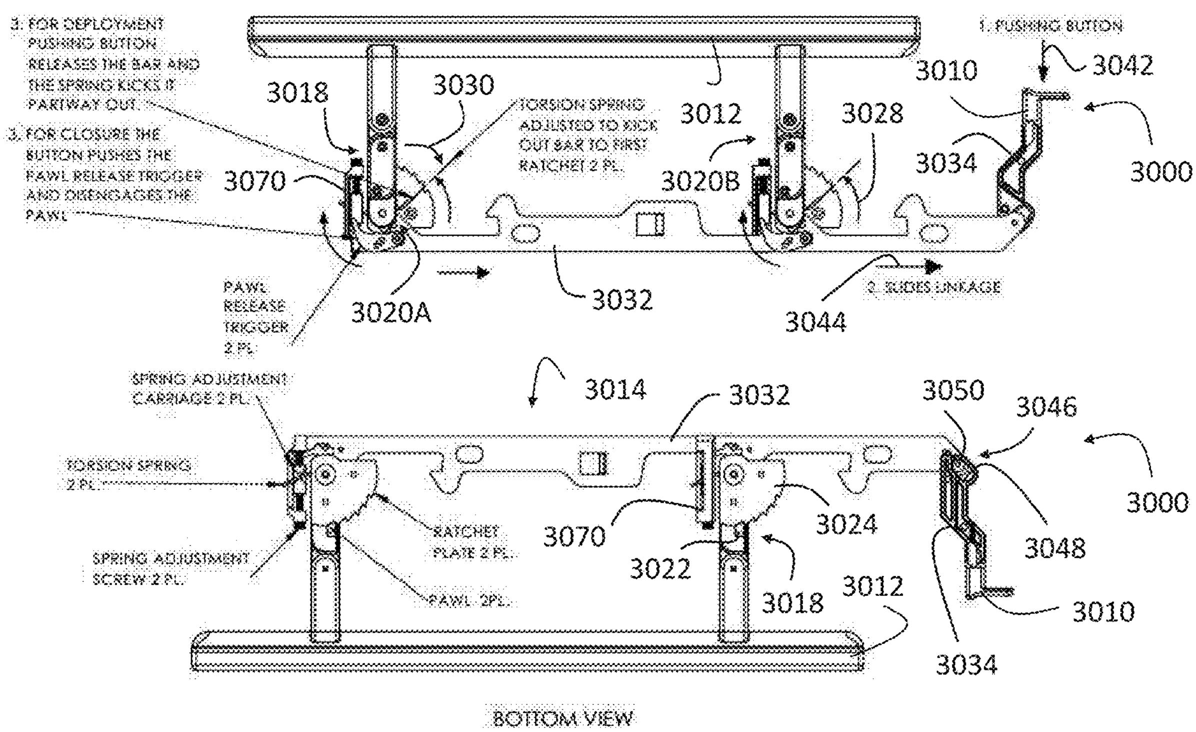
A47K 17/02 (2006.01)

(52) **U.S. Cl.**
CPC ***A47K 17/024*** (2013.01)

(57) **ABSTRACT**

Deployable handle devices are disclosed. Examples of handle devices having a handle configured to move between a stowed configuration and a deployed configuration, a handle housing, a handle arm, a pawl configured to move independently from the handle arm between a first pawl position and a second pawl position, a ratchet, a linkage configured to move the pawl from the first pawl position to the second pawl position, and an actuator configured to move the linkage from the first linkage position to the second linkage position when actuated.



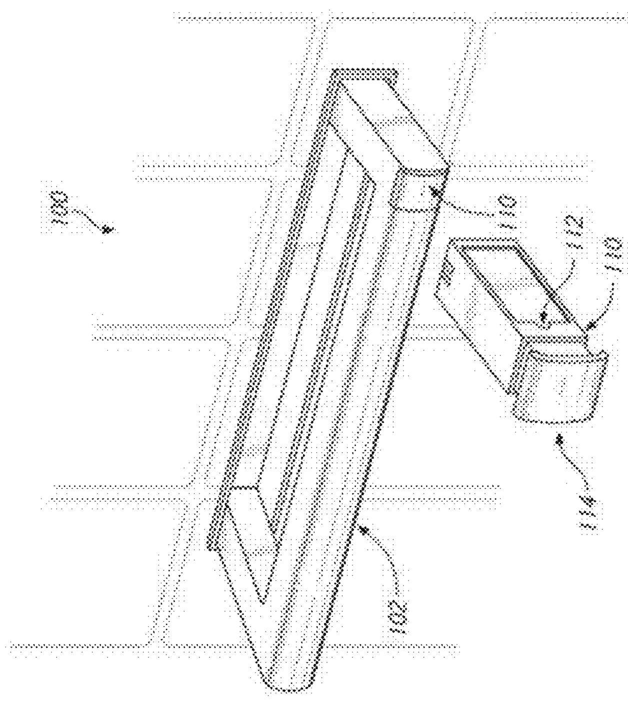


FIG. 1

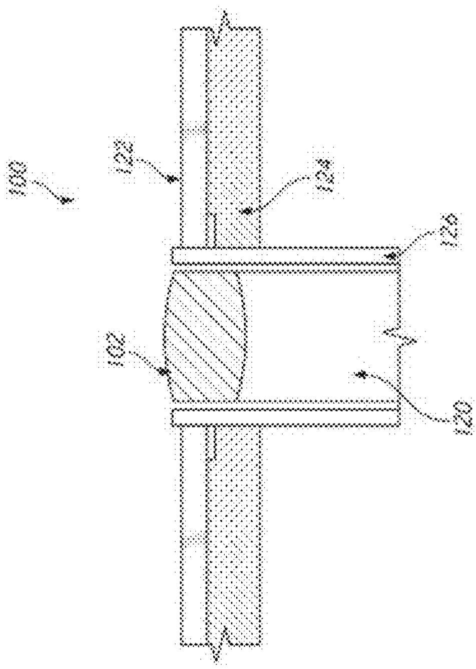


FIG. 2

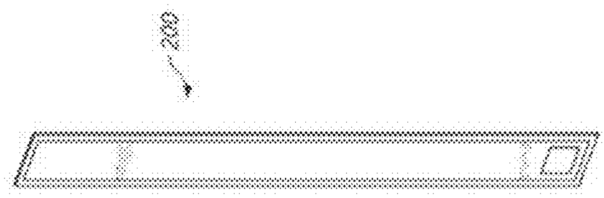


FIG. 3B

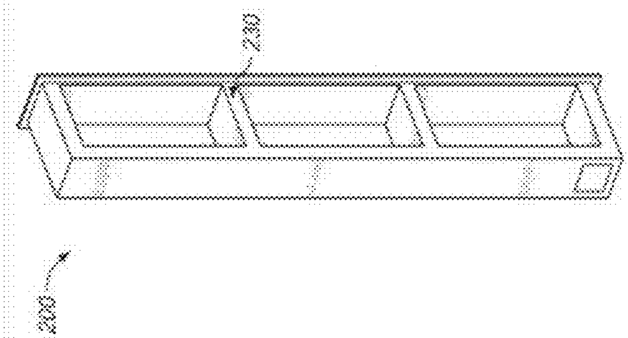


FIG. 3A

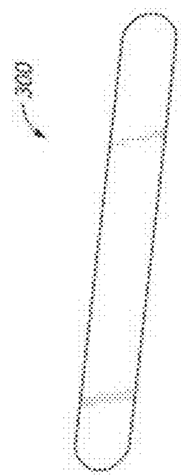


FIG. 4A

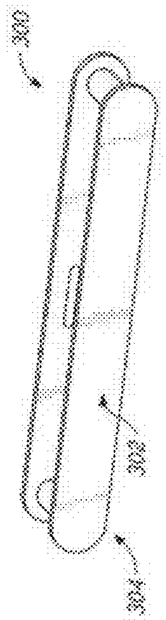


FIG. 4B

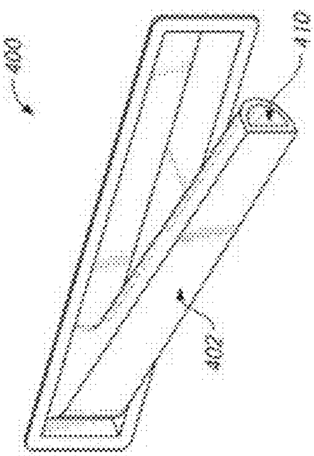


FIG. 5

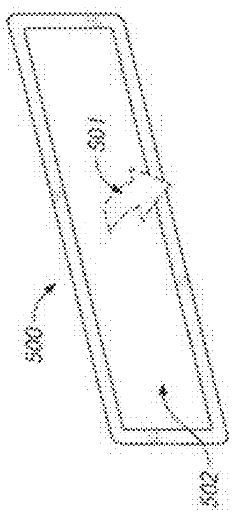


FIG. 6A

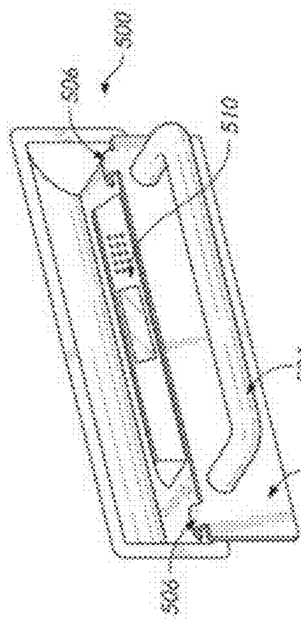


FIG. 6B

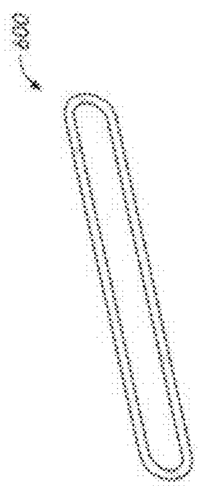


FIG. 7A

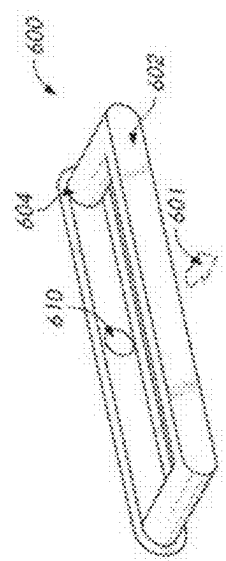


FIG. 7B

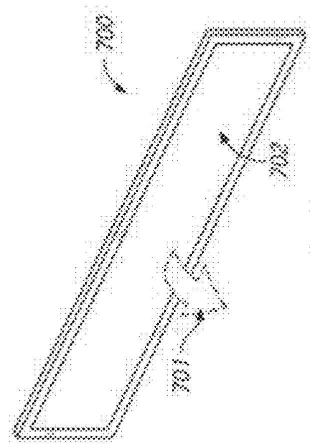


FIG. 8A

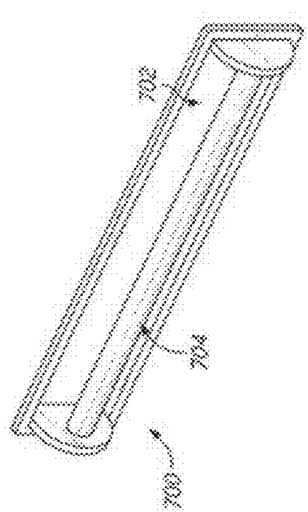


FIG. 8B

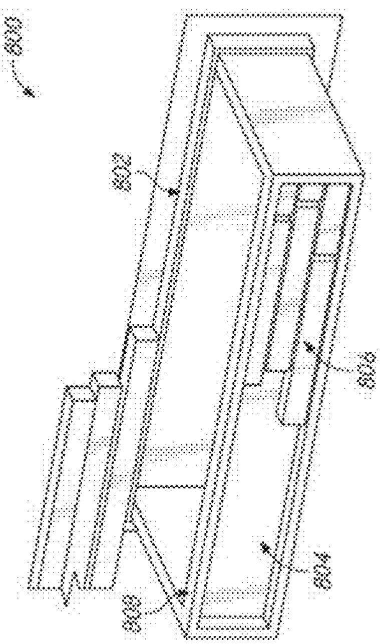


FIG. 9

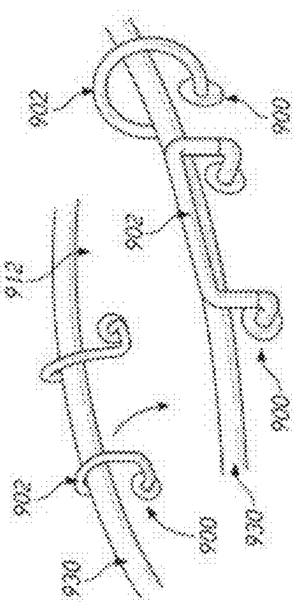


FIG. 10

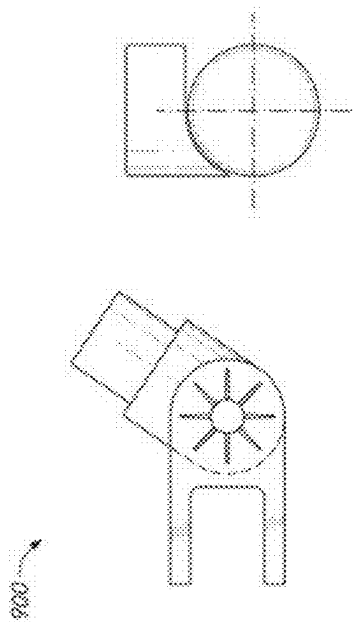


FIG. 13

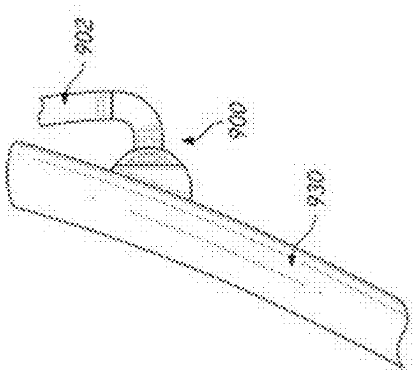


FIG. 11

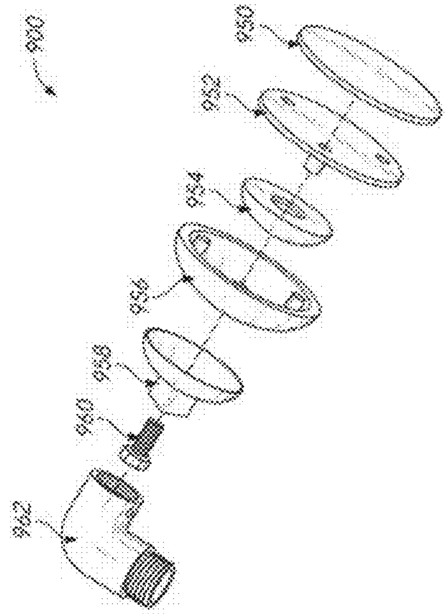


FIG. 12

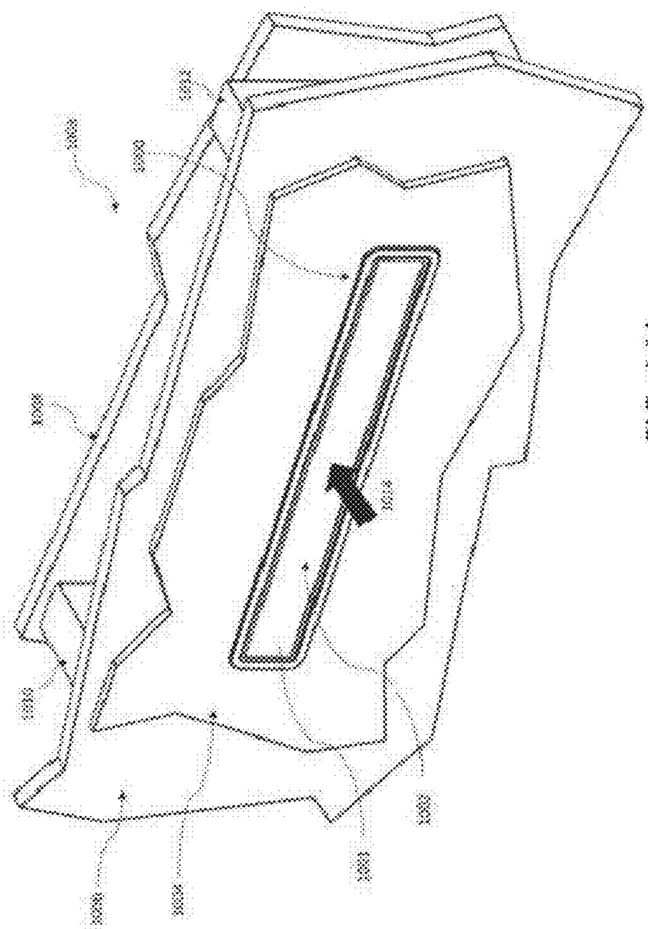


FIG. 14A

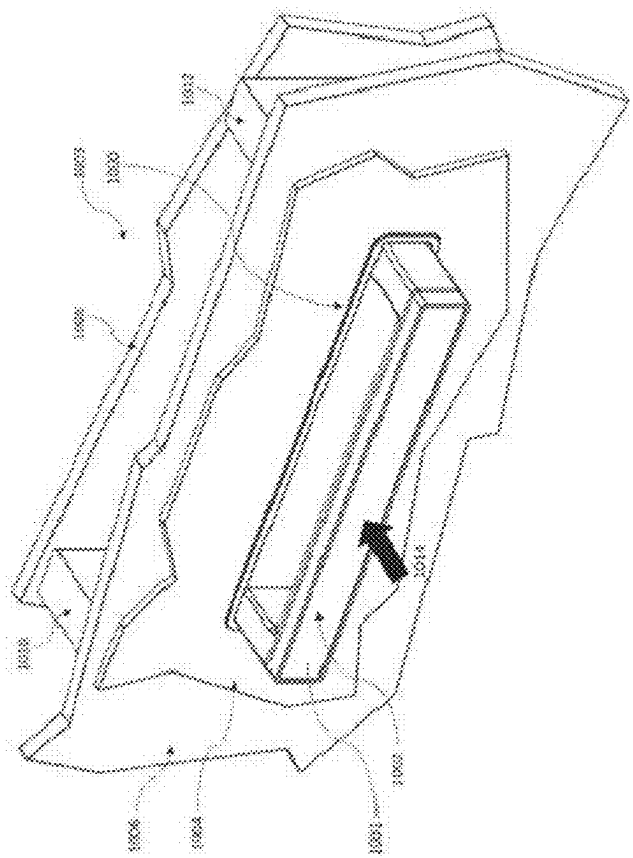


FIG. 14B

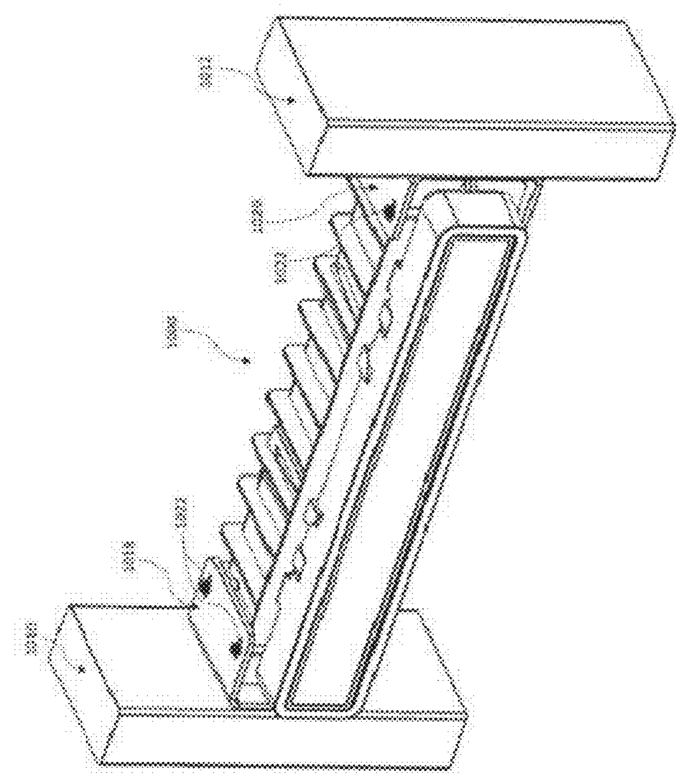


FIG. 15A

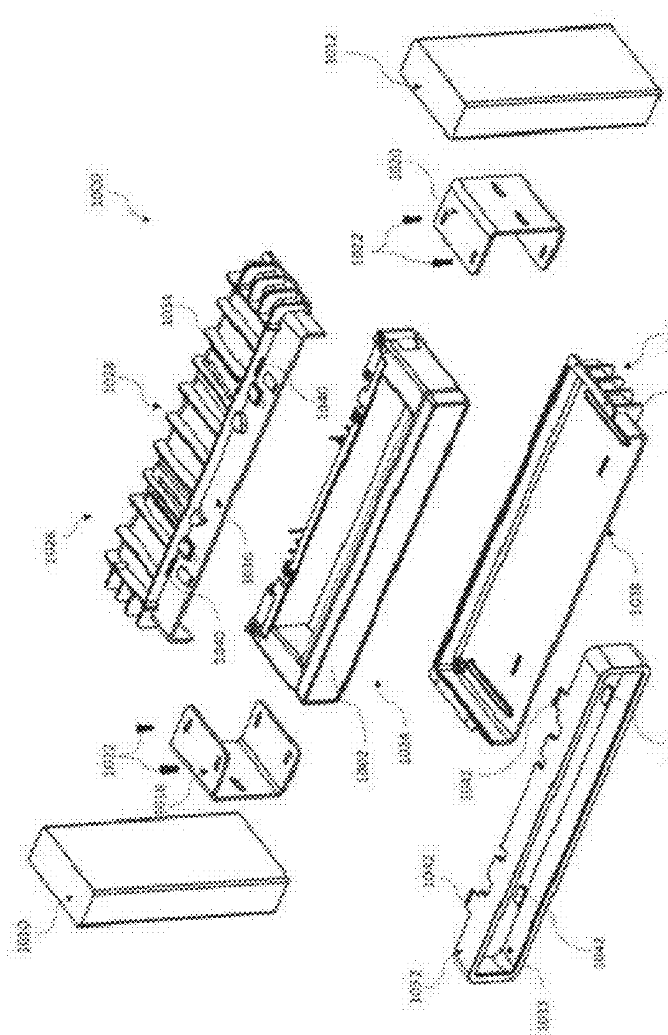


FIG. 15B

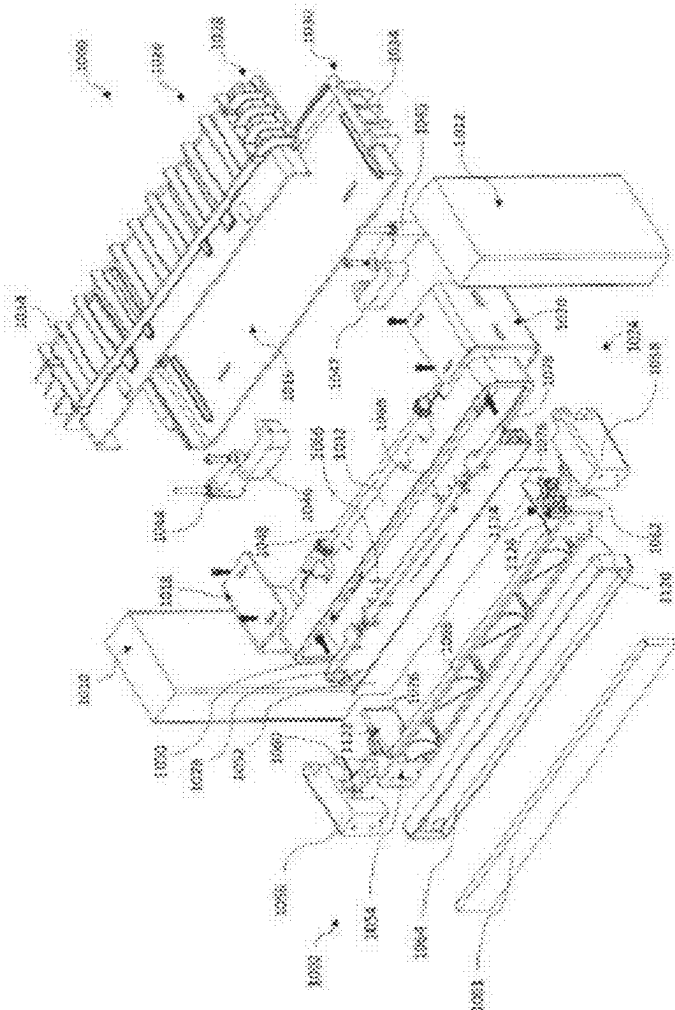


FIG. 15C

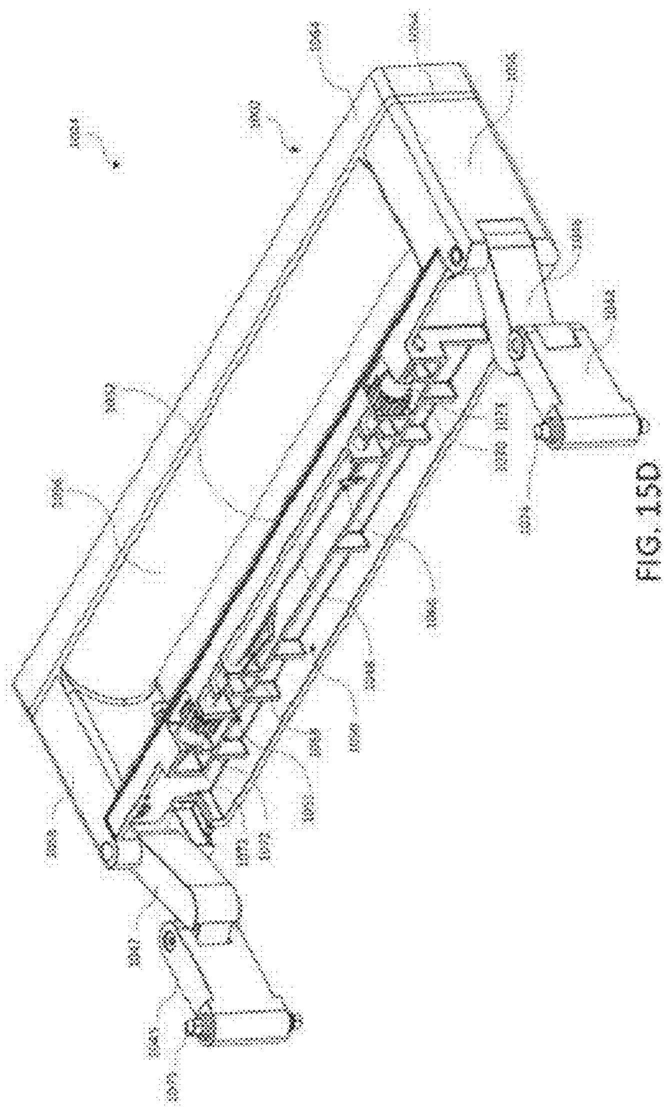


FIG. 15D

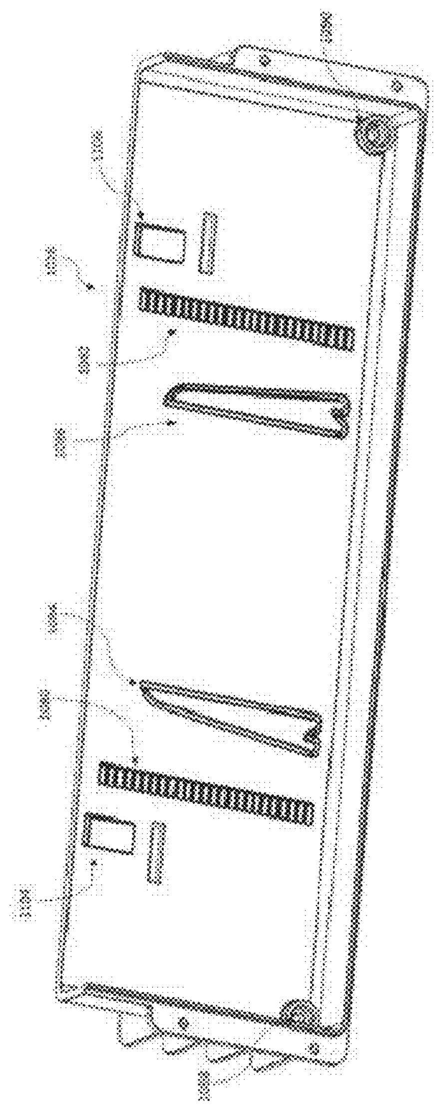


FIG. 16A

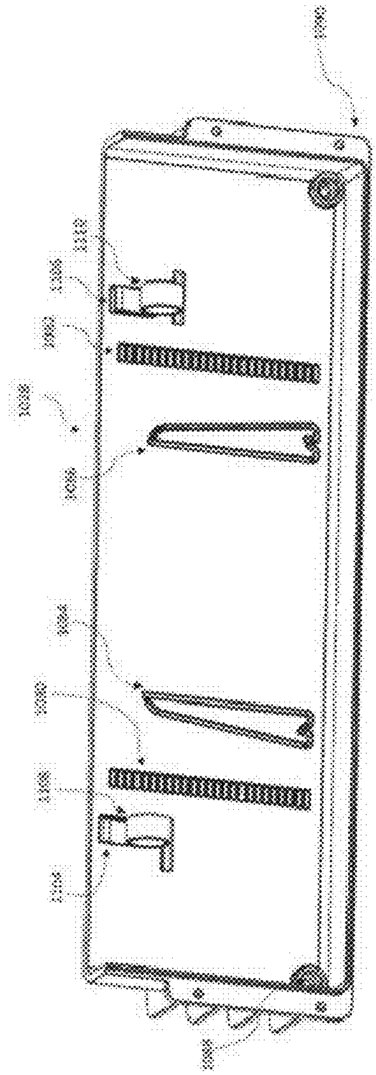


FIG. 16B

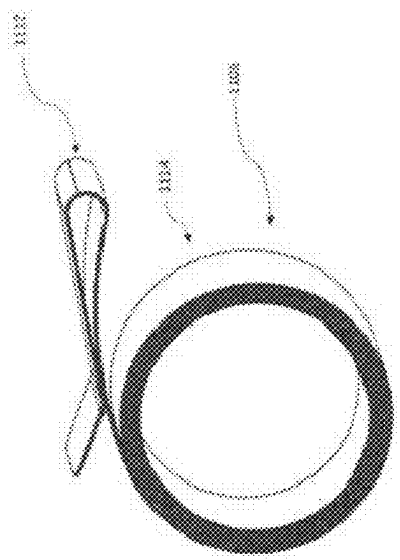


FIG. 16C

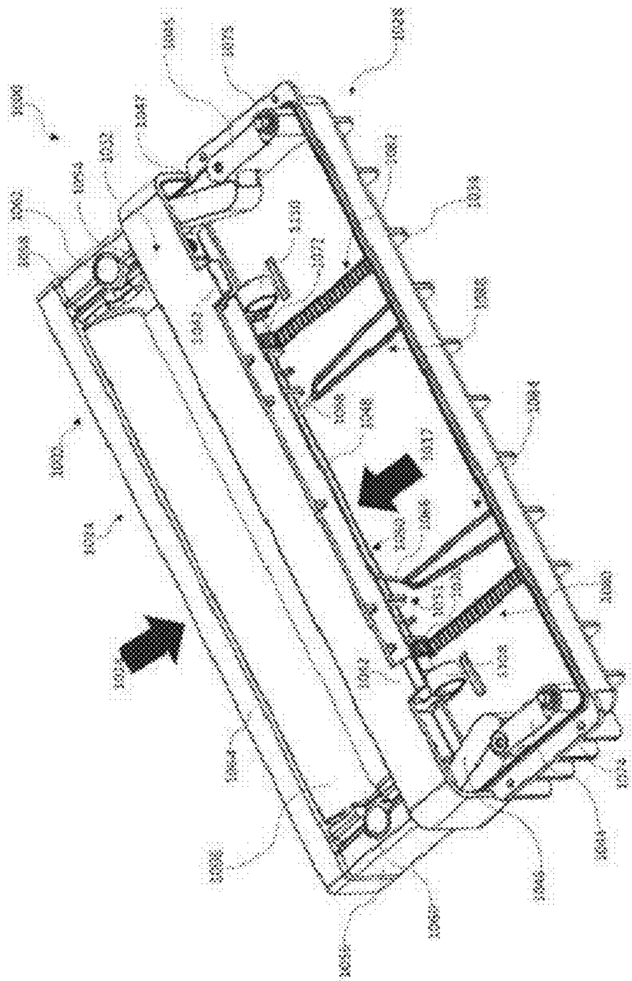


FIG. 17A

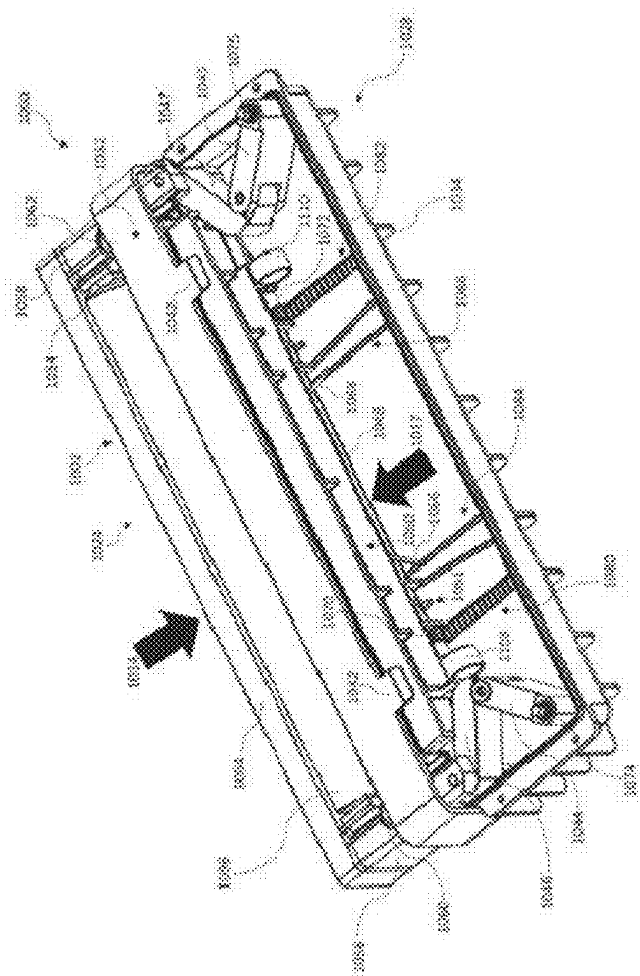


FIG. 17B

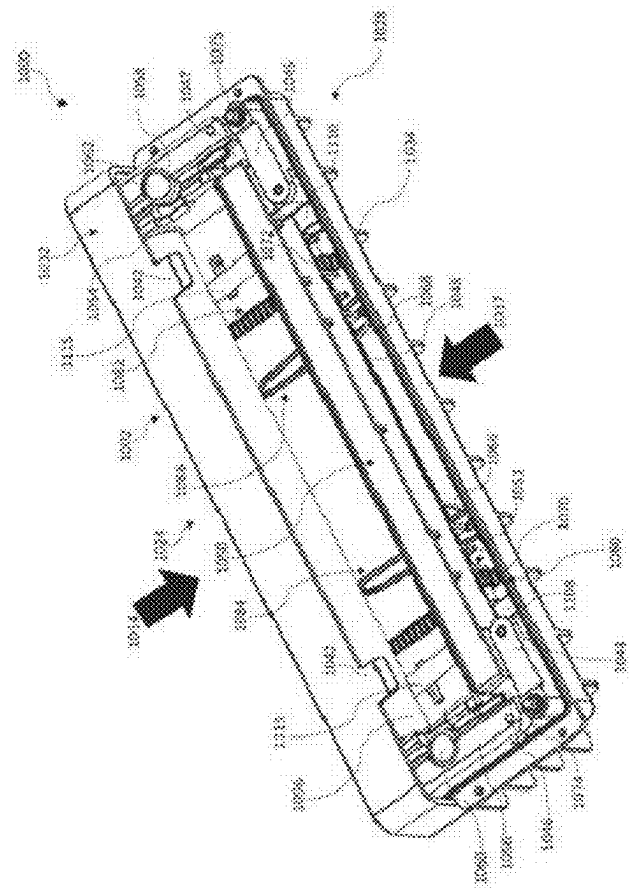


FIG. 17C

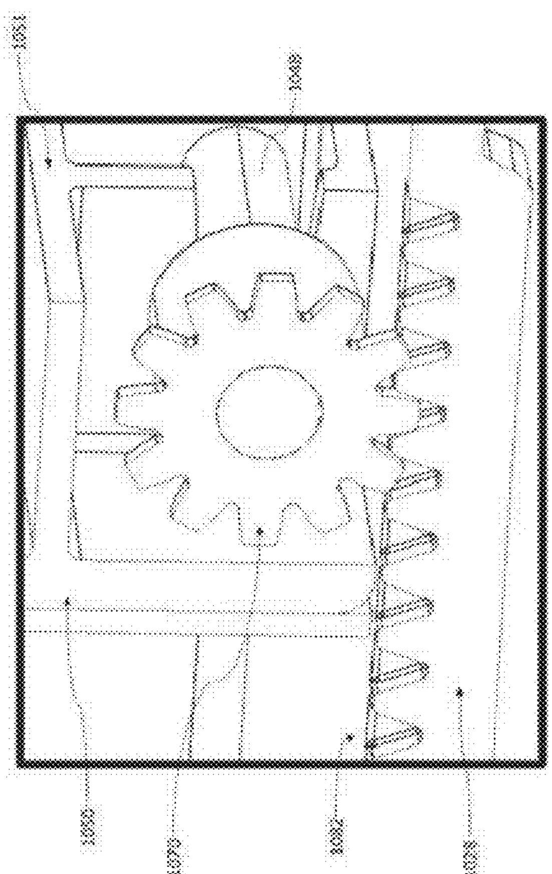


FIG. 17D

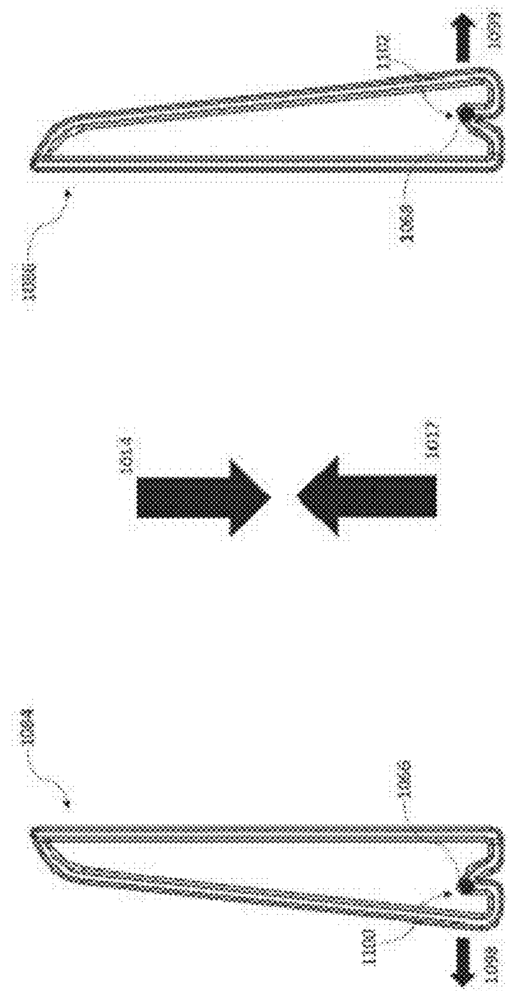


FIG. 18A

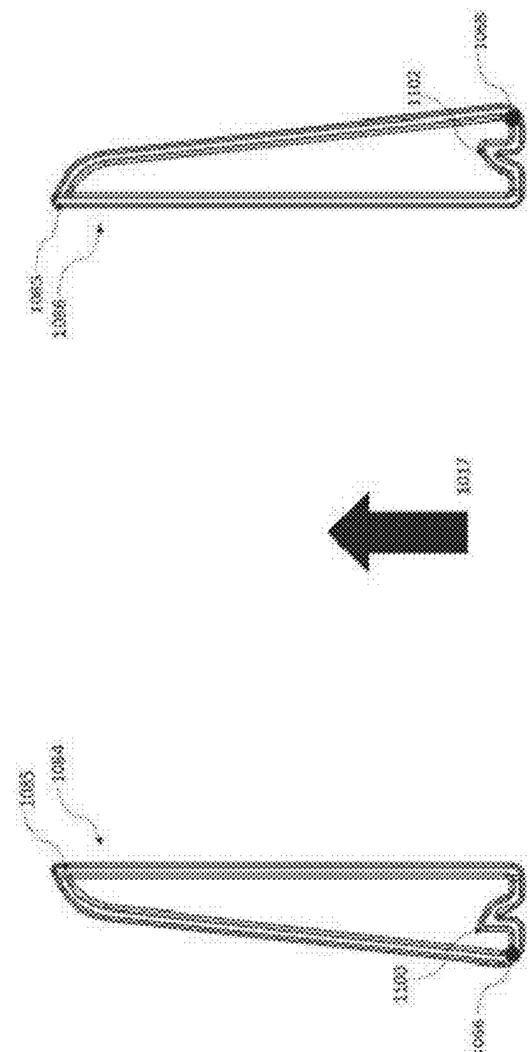


FIG. 18B

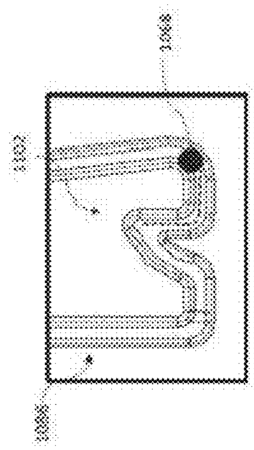


FIG. 18C-2

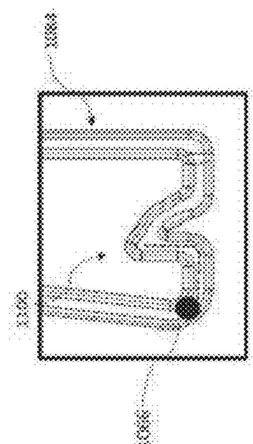


FIG. 18C-1

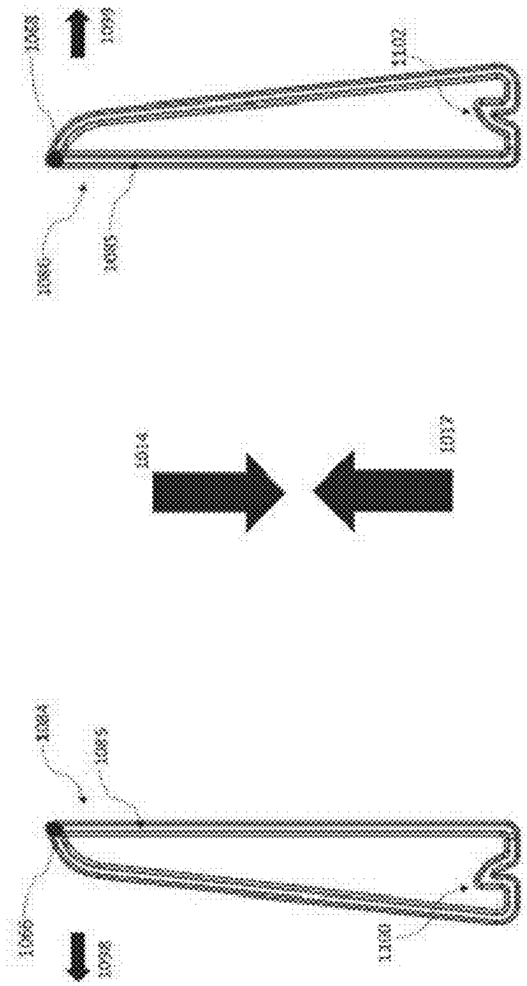


FIG. 18D

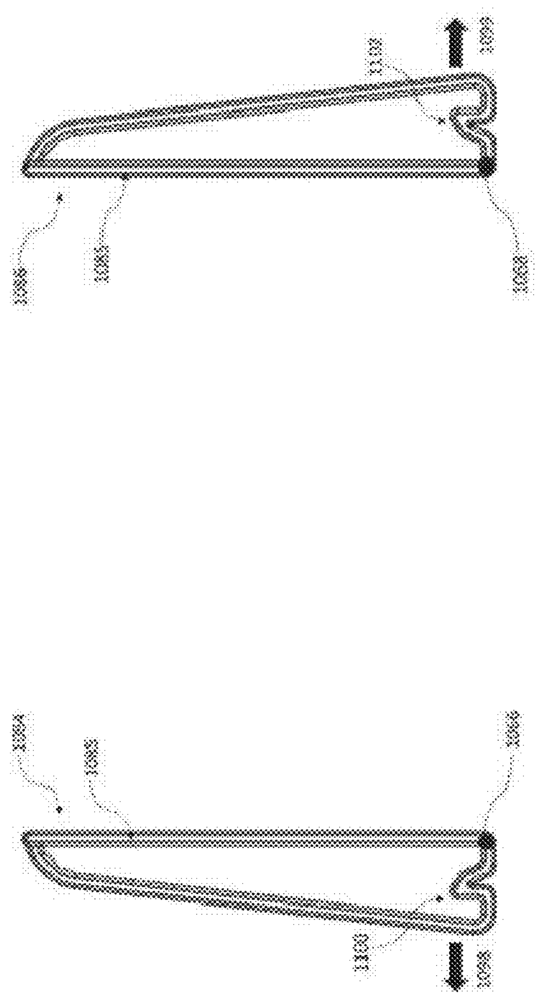


FIG. 18E

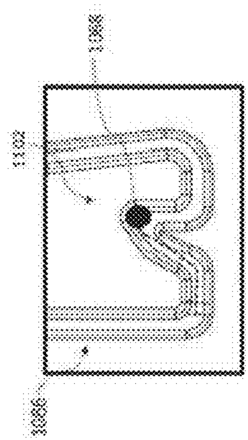


FIG. 18F-2

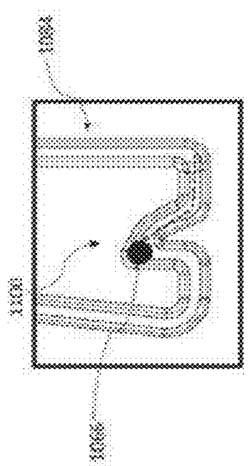


FIG. 18F-1

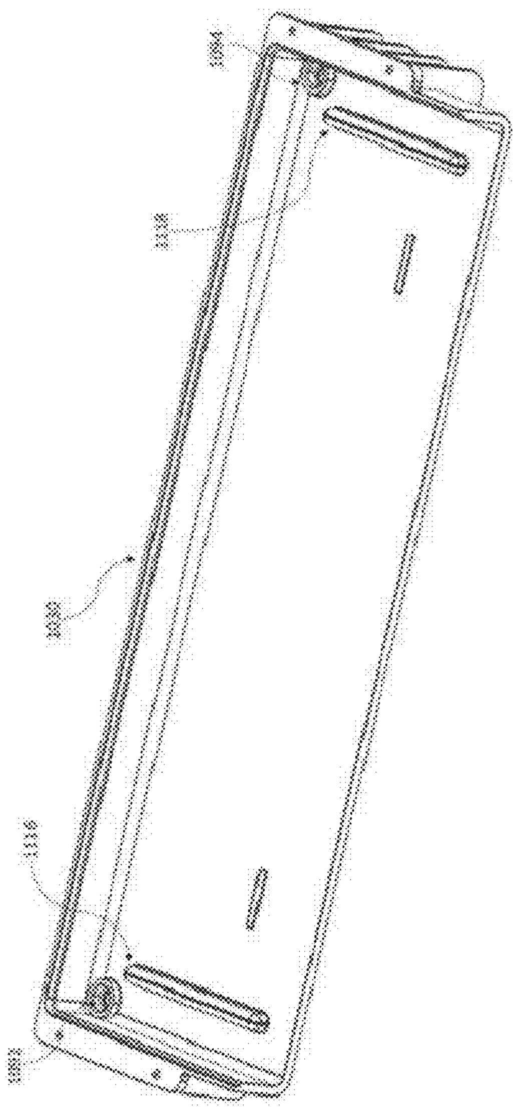


FIG. 19

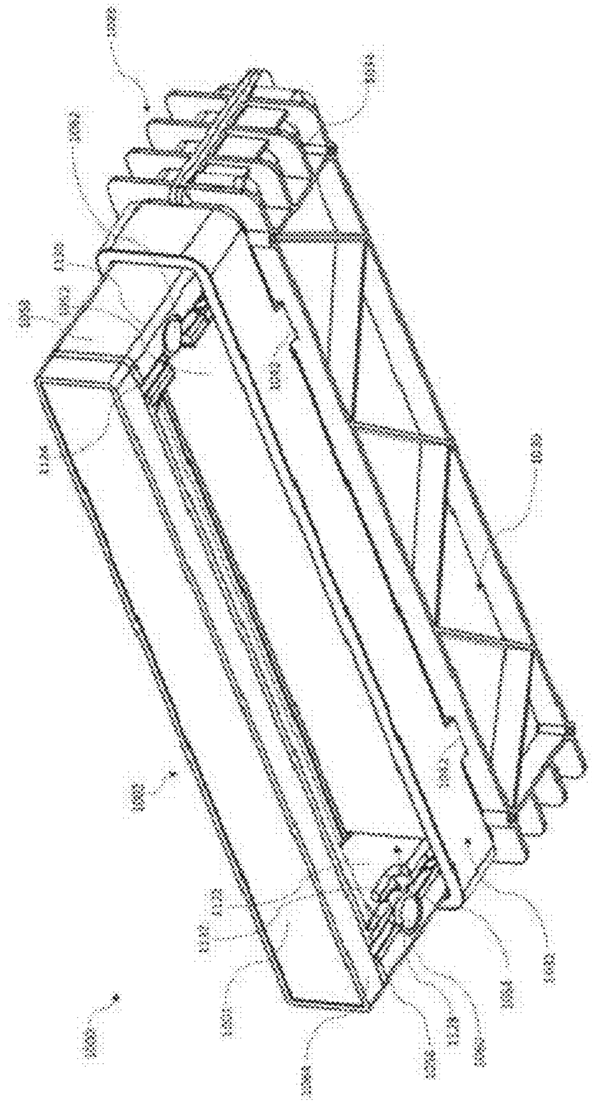


FIG. 20A

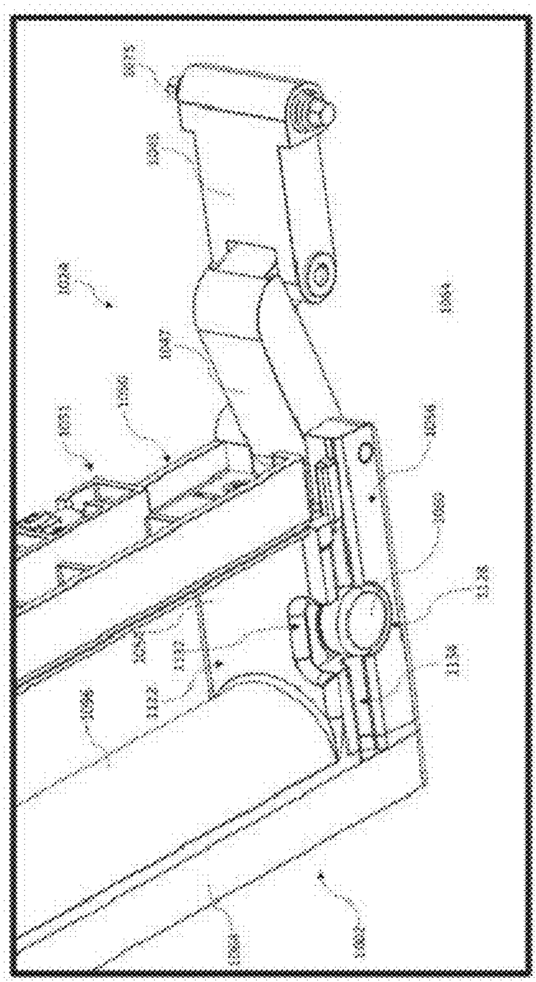


FIG. 20B

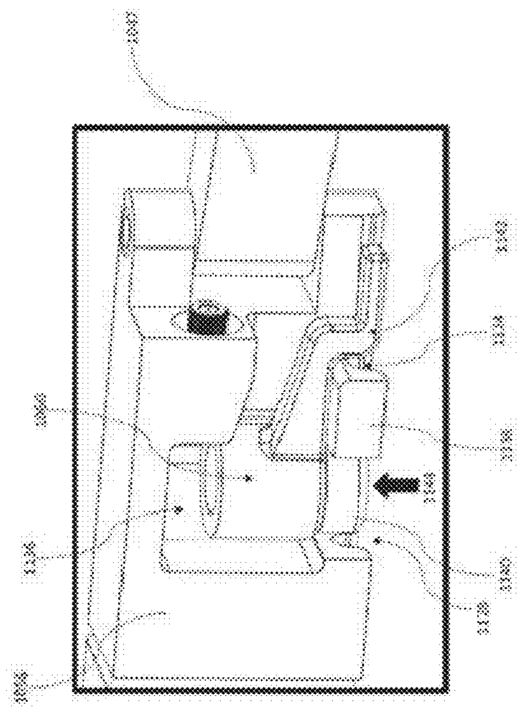


FIG. 20C

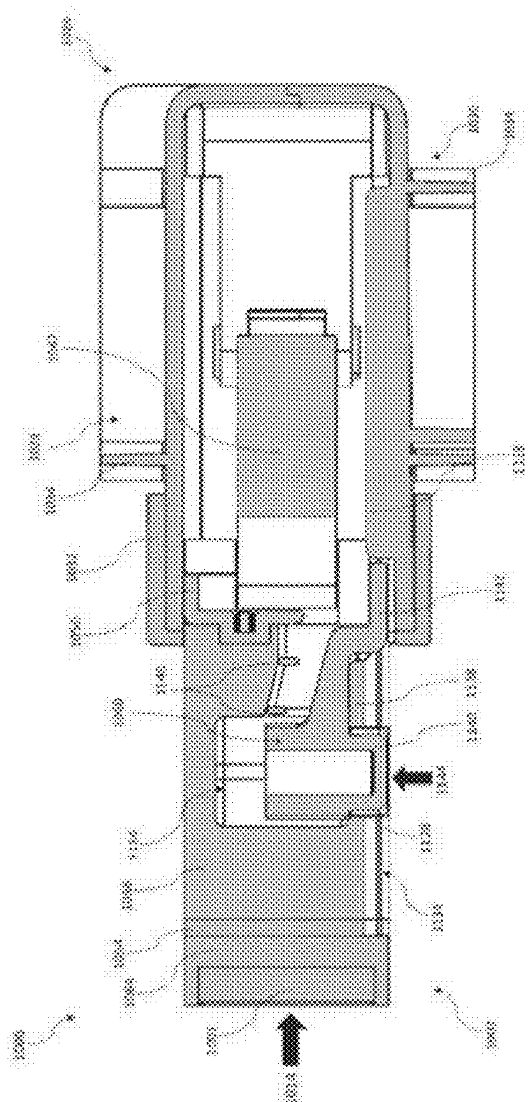


FIG. 21A

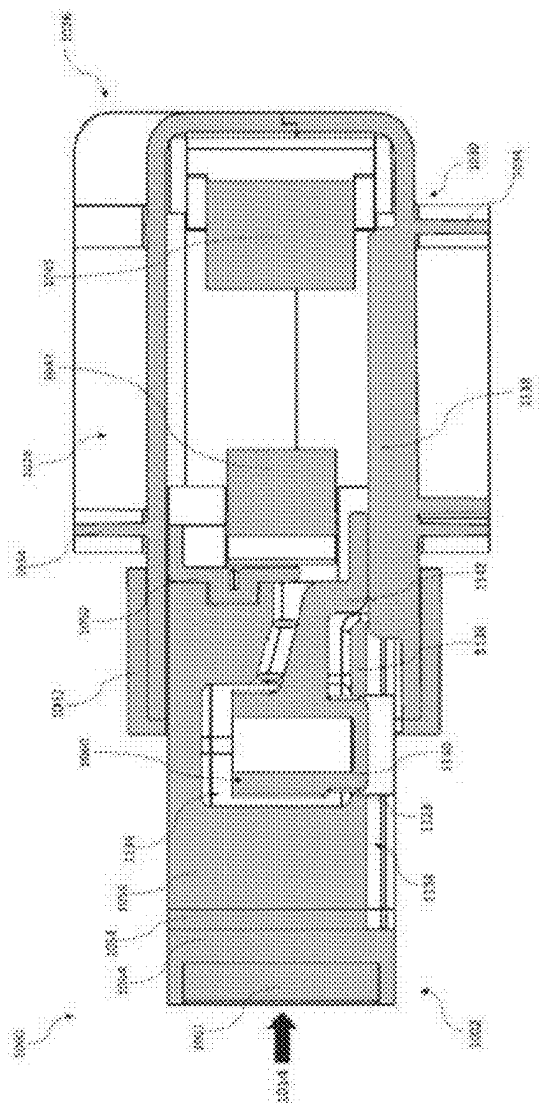


FIG. 21B

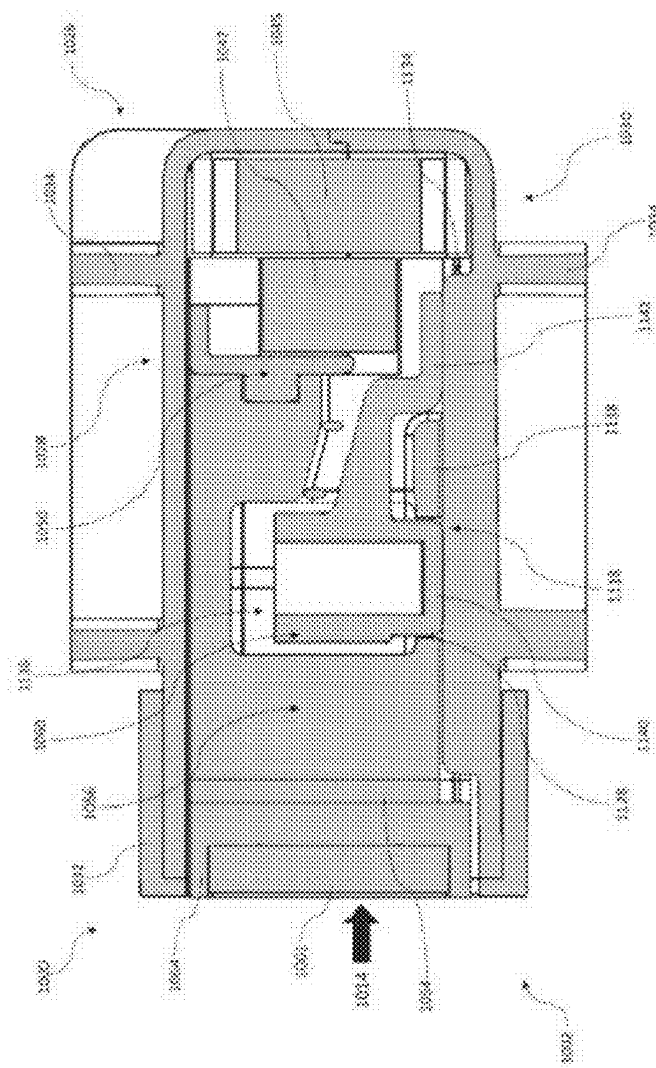


FIG. 21C

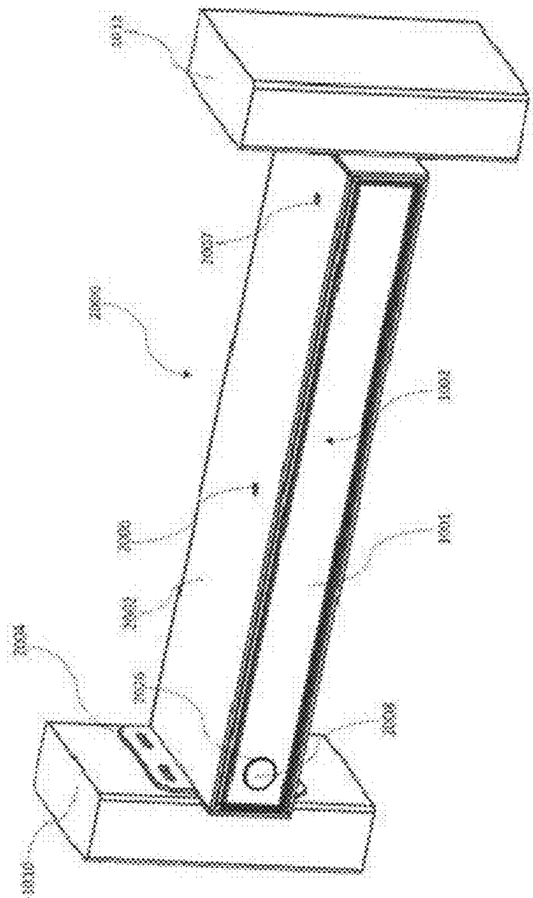


FIG. 22A

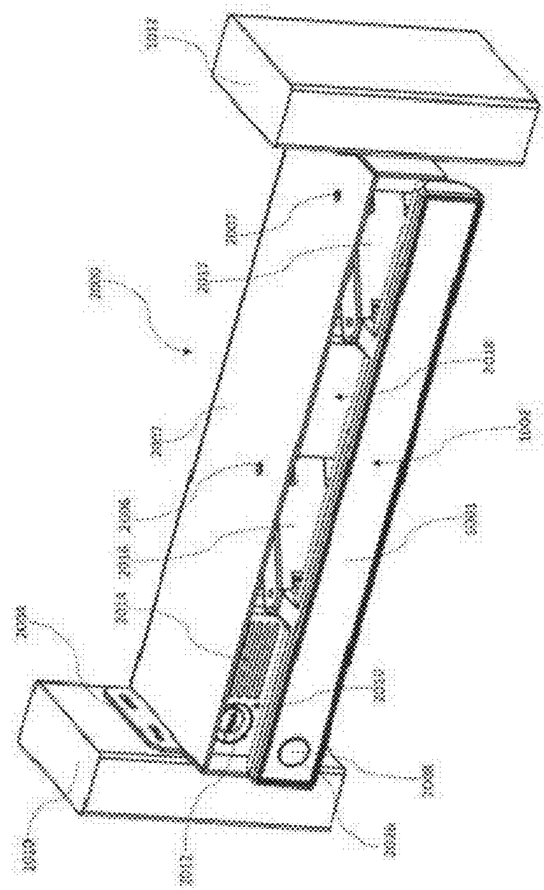


FIG. 22B

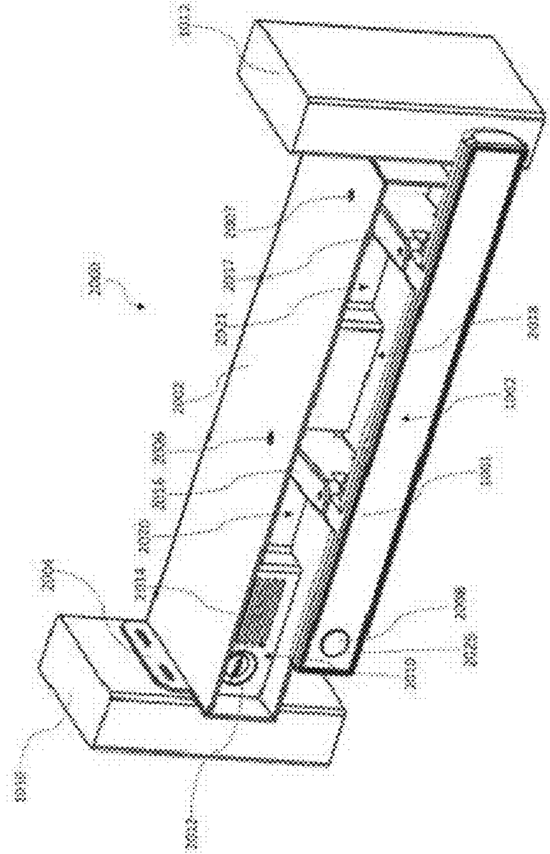


FIG. 22C

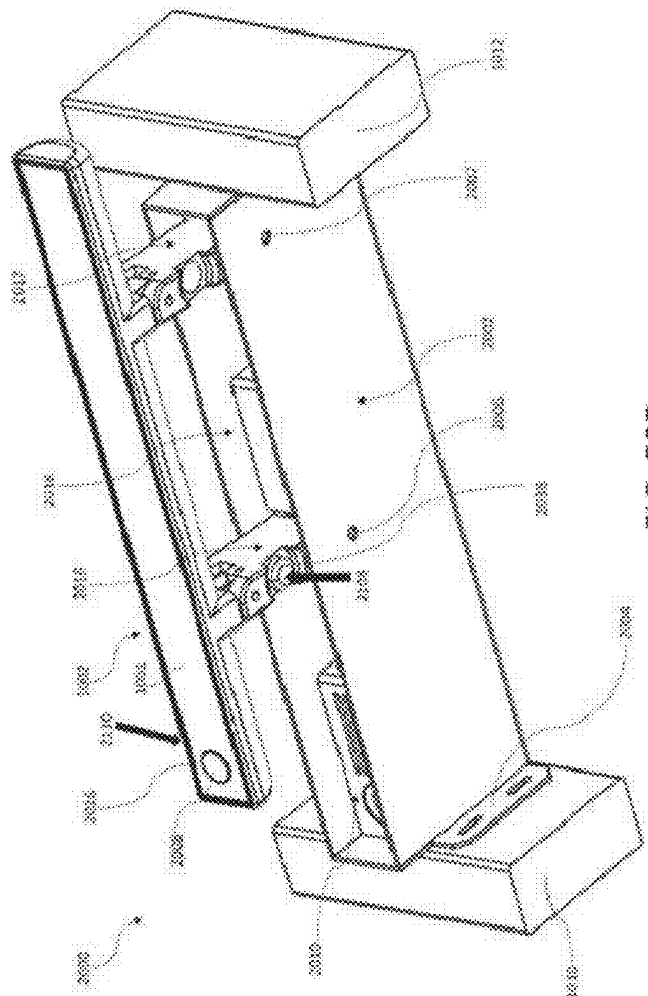


FIG. 22D

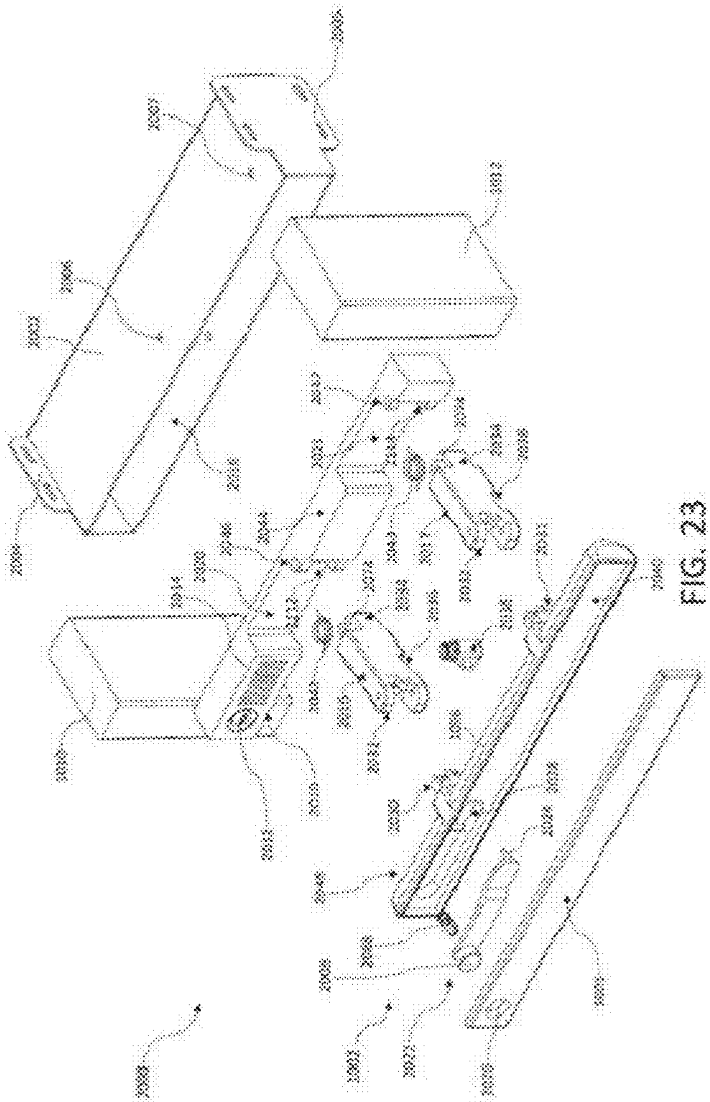


FIG. 23

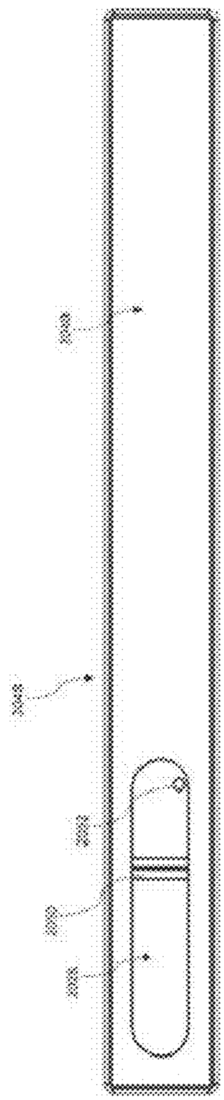


FIG. 24A

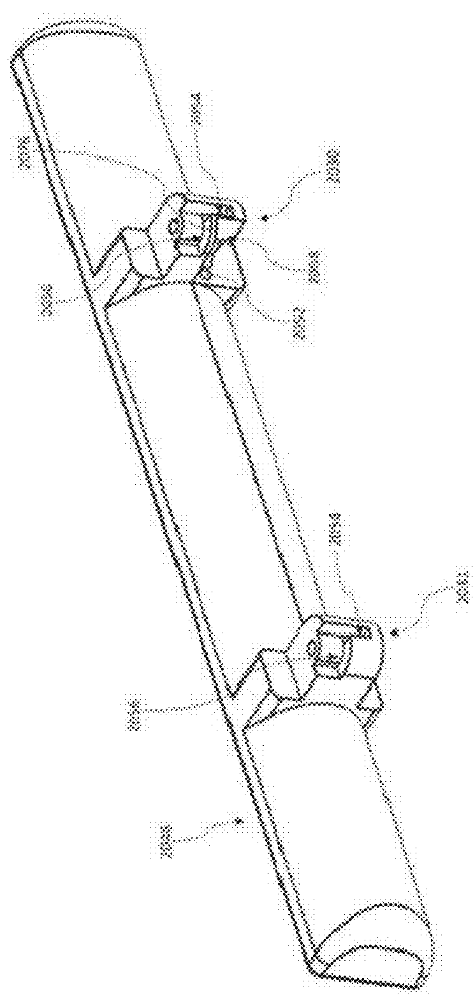


FIG. 24B

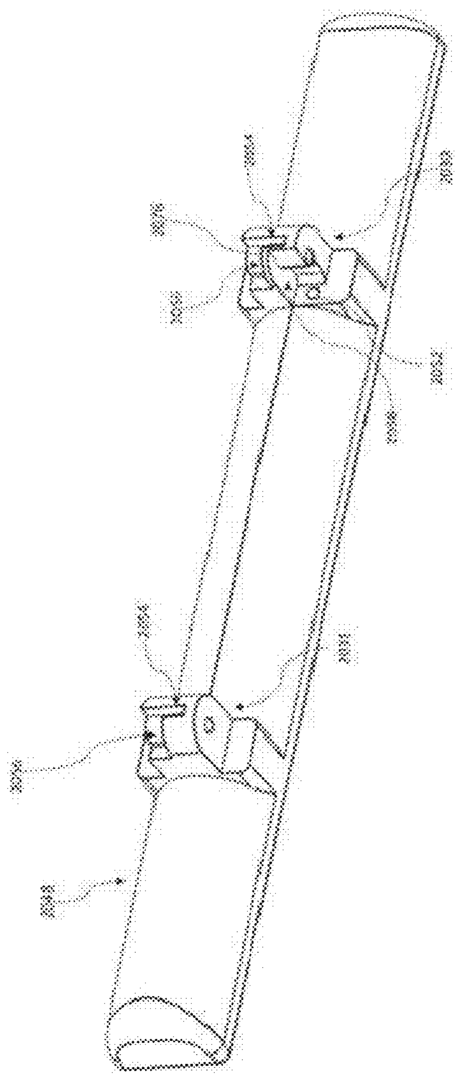


FIG. 24C

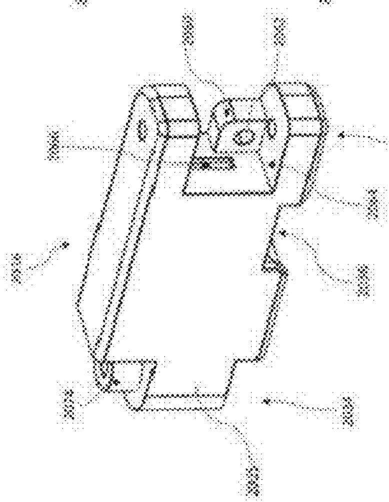


FIG. 25A

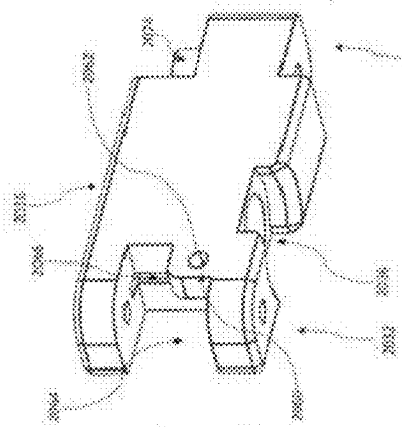


FIG. 25B

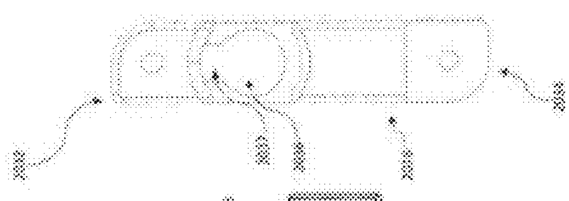


FIG. 25C

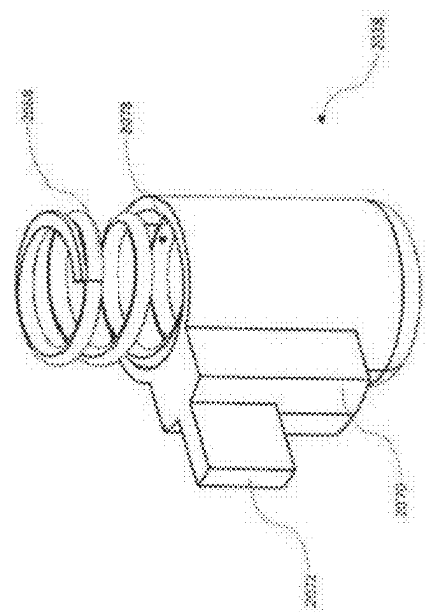


FIG. 26

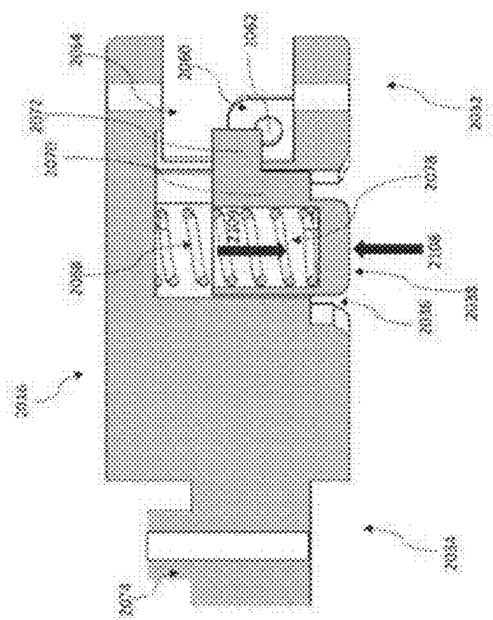


FIG. 27B

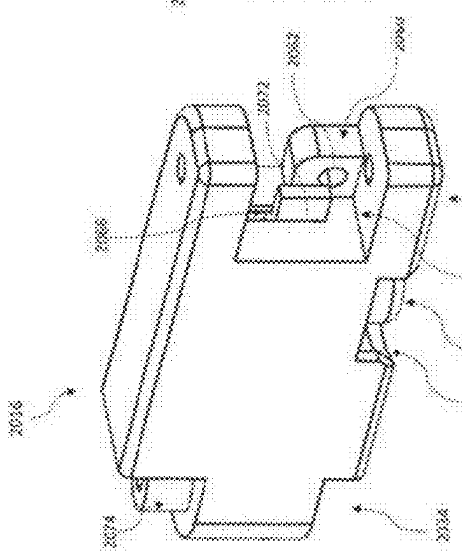


FIG. 27A

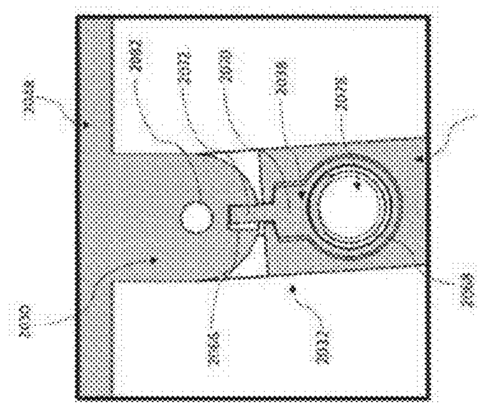


FIG. 27D

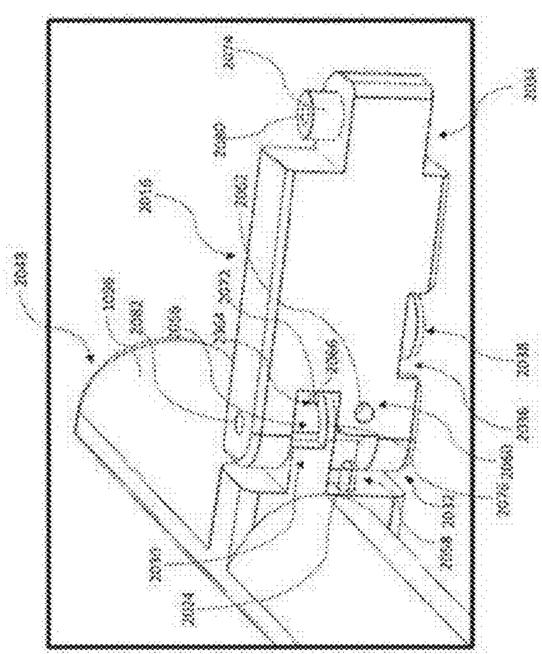


FIG. 27C

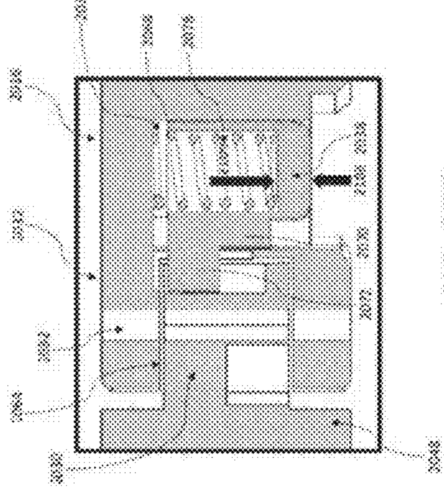


FIG. 27F

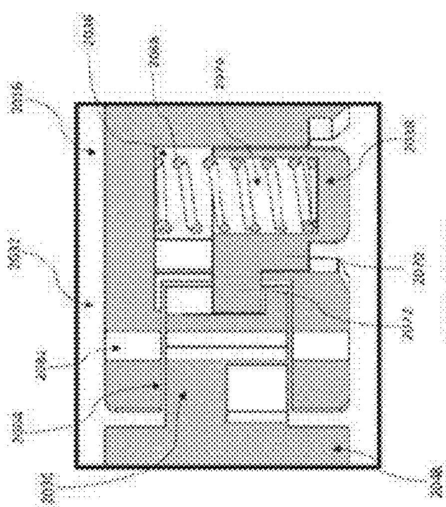


FIG. 27E

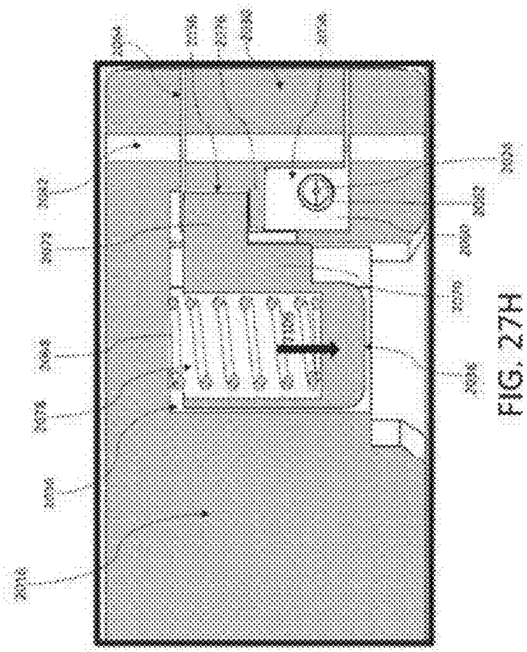


FIG. 27H

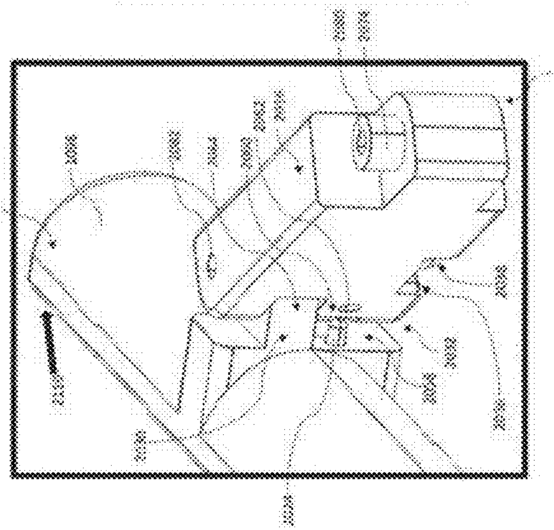


FIG. 27G

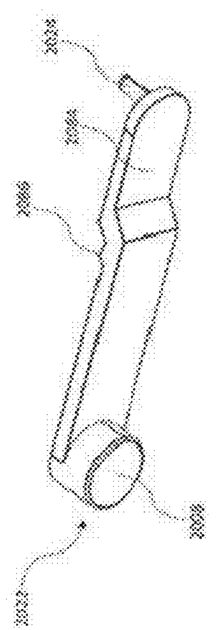


FIG. 28

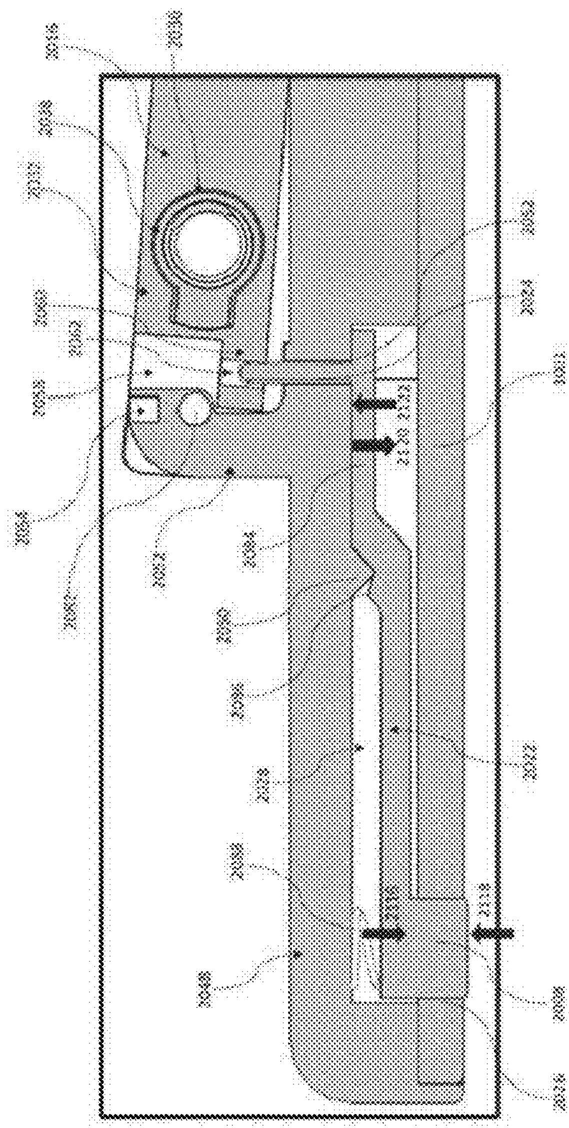


FIG. 29

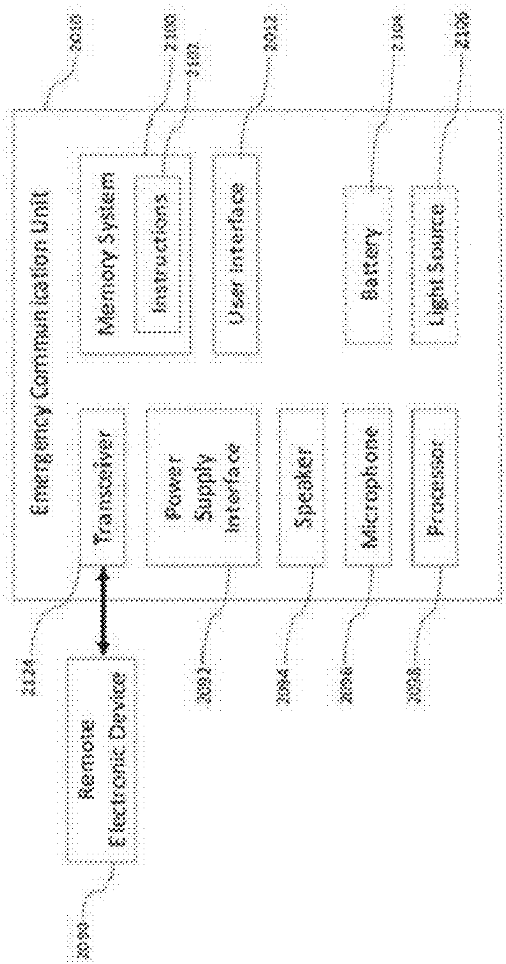

Emergency Communication Unit
Transceiver
Power Supply Interface
Speaker
Microphone
Processor
Memory System
Instructions
User Interface
Battery
Light Source
Remote Electronic Device

FIG. 30

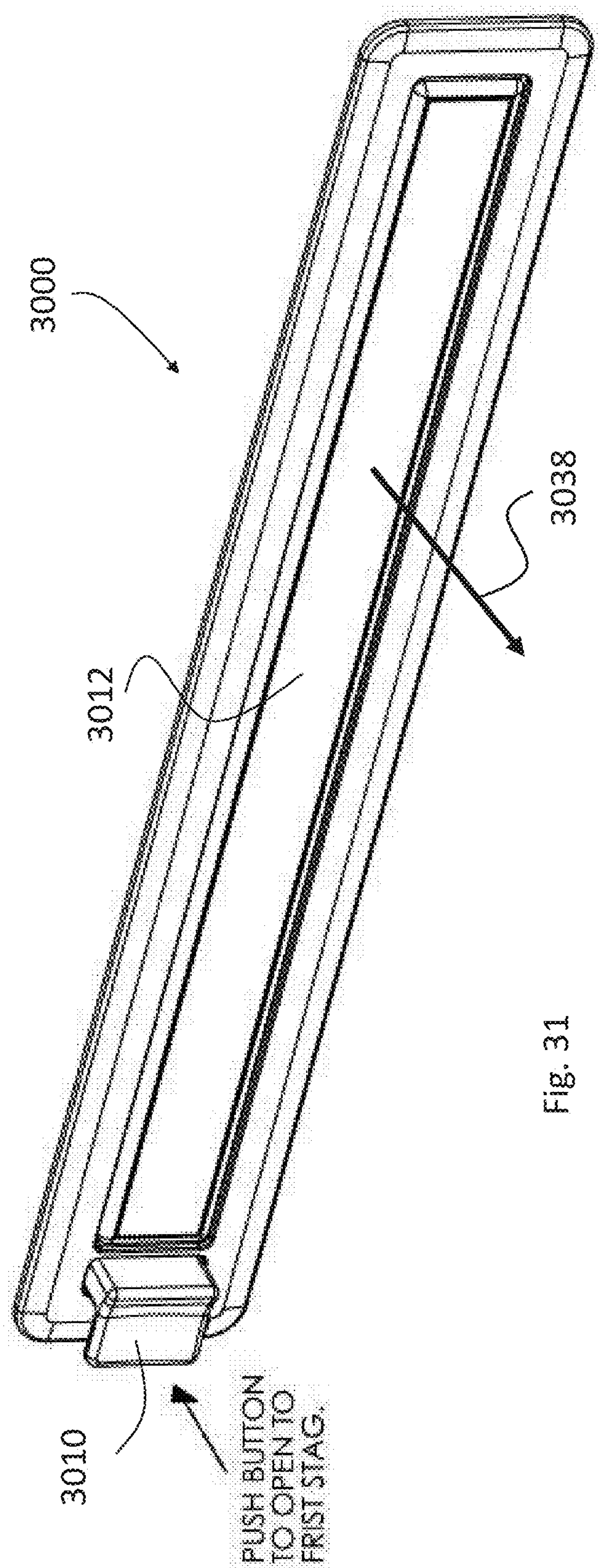


Fig. 31

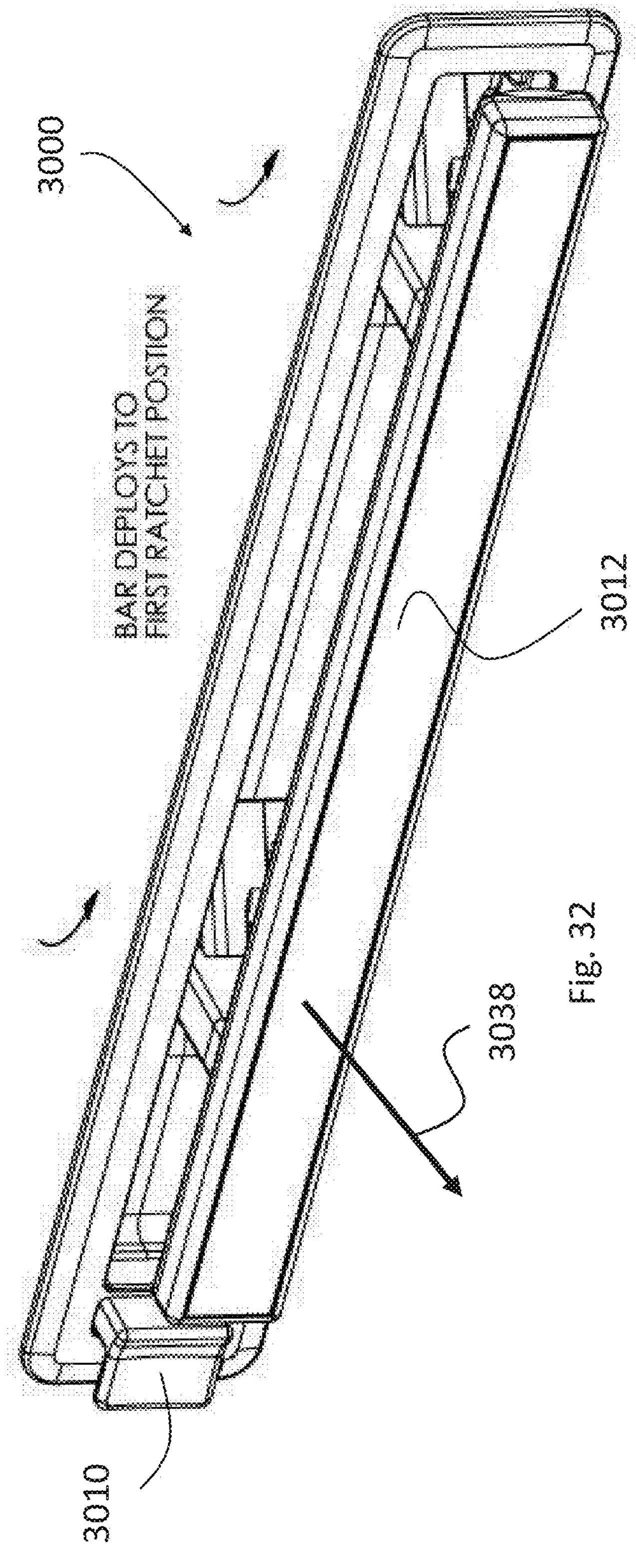


Fig. 32

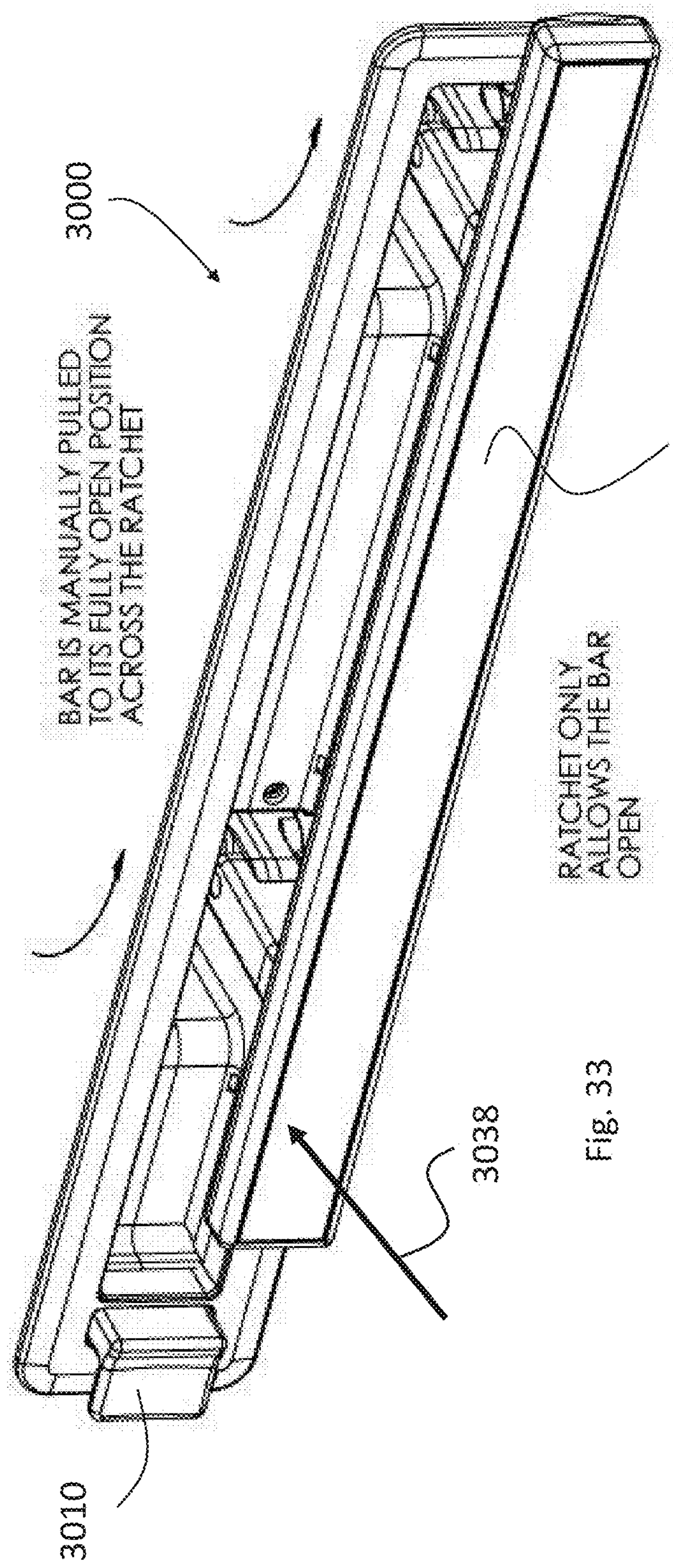

3000
BAR IS MANUALLY PULLED
TO ITS FULLY OPEN POSITION
ACROSS THE RATCHET
3012
RATCHET ONLY
ALLOWS THE BAR
OPEN
3038
3010

Fig. 33

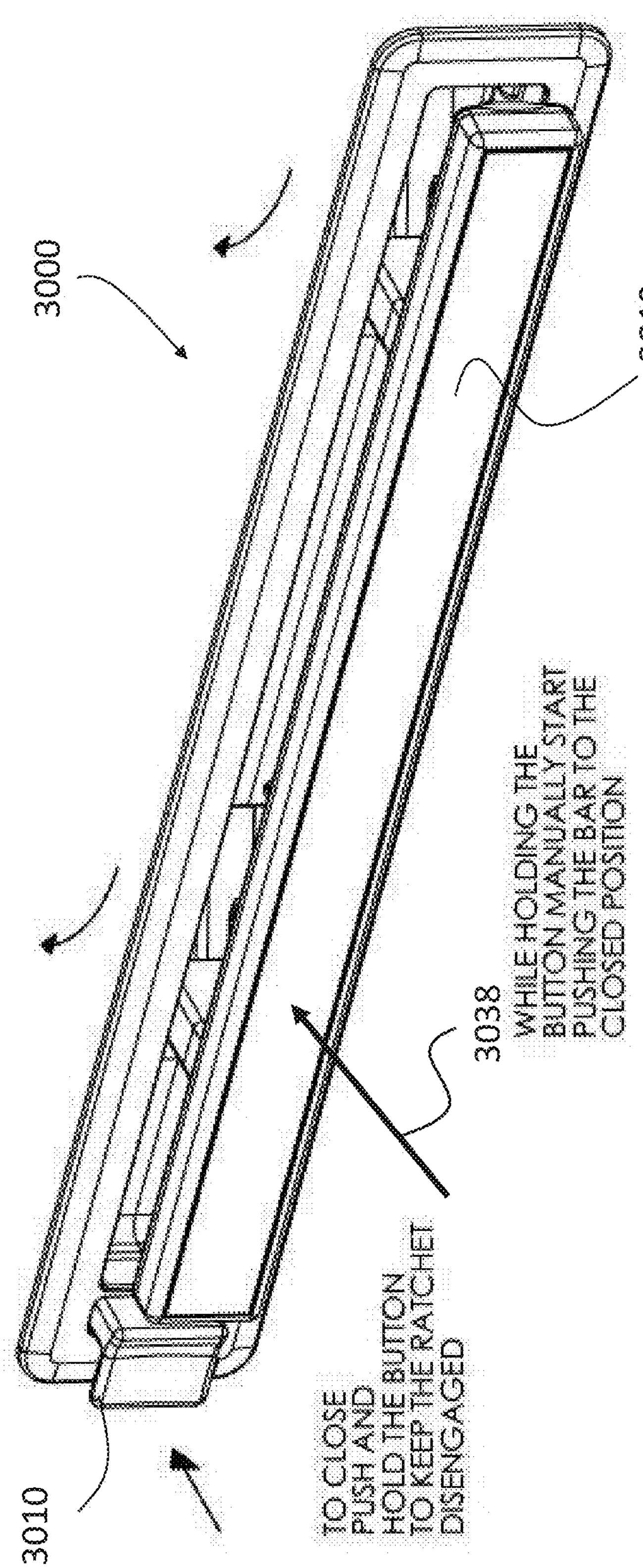

3000
3012
WHILE HOLDING THE BUTTON MANUALLY START PUSHING THE BAR TO THE CLOSED POSITION
3038
TO CLOSE PUSH AND HOLD THE BUTTON TO KEEP THE RATCHET DISENGAGED
3010

Fig. 34

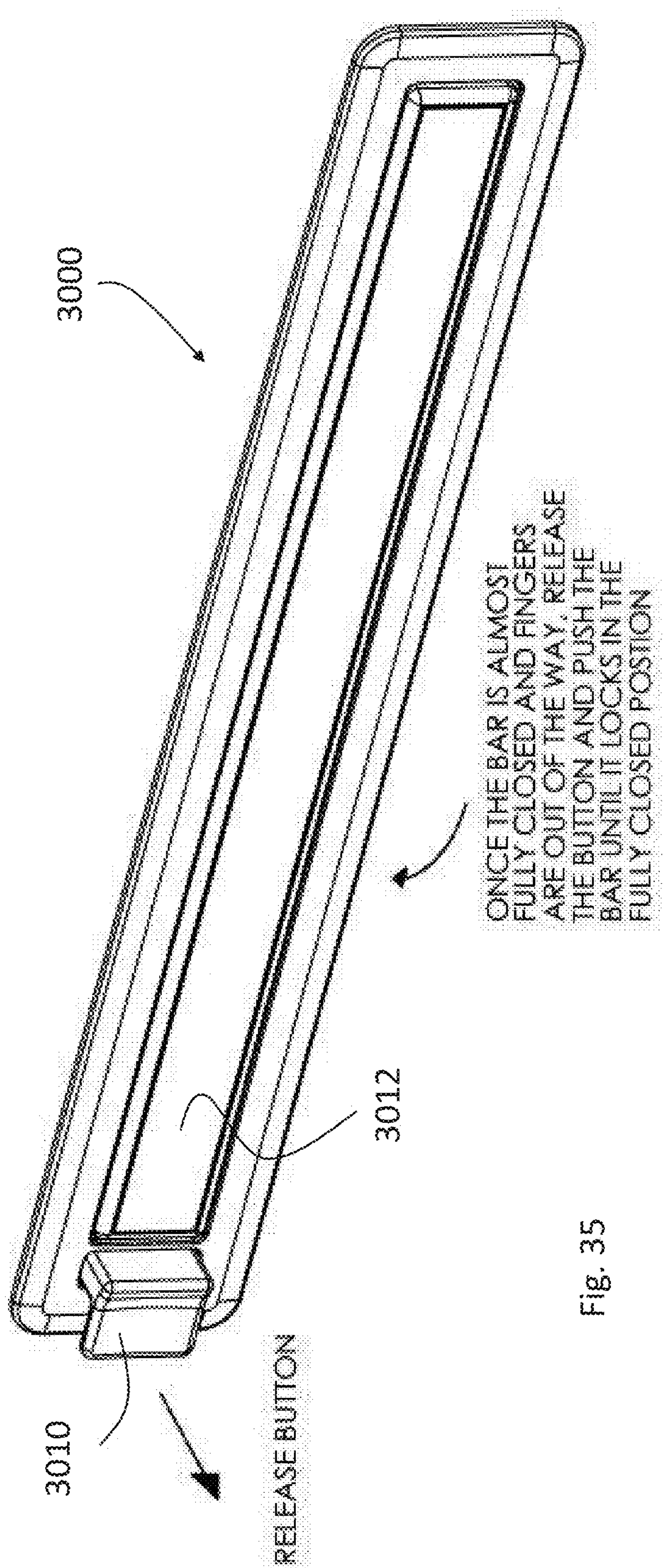

3000
ONCE THE BAR IS ALMOST FULLY CLOSED AND FINGERS ARE OUT OF THE WAY, RELEASE THE BUTTON AND PUSH THE BAR UNTIL IT LOCKS IN THE FULLY CLOSED POSTION
3012
3010
RELEASE BUTTON

Fig. 35

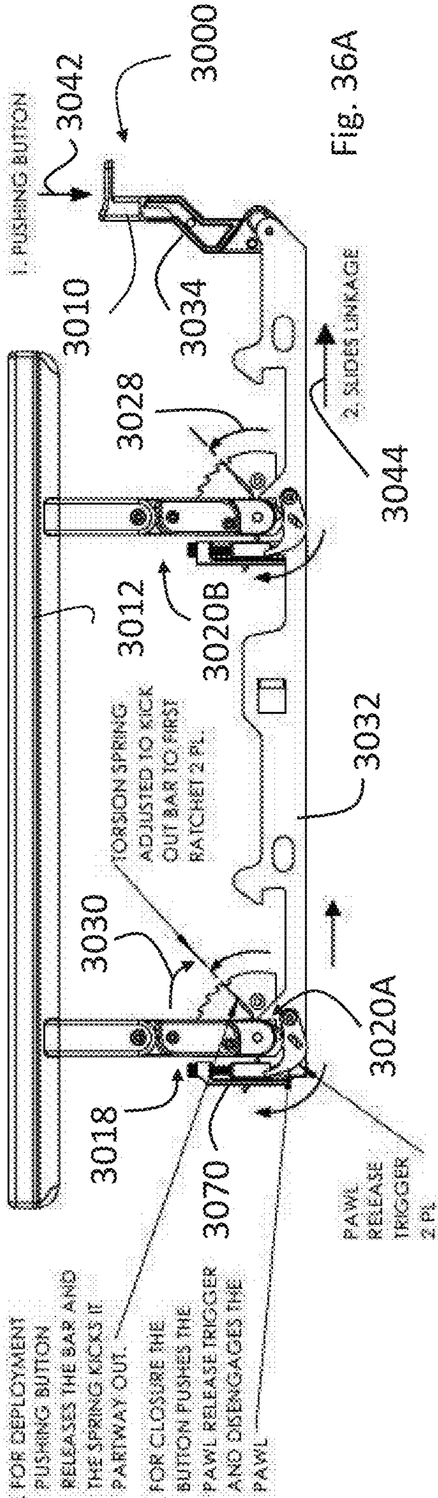


Fig. 36A

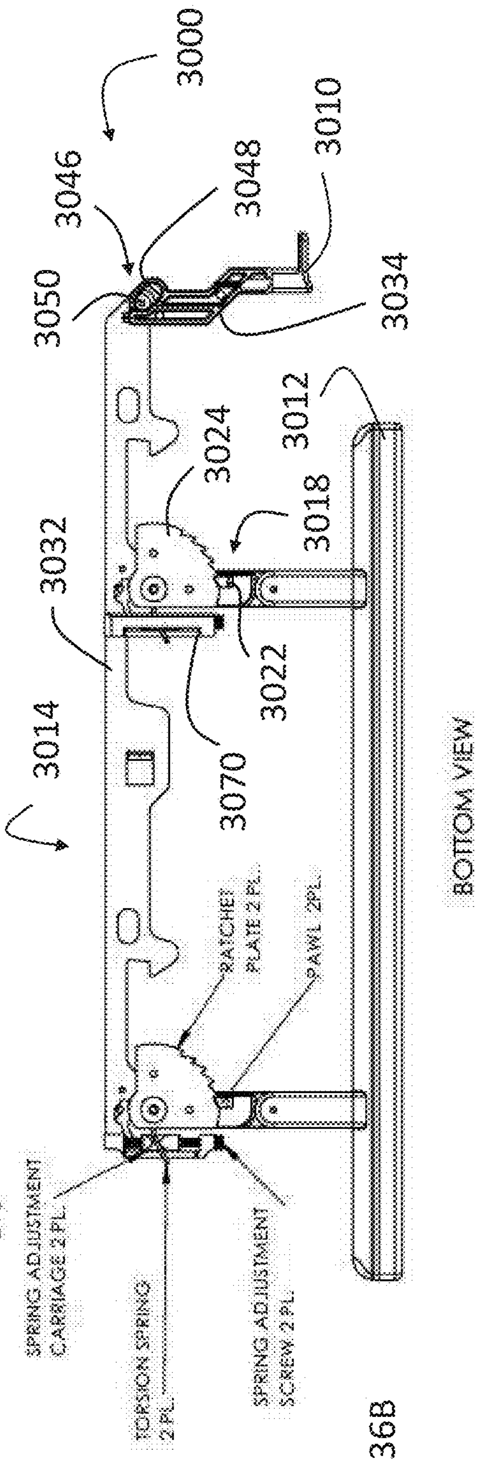


Fig. 36B

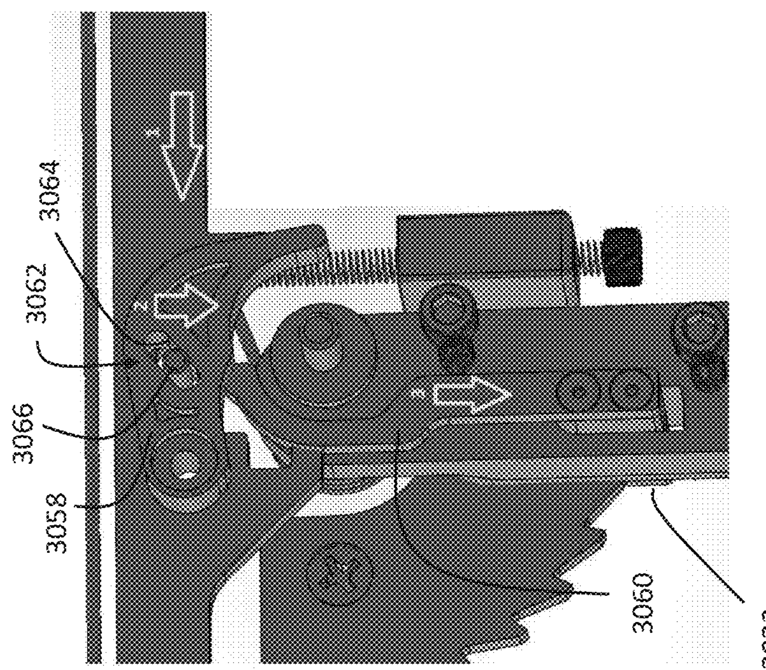


Fig. 37A

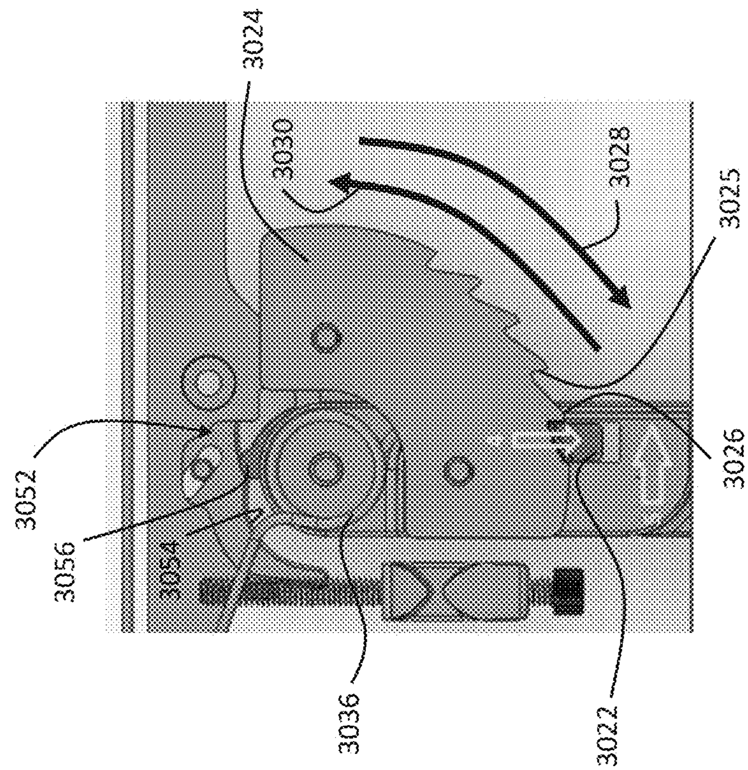
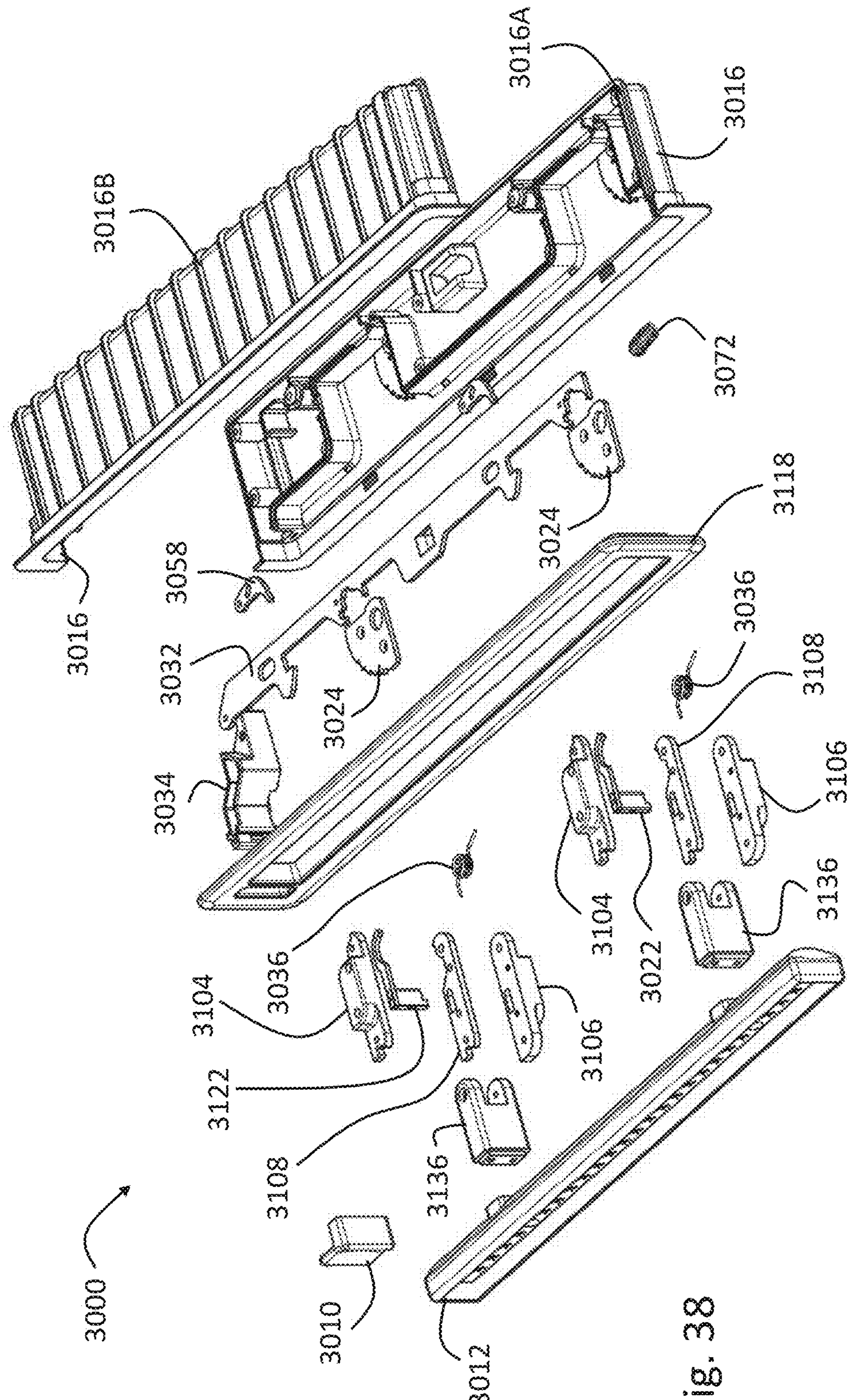


Fig. 37B


3016B
3016A
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3016
3058
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3118
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3104
3104
3022
3136
3122
3106
3108
3136
3000
3010
3012

Fig. 38

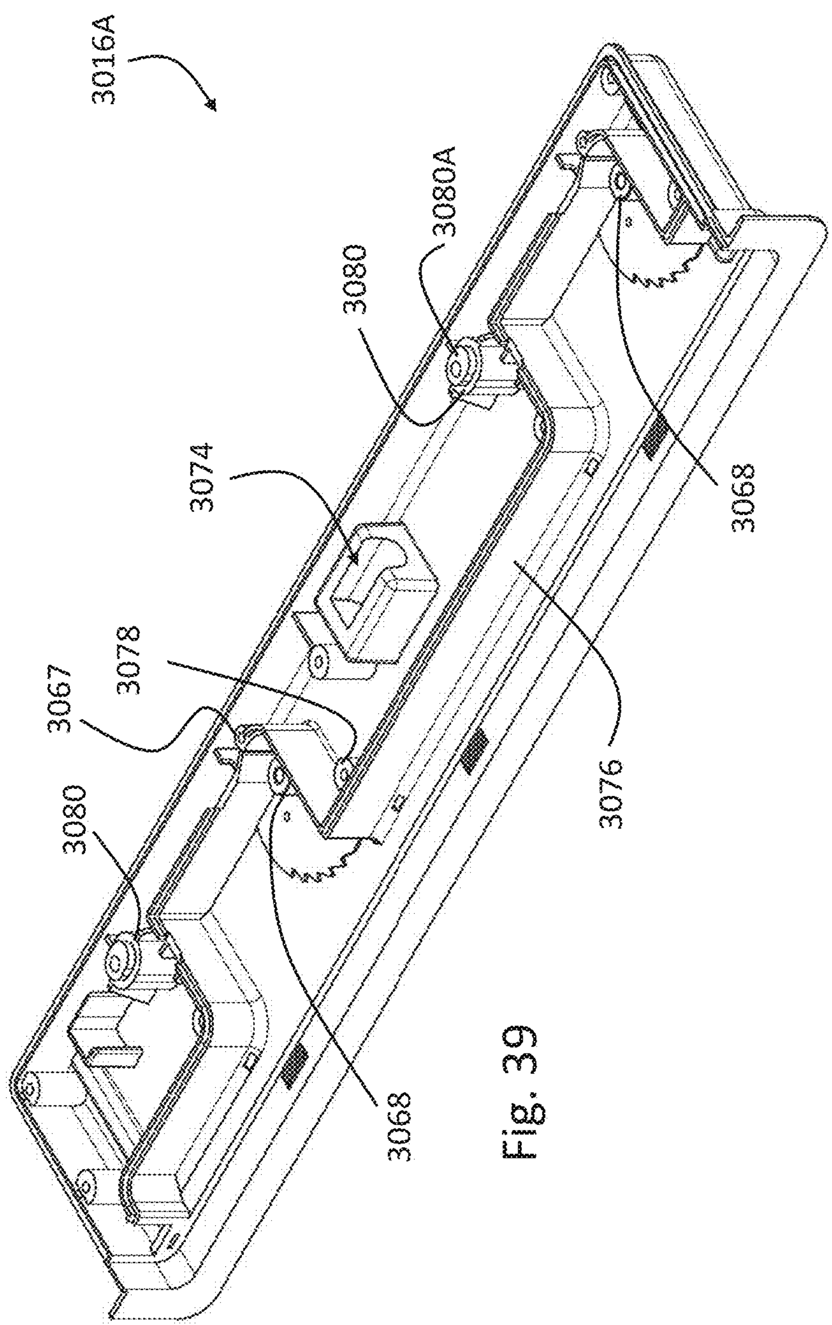


Fig. 39

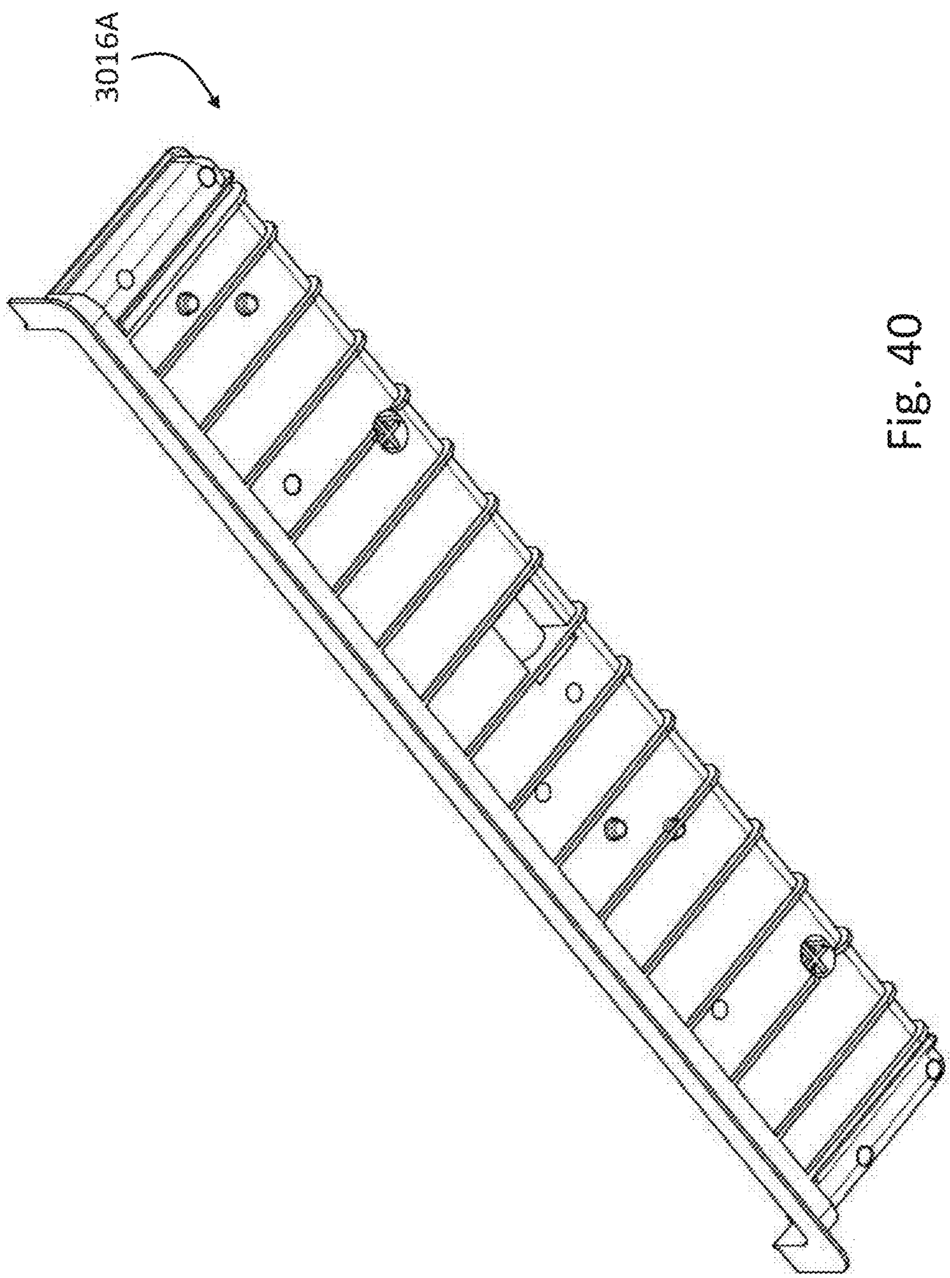


Fig. 40

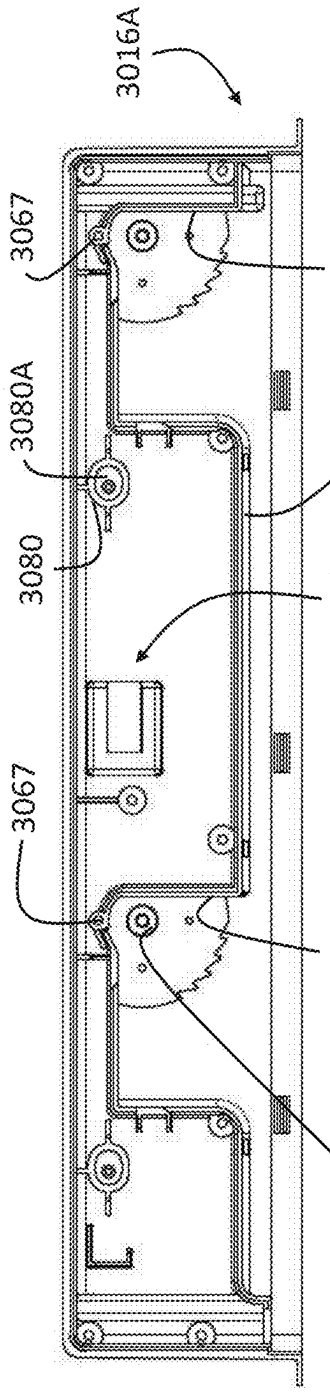


Fig. 41

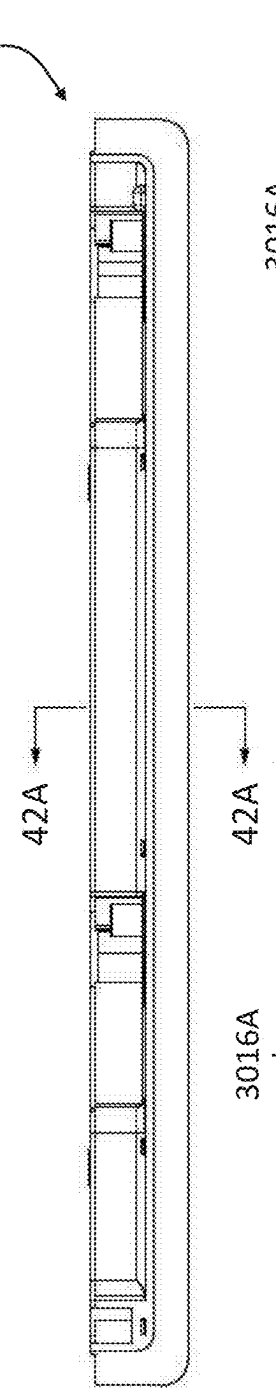


Fig. 42

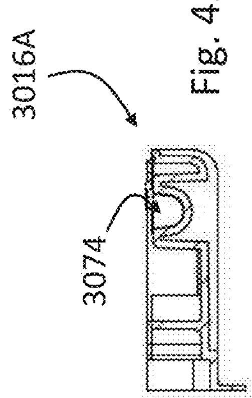


Fig. 42A

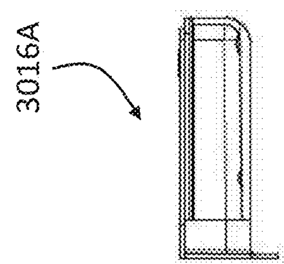


Fig. 43

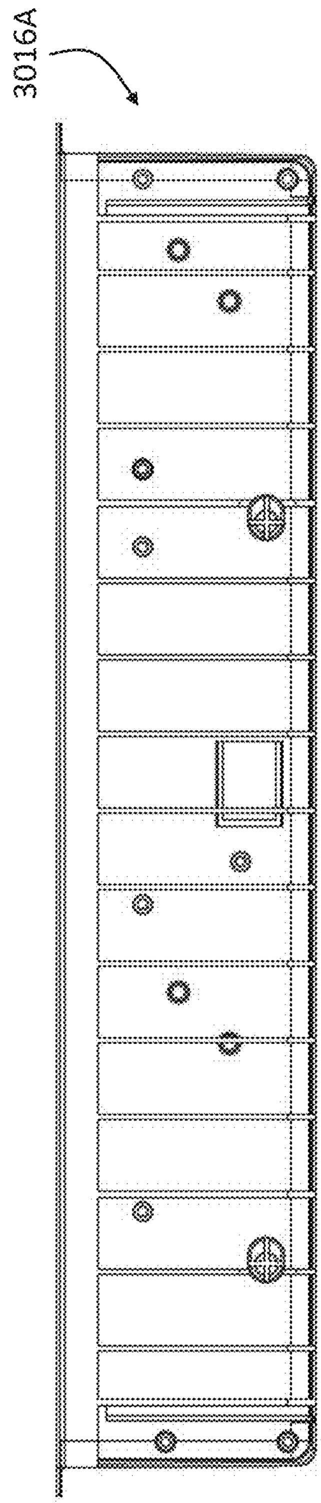


Fig. 44

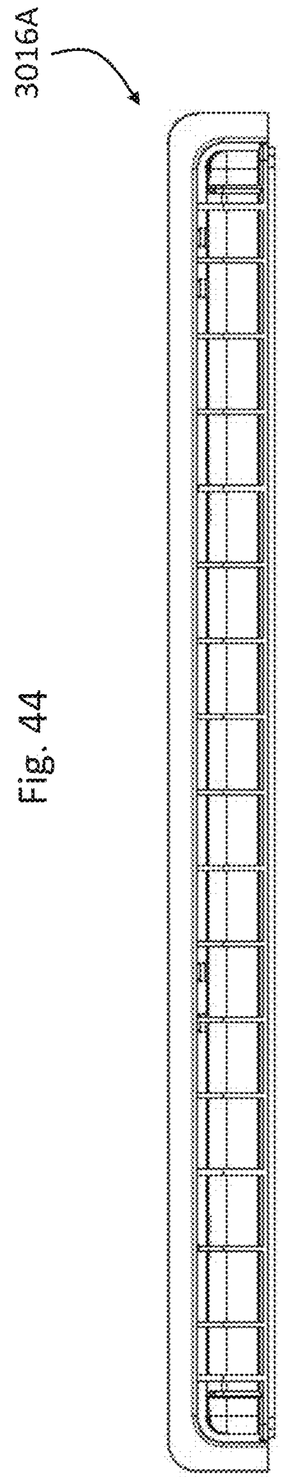


Fig. 45

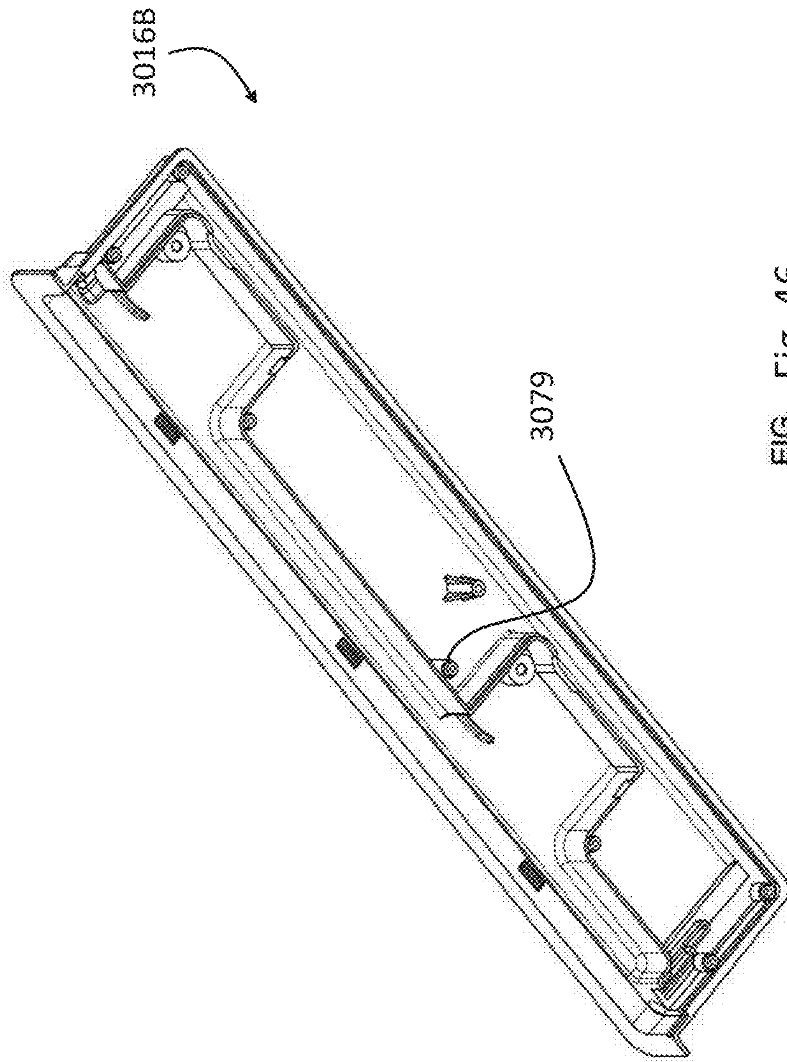


FIG. Fig. 46

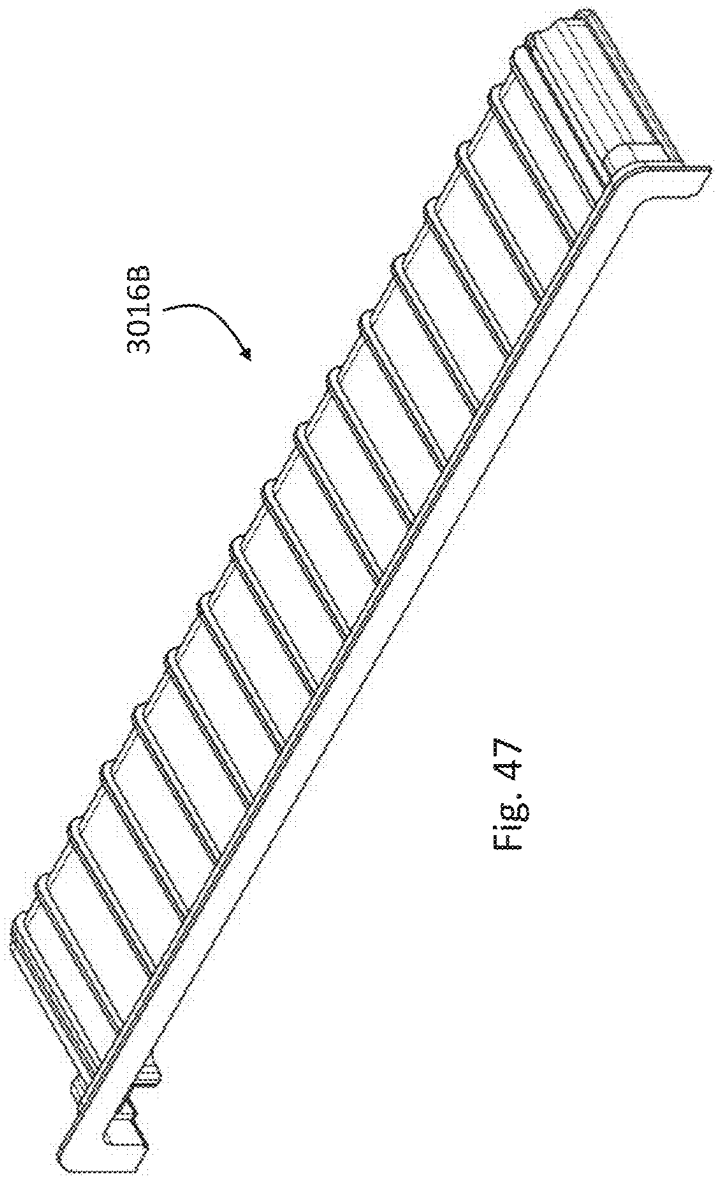


Fig. 47

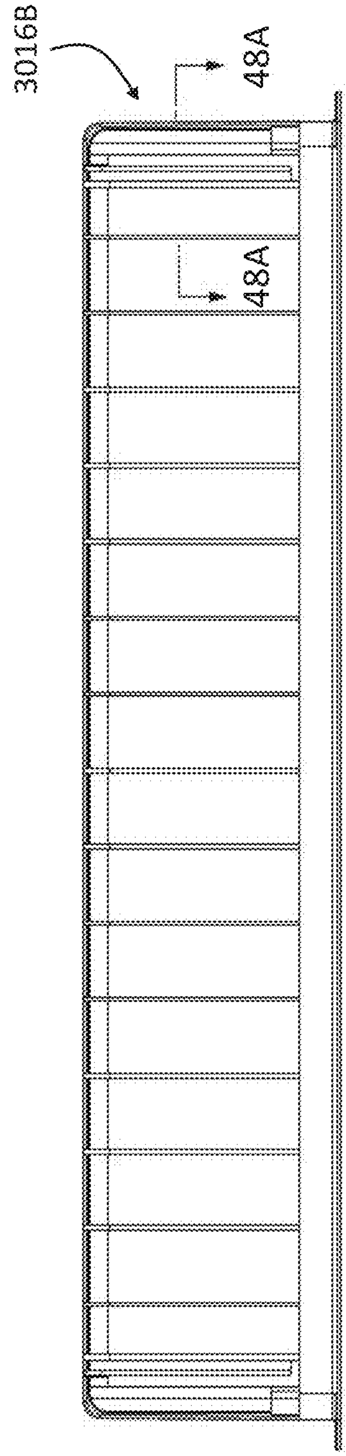


Fig. 48

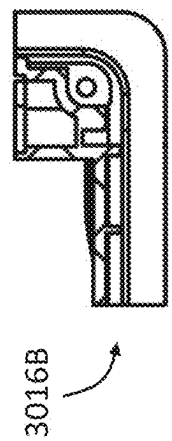


Fig. 48A

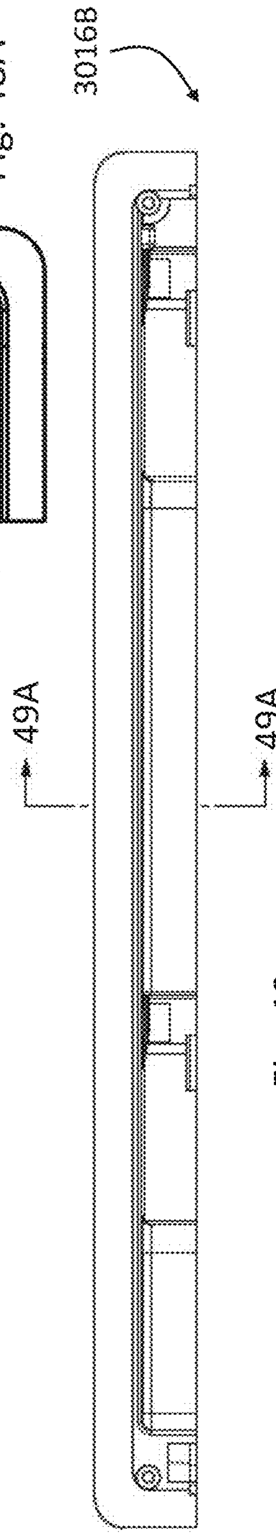


Fig. 49

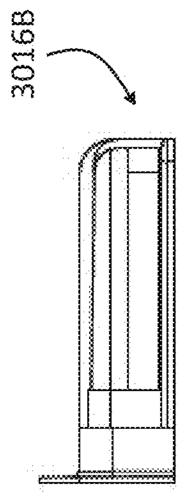


Fig. 50

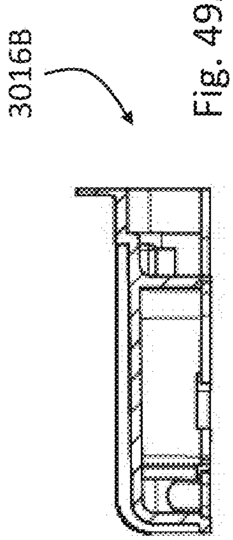


Fig. 49A

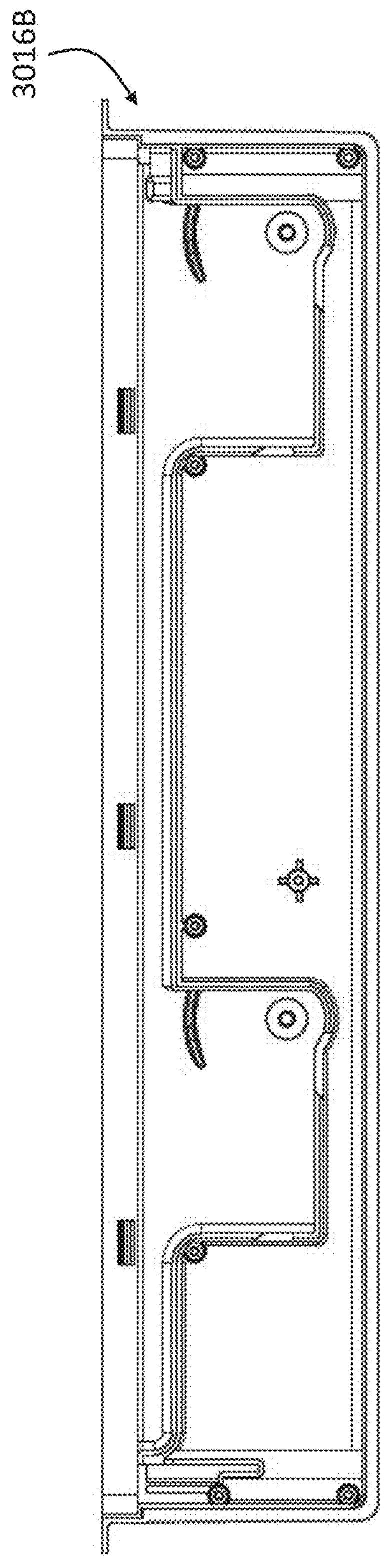


Fig. 51

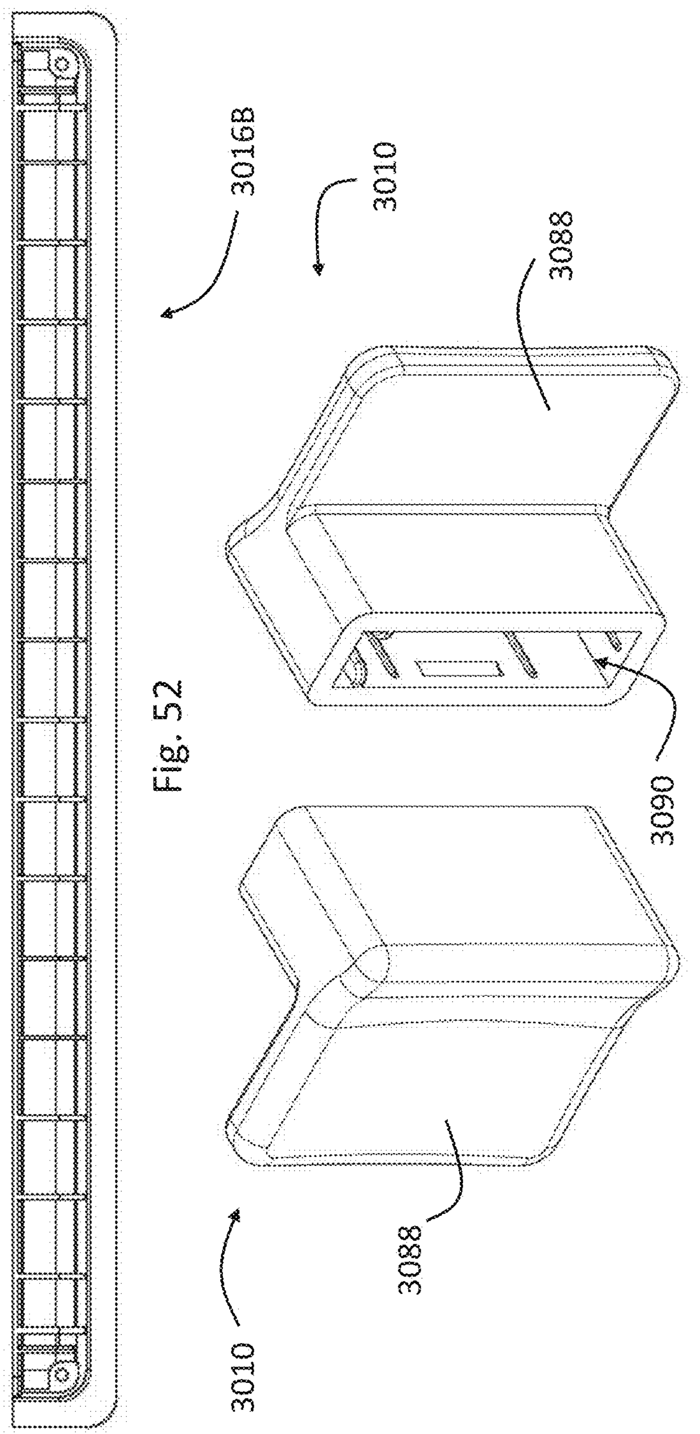


Fig. 52

Fig. 53B

Fig. 53A

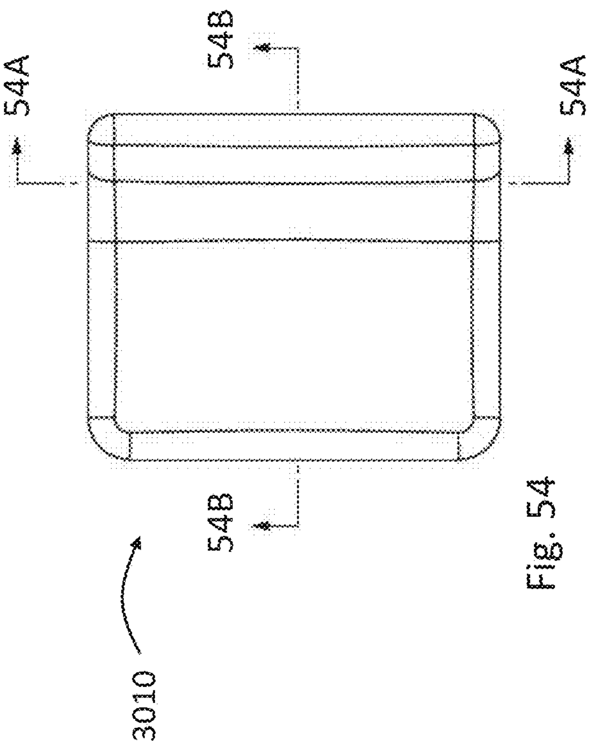


Fig. 54

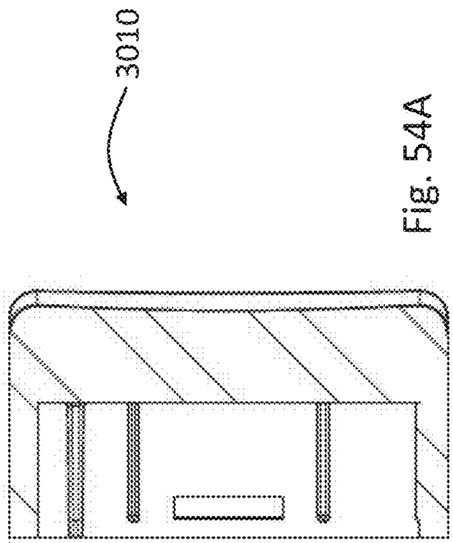


Fig. 54A

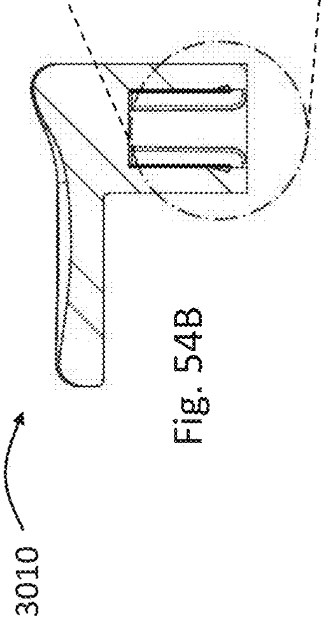


Fig. 54B

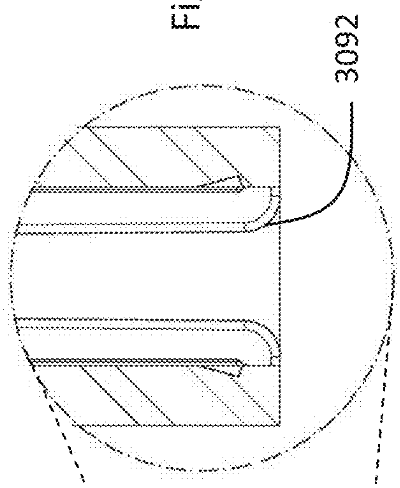


Fig. 54C

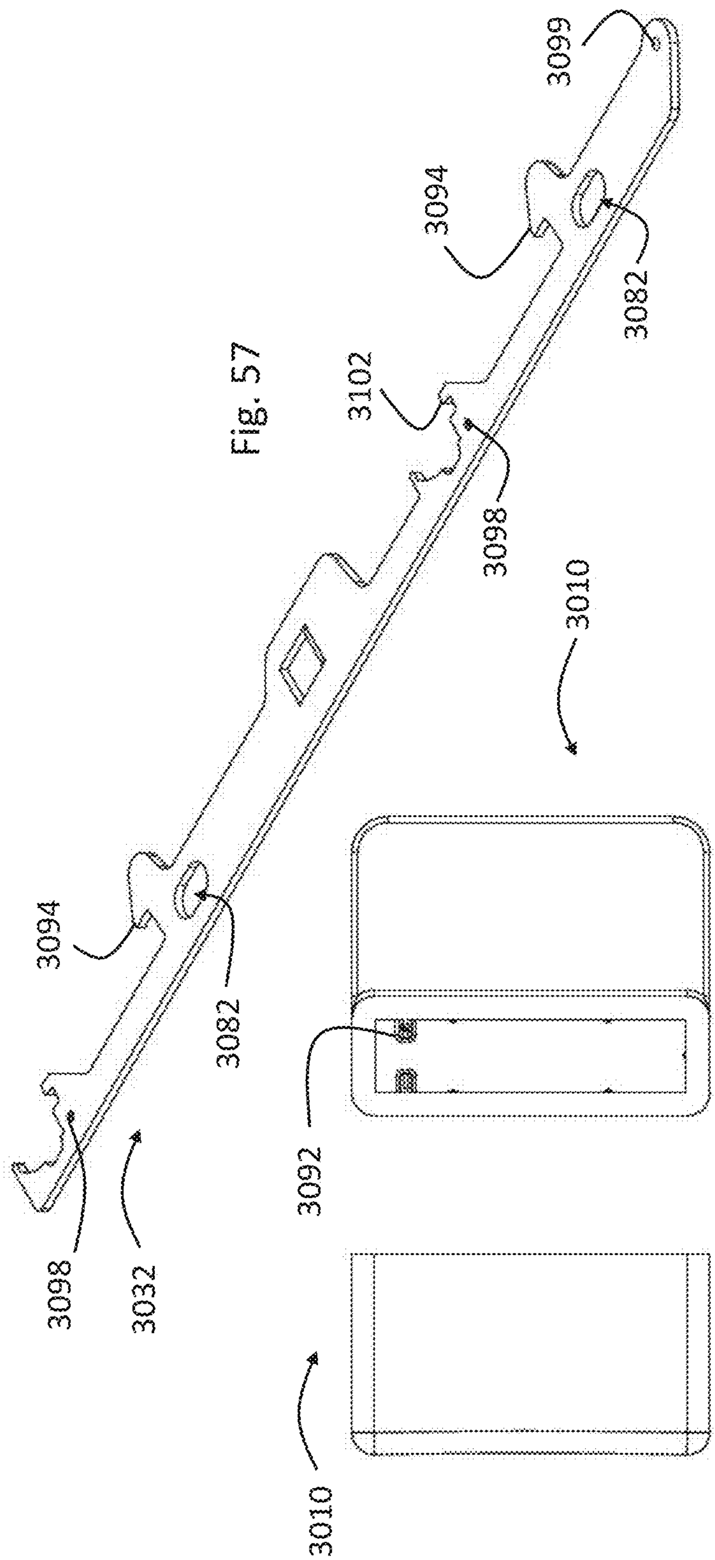


Fig. 57

Fig. 56

Fig. 55

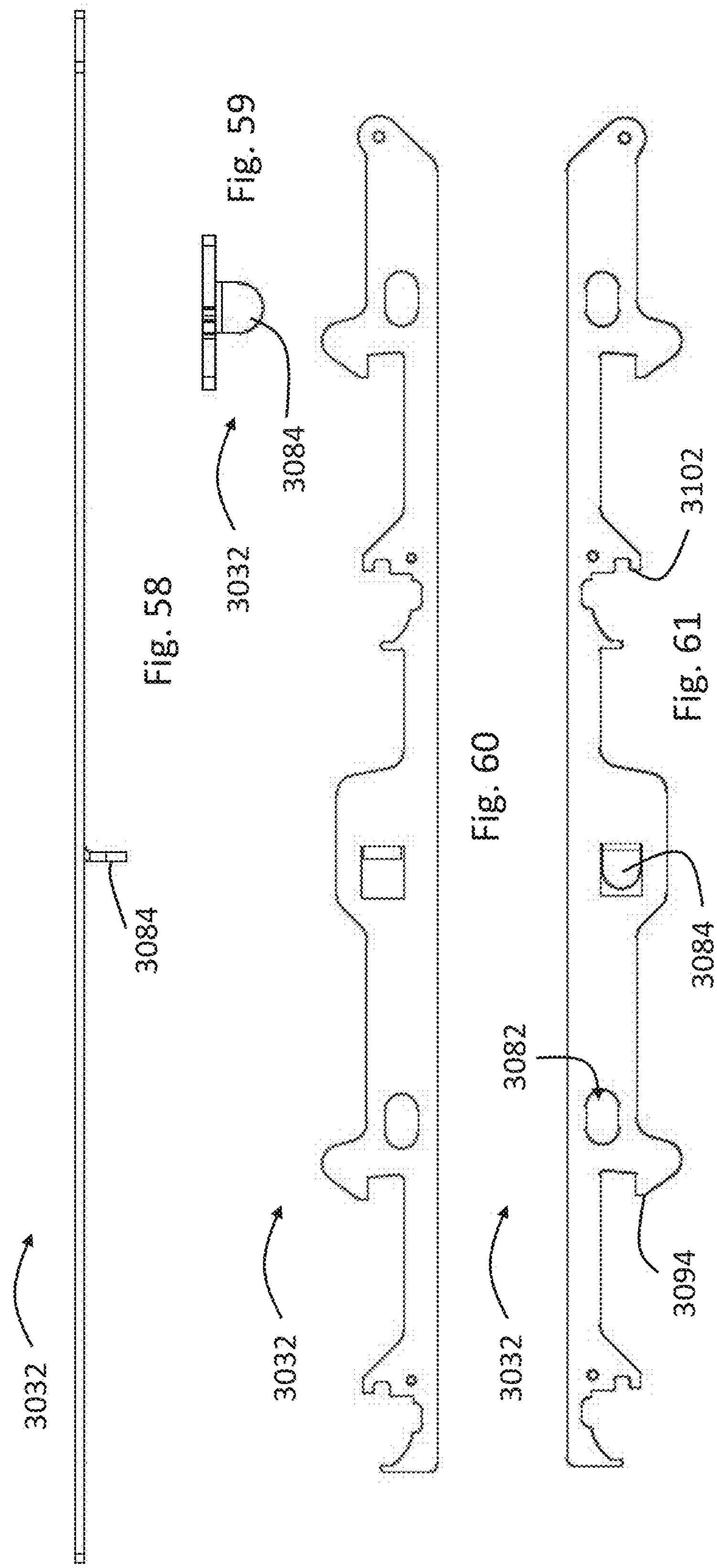


Fig. 58

Fig. 59

Fig. 60

Fig. 61

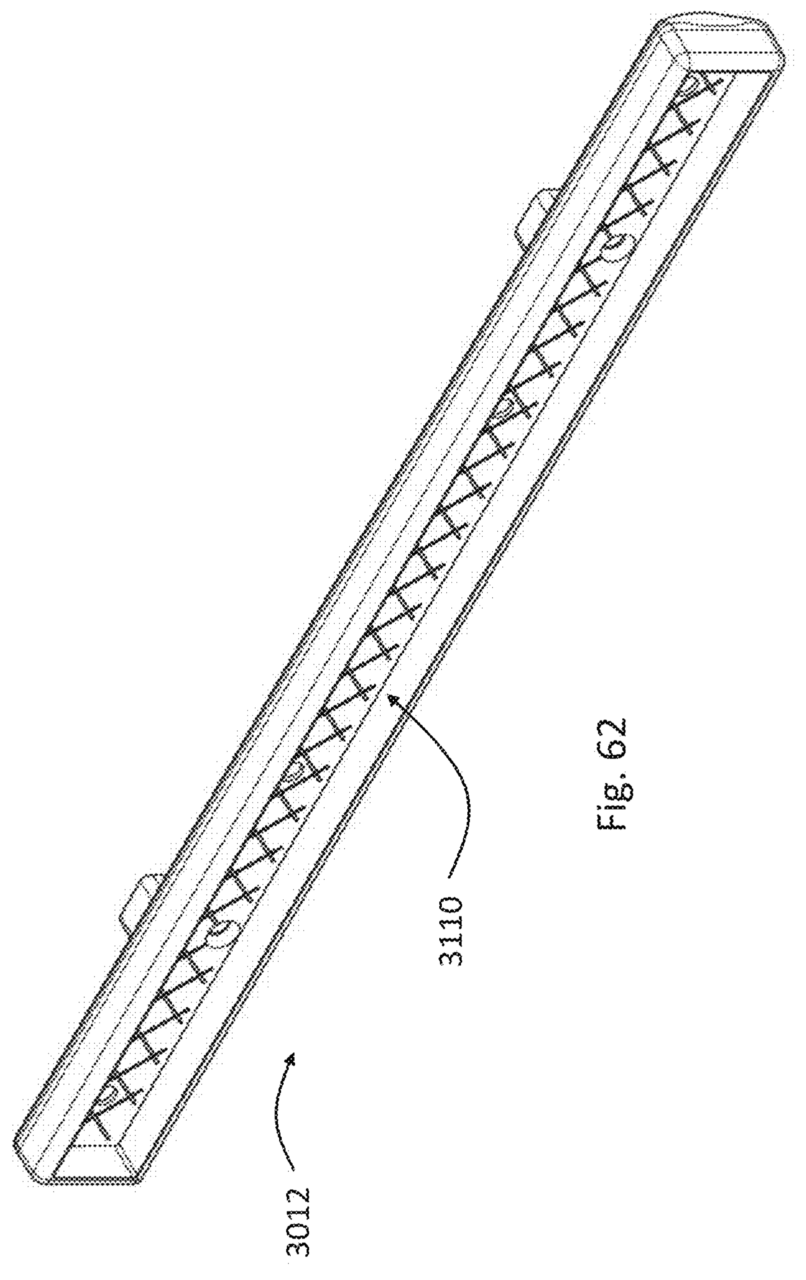


Fig. 62

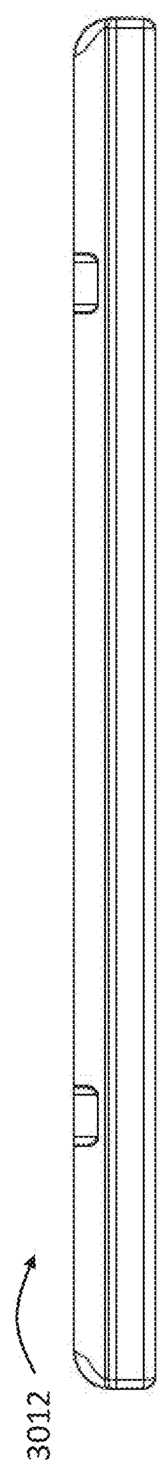


Fig. 63

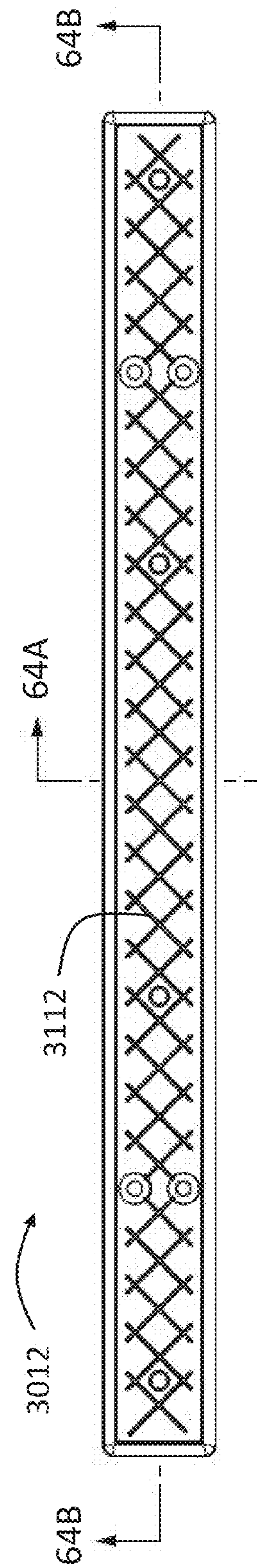


Fig. 64

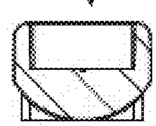


Fig. 64A

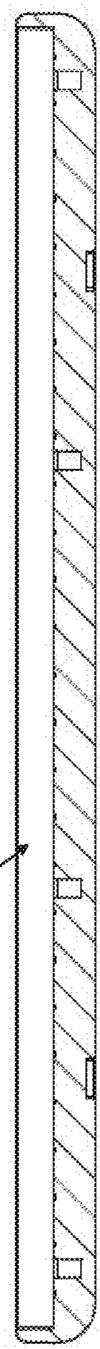


Fig. 64B

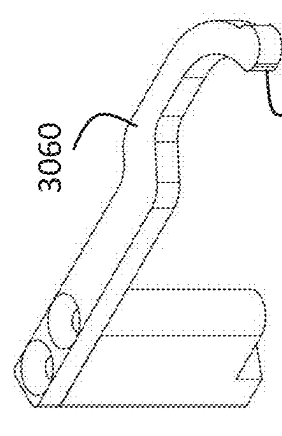


Fig. 65

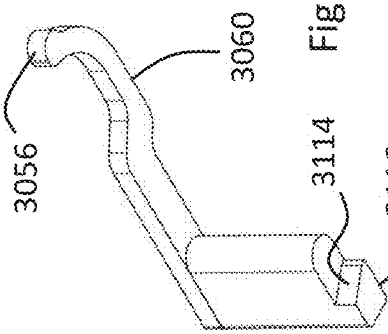
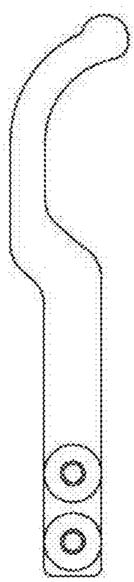
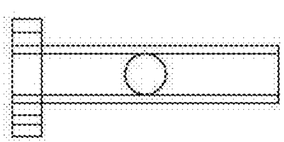


Fig. 66



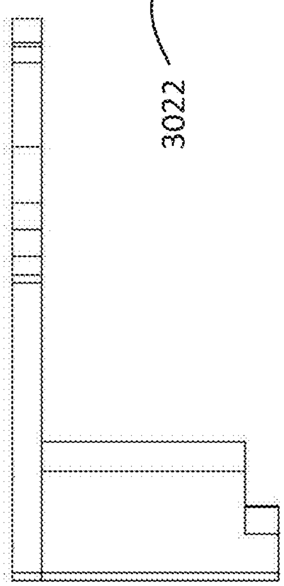
3022

Fig. 67



3022

Fig. 68



3022

Fig. 69

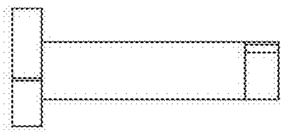


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Fig. 70

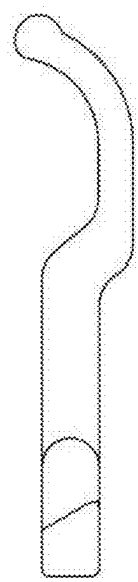

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Fig. 71

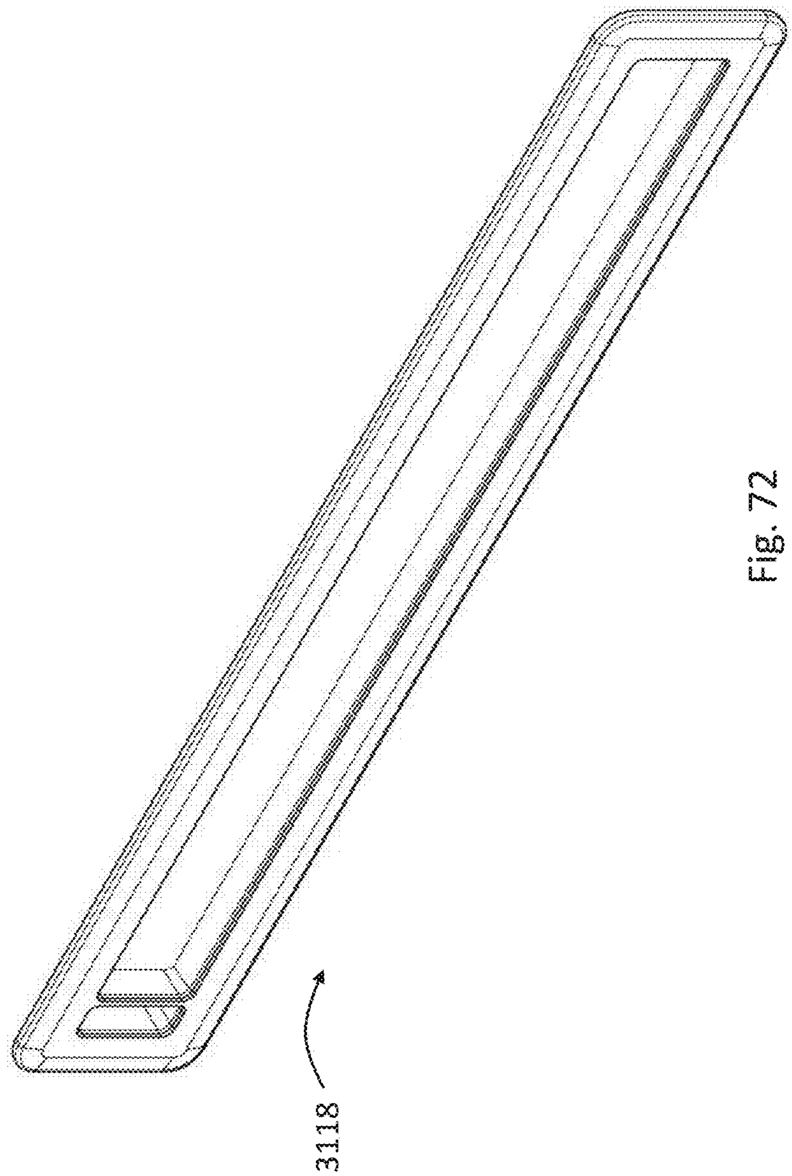


Fig. 72

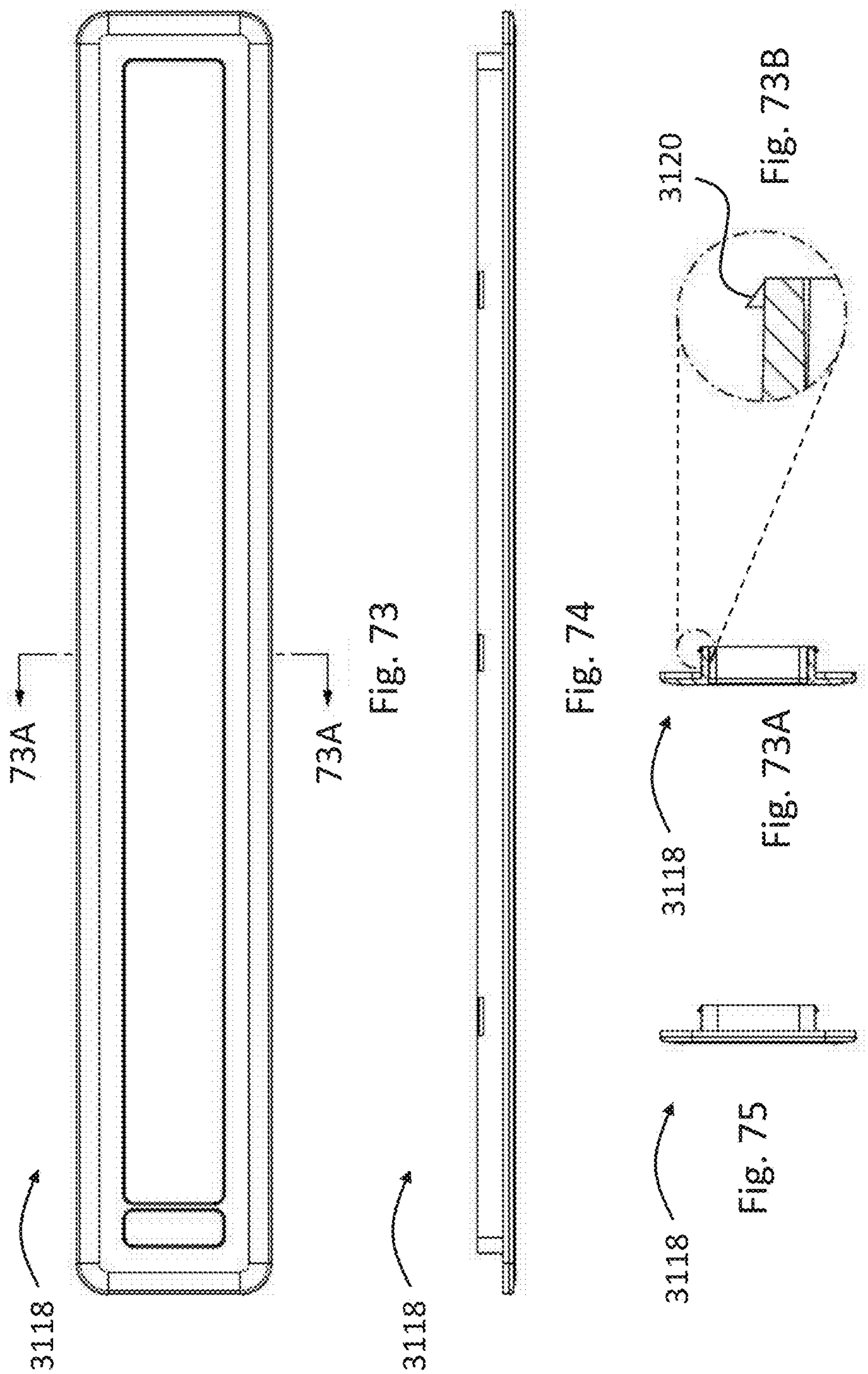


Fig. 73
73A
73A
3118
Fig. 74
3118
Fig. 73B
3120
Fig. 73A
3118
Fig. 75
3118

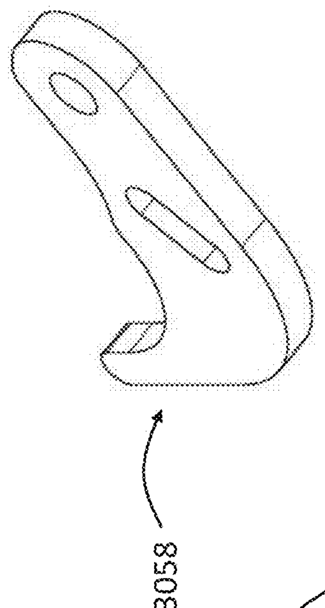


Fig. 77

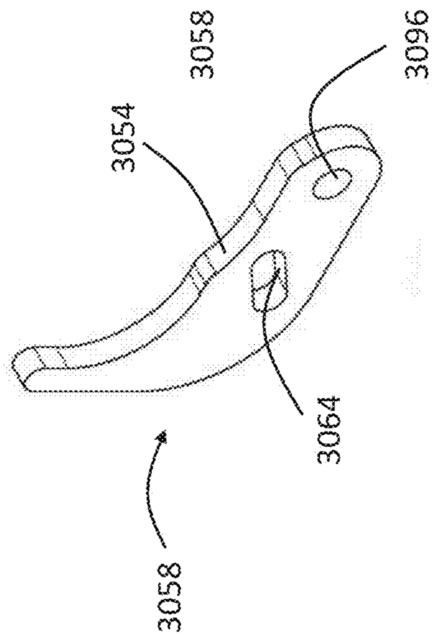


Fig. 76

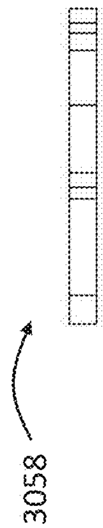


Fig. 79

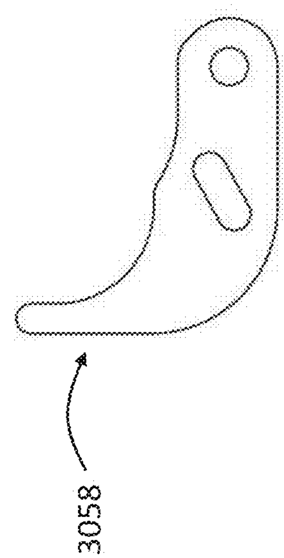


Fig. 78

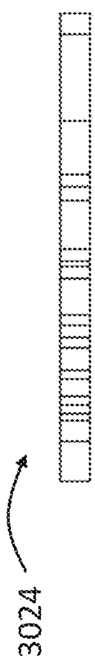


Fig. 81

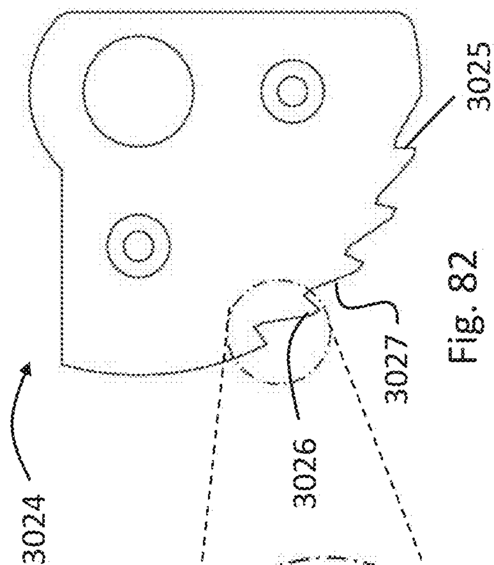


Fig. 82

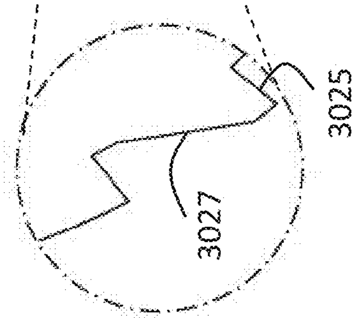


Fig. 82A

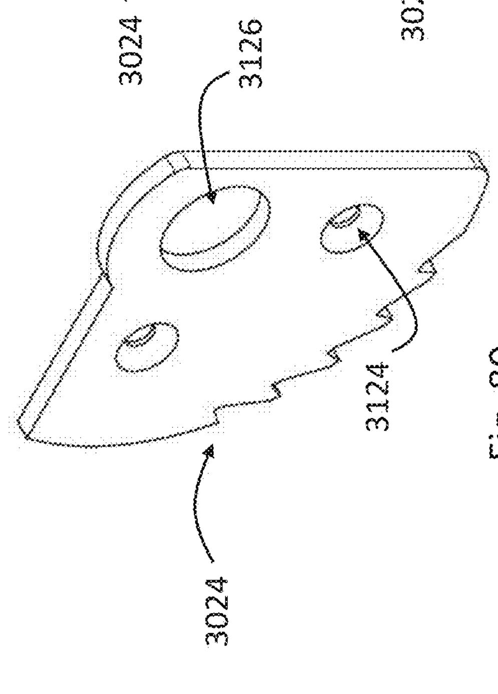


Fig. 80

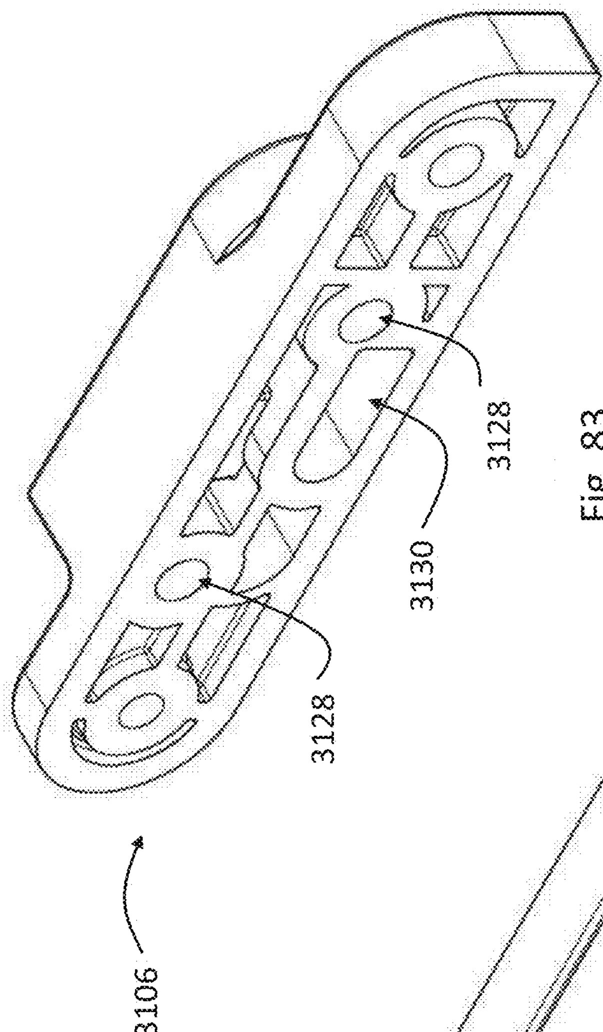


Fig. 83

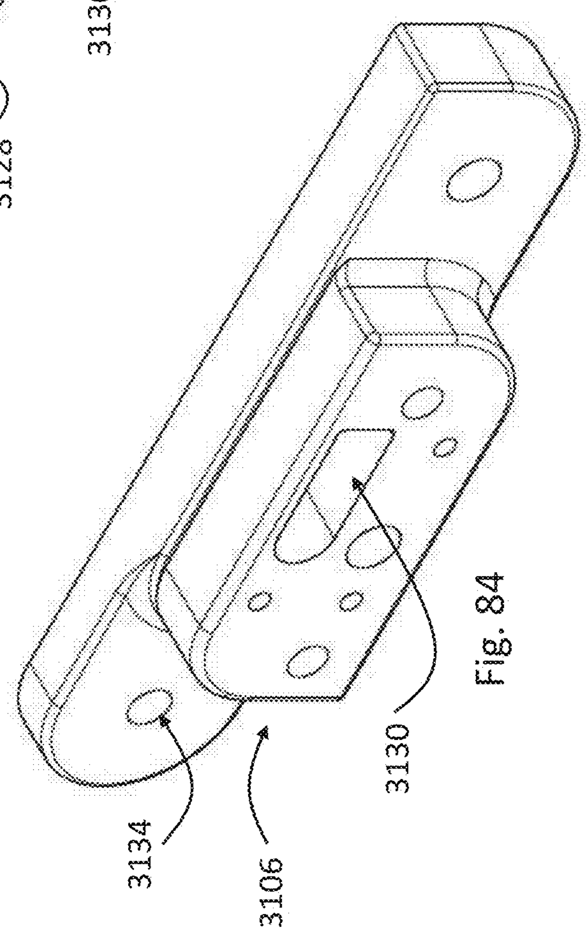


Fig. 84

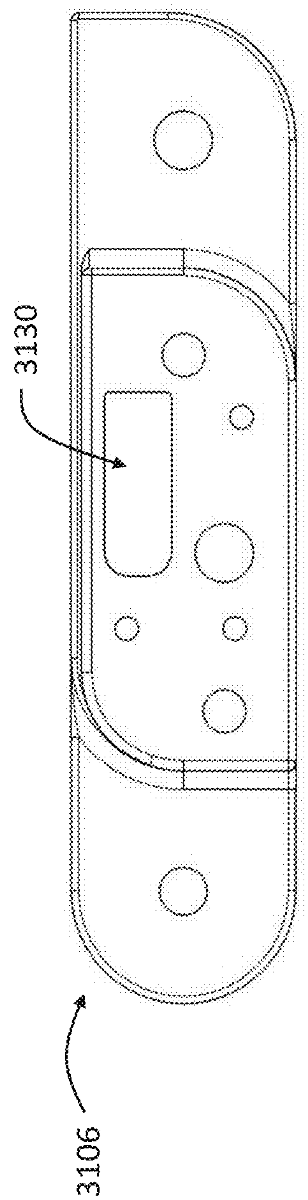


Fig. 85

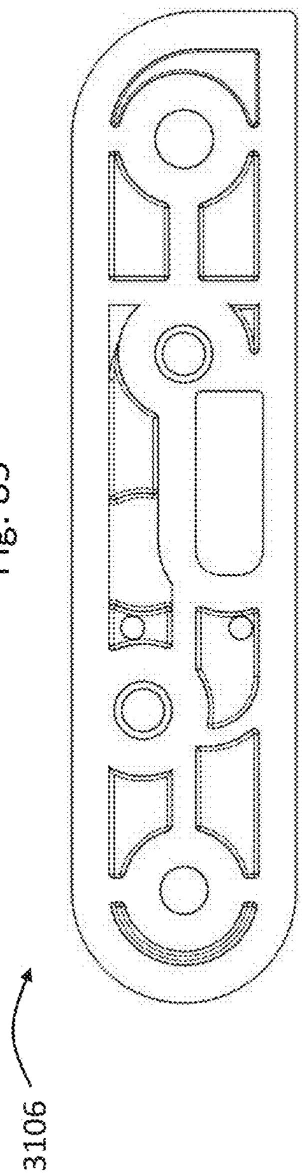


Fig. 86

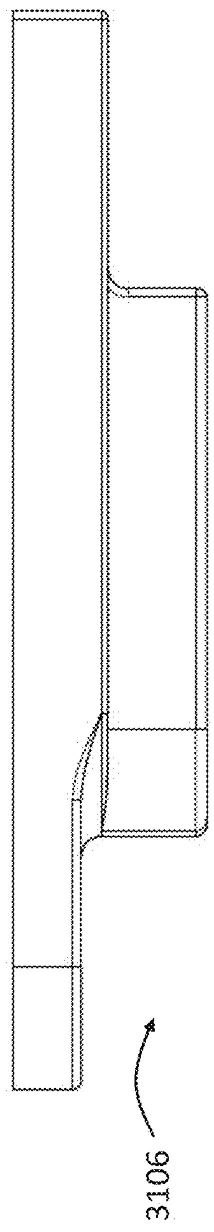


Fig. 87

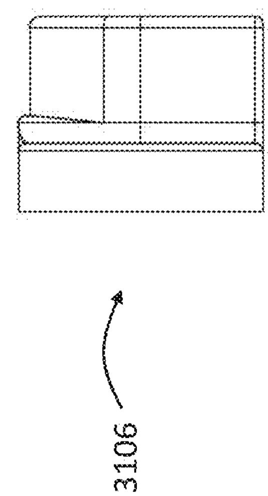


Fig. 88

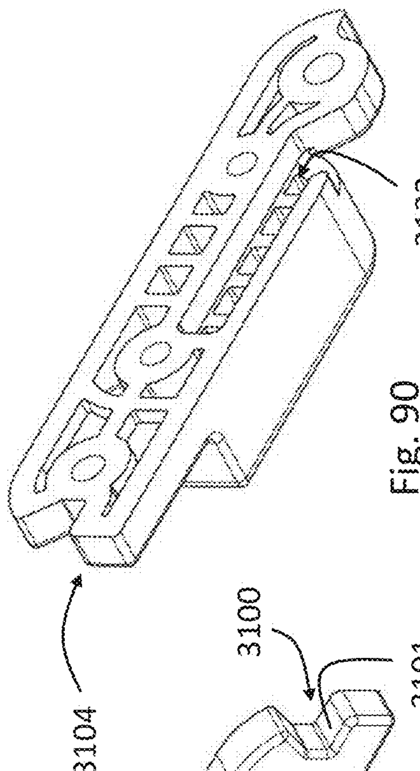


Fig. 90

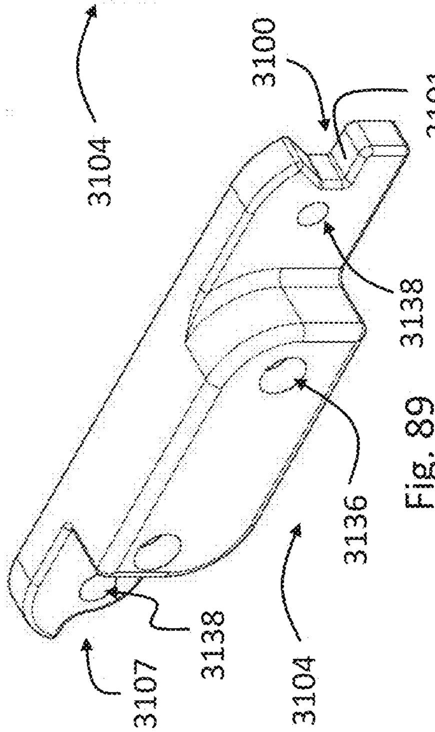


Fig. 89

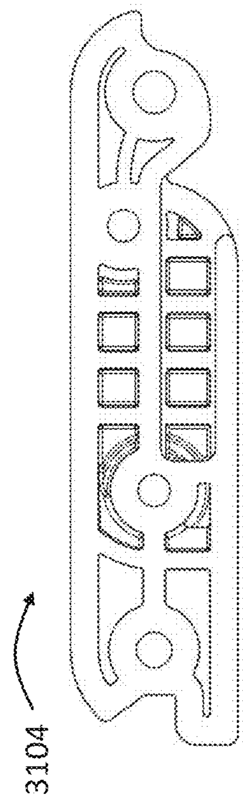


Fig. 92

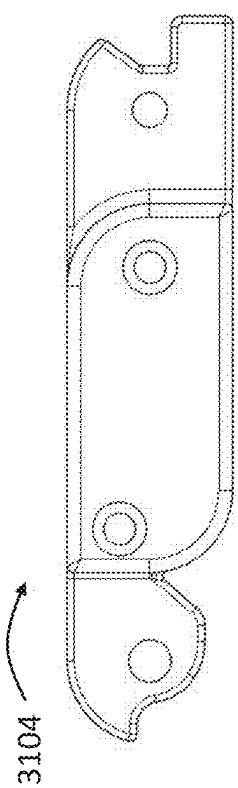


Fig. 91

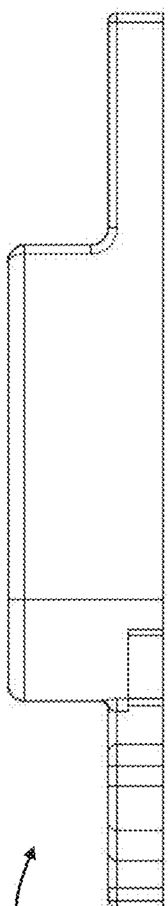


Fig. 93

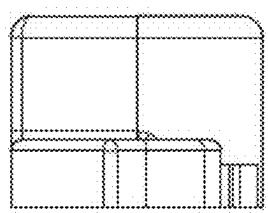


Fig. 94

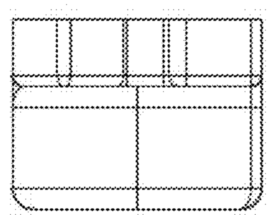


Fig. 95

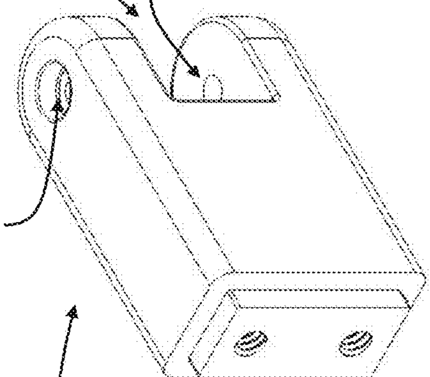
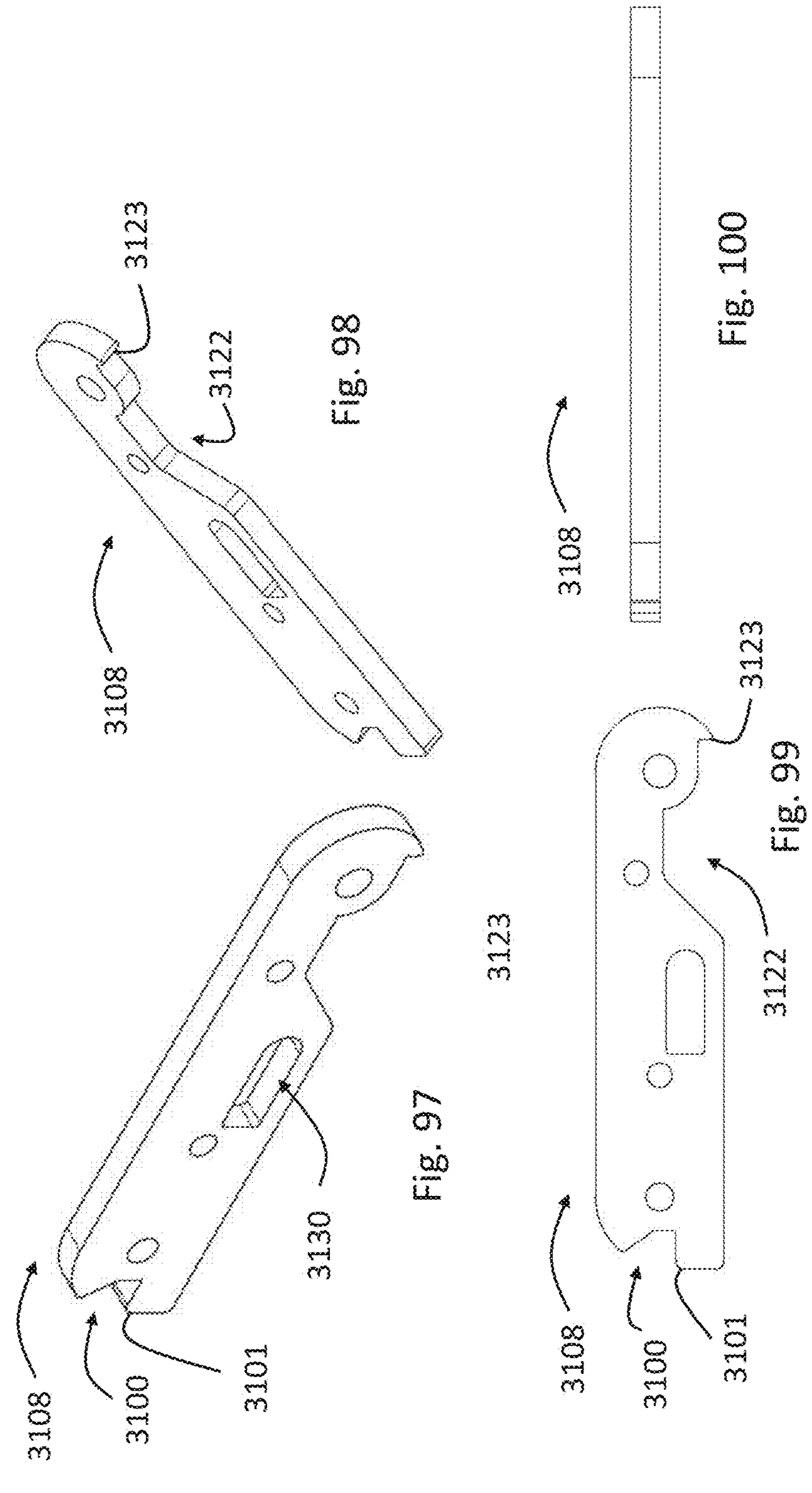


Fig. 96


3123
3122
3108
Fig. 98
3108
Fig. 100
3123
3130
3108
3100
3101
Fig. 97
3123
Fig. 99
3122
3108
3100
3101

DEPLOYABLE HANDLE DEVICES

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Application No. 63/557,829 filed Feb. 26, 2024 entitled Deployable Handle Devices, the entirety of this application being incorporated herein by reference.

FIELD

[0002] This disclosure relates to components for improved safety devices such as handle devices, handle assemblies, devices for alerting emergency services, and systems and methods of manufacture and use.

BACKGROUND

[0003] Household or other building structure environments can often be hazardous without handle supports being available. For example, bathing environments are often slippery due to wet conditions, which can result in injury of a person. Due to the private nature of household or bathing environments, injuries can become more dangerous than would otherwise be the case because help from individuals is not readily available. This can be especially hazardous for older or disabled users.

[0004] This background information is provided for the purpose of making known information believed by the applicant to be of possible relevance to the present disclosure. No admission is necessarily intended, nor should be construed, that any of the preceding information constitutes prior art against the present disclosure.

SUMMARY

[0005] It is desirable to improve the safety in building structures, including household and bathing environments. Handles are common to improve the safety of, for example, bathing environments but handles are not aesthetic and can disrupt the walls of a bathing environment. This disclosure provides handle devices or grab bars that can be stored in the walls of a building, household, bathing, or other environment and can be selectively deployed when the environment is in use, improving the aesthetic of the environment. The handle devices can be used in any environment where a stowable/deployable handle device may be desired, in particular for safety. This disclosure provides handle devices that can include emergency contact capabilities such that a user can contact emergency contacts (e.g., services) if injured or in need of assistance. In some variants, this disclosure provides a universal handle device that can attached to the contoured surfaces of a free standing tub, improving safety despite the inability to attach handles to the walls of a bathing environment.

[0006] According to an aspect of the present disclosure there is provided, which may be used alone or in combination with any other aspects described herein, a deployable handle device comprising:

[0007] a handle configured to be grasped by a user, the handle configured to move between a stowed configuration and a deployed configuration, wherein in the stowed configuration, the handle is configured to be substantially flush with a surrounding wall, and in the deployed configuration, the handle is configured to protrude from the surrounding wall for the user to grasp the handle;

[0008] a handle housing configured to be positioned in the surrounding wall with the handle at least partially positioned in the handle housing;

[0009] a handle arm pivotably connected to the handle housing at a first end and pivotably connected to the handle at a second, opposing end;

[0010] a pawl connected to the handle arm such that the pawl moves with the handle arm, the pawl being configured to move independently from the handle arm between a first pawl position and a second pawl position;

[0011] a ratchet mounted within the housing, the ratchet defining teeth biased to, when the pawl is in the first pawl position, allow movement of the pawl in a first direction and restrict movement of the pawl in a second direction;

[0012] a linkage movably connected to the handle housing, the linkage being configured to move from a first linkage position to a second linkage position, the linkage being configured to move the pawl from the first pawl position to the second pawl position as the linkage moves from the first linkage position to the second linkage position;

[0013] an actuator connected to the linkage, the actuator being configured to move the linkage from the first linkage position to the second linkage position when actuated;

[0014] wherein the pawl is engaged with the ratchet in the first pawl position and the pawl is disengaged with the ratchet in the second position such that the pawl can move in the second direction; and

[0015] wherein the handle arm and pawl are configured to move in the first direction as the handle moves from the stowed configuration to the deployed configuration, and the handle arm and pawl are configured to move in the second direction as the handle moves from the deployed configuration to the stowed configuration.

[0016] In an aspect, the linkage comprises a first linear cam, the pawl comprises a respective first follower, and movement of the pawl from the first pawl position to the second pawl position is caused by movement of the first follower along the first linear cam.

[0017] In an aspect, the first linear cam is a cam surface of a pawl trigger connected to the linkage.

[0018] In an aspect, the pawl trigger defines a second linear cam, and the linkage comprises a respective second follower, and movement of the linkage from the first linkage position to the second linkage position actuates the pawl trigger to move the pawl from the first pawl position to the second pawl position.

[0019] In an aspect, the actuator defines a third linear cam, the linkage comprises a respective third follower, and movement of the third linear cam drives the third follower and the linkage from the first linkage position to the second linkage position.

[0020] In an aspect, the linkage comprises a locking clip configured to mate with a corresponding ledge of the handle arm when the handle is in the stowed configuration, and movement of the linkage from the first linkage position to the second linkage position disengages the locking clip from the ledge.

[0021] In an aspect, the handle device further comprises a spring connected to the handle housing and configured to bias the handle toward the deployed configuration.

[0022] In an aspect, the handle device further comprises a spring connected to the handle housing and configured to bias the linkage arm toward the first position.

[0023] In an aspect, the handle device further comprises a button connected to the actuator such that force exerted on the button actuates the actuator.

[0024] In an aspect, the handle defines a recess configured for receiving an insert.

[0025] Methods of using the system(s) disclosed herein (including device(s), apparatus(es), assembly(ies), structure (s), and/or the like) are included; the methods of use can include using or assembling any one or more of the features disclosed herein to achieve functions and/or features of the system(s) as discussed in this disclosure. Methods of manufacturing the system(s) disclosed herein are included; the methods of manufacture can include providing, making, connecting, assembling, and/or installing any one or more of the features of the system(s) disclosed herein to achieve functions and/or features of the system(s) as discussed in this disclosure.

[0026] This Summary is provided to introduce a selection of concepts in a simplified form. The concepts are further described in the Detailed Description section. Elements or steps other than those described in this Summary are possible, and no element or step is necessarily required. This Summary is not intended to identify key features or essential features of the claimed subject matter. The claimed subject matter is not limited to implementations that solve any or all disadvantages noted in any part of this disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] The abovementioned and other features of the embodiments disclosed herein are described below with reference to the drawings of the embodiments. The illustrated embodiments are intended to illustrate, but not to limit, the scope of protection. Various features of the different disclosed embodiments can be combined to form further embodiments, which are part of this disclosure.

[0028] FIG. **1** illustrates an example handle device with an emergency call button.

[0029] FIG. **2** illustrates a sectional view of an example handle device in a stowed position.

[0030] FIG. **3**A illustrates an example vertically oriented handle device with multiple rungs in a deployed configuration.

[0031] FIG. **3**B illustrates the example handle device of FIG. **3**A in the stowed configuration.

[0032] FIG. **4**A illustrates another example handle device in a stowed configuration.

[0033] FIG. **4**B illustrates the example handle device of FIG. **4**A in a deployed configuration.

[0034] FIG. **5** illustrates an example handle device that is hinged on one side in the deployed configuration.

[0035] FIG. **6**A illustrates an example handle device that is hinged on two sides in the stowed configuration.

[0036] FIG. **6**B illustrates the example handle device of FIG. **6**A in the deployed configuration.

[0037] FIG. **7**A illustrates an example handle device in the stowed configuration with an emergency call button that can be stored in the wall of a bathing environment such that the handle is flush with the wall.

[0038] FIG. **7**B illustrates the example handle device of FIG. **7**A in the deployed positon.

[0039] FIG. **8**A illustrates an example handle device that is rotatably deployed from within the wall of a bathing environment.

[0040] FIG. **8**B illustrates the example handle device of FIG. **8**A in the deployed configuration.

[0041] FIG. **9** illustrates an example handle device that has a customizable front and can be stored in the wall of a bathing environment such that the handle is flush with the wall of the bathing environment.

[0042] FIG. **10** illustrates example universal handle devices mounted in various positions on a free standing tub.

[0043] FIG. **11** illustrates an enlarged view of an example universal handle device mounted to the contoured surface of a free standing tube.

[0044] FIG. **12** illustrates an exploded view of the example universal handle device of FIG. **11**.

[0045] FIG. **13** schematically illustrates the example universal handle device in different configurations.

[0046] FIG. **14**A illustrates an example handle device in a stowed configuration.

[0047] FIG. **14**B illustrates the handle device of FIG. **14**A in a deployed configuration.

[0048] FIG. **15**A illustrates the handle device suspended between two supports.

[0049] FIG. **15**B illustrates an exploded view of the handle device.

[0050] FIG. **15**C illustrates an exploded view of the handle device.

[0051] FIG. **15**D illustrates an example handle assembly.

[0052] FIG. **16**A illustrates an example upper housing portion.

[0053] FIG. **16**B illustrates an example upper housing portion with constant force springs.

[0054] FIG. **16**C illustrates an example constant force spring.

[0055] FIG. **17**A illustrates the handle device in a deployed configuration.

[0056] FIG. **17**B illustrates the handle device in a transitional configuration.

[0057] FIG. **17**C illustrates the handle device in a stowed configuration.

[0058] FIG. **17**D illustrates a gear and gear rack.

[0059] FIG. **18**A illustrates ends of spring arms in guide channels in the stowed configuration.

[0060] FIG. **18**B illustrates ends of spring arms in guide channels in a transition configuration.

[0061] FIGS. **18**C-**1** and **18**C-**2** illustrates ends of spring arms in guide channels in a transition configuration.

[0062] FIG. **18**D illustrates ends of spring arms in guide channels in the deployed configuration.

[0063] FIG. **18**E illustrates ends of spring arms in guide channels in a transition configuration.

[0064] FIGS. **18**F-**1** and **18**F-**2** illustrates ends of spring arms in guide channels in the stowed configuration.

[0065] FIG. **19** illustrates an example lower housing portion.

[0066] FIG. **20**A illustrates a perspective view of the handle device.

[0067] FIG. **20**B illustrates an example lock housing and lock mechanism of the handle device.

[0068] FIG. **20**C illustrates the lock housing and lock mechanism of the handle device.

[0069] FIG. **21**A illustrates a section view of the handle device in the deployed configuration.

[0070] FIG. **21**B illustrates a section view of the handle device in a transitional configuration.

[0071] FIG. **21**C illustrates a section view of the handle device in the stowed configuration.

[0072] FIG. **22**A illustrates an example handle device in the stowed configuration.

[0073] FIG. **22**B illustrates the handle device in a transitional configuration.

[0074] FIG. **22**C illustrates the handle device in a deployed configuration.

[0075] FIG. **22**D illustrates the handle device in a deployed configuration.

[0076] FIG. **23** illustrates an exploded view of the handle device.

[0077] FIGS. **24**A, **24**B, and **24**C illustrate views of the handle frame.

[0078] FIGS. **25**A, **25**B, and **25**C illustrate views of an example link.

[0079] FIG. **26** illustrates an example deployment locking mechanism.

[0080] FIG. **27**A illustrates the deployment locking mechanism within the link.

[0081] FIG. **27**B illustrates a section view of the deployment locking mechanism within the link.

[0082] FIG. **27**C illustrates the deployment locking mechanism within the link coupled to the handle frame in the locked deployed configuration.

[0083] FIG. **27**D illustrates a section view of the deployment locking mechanism within the link coupled to the handle frame in the locked deployed configuration.

[0084] FIG. **27**E illustrates a side section view of the deployment locking mechanism within the link coupled to the handle frame in the locked deployed configuration.

[0085] FIG. **27**F illustrates a side section view of the deployment locking mechanism within the link coupled to the handle frame with a tab of the deployment locking mechanism moved out of the locking position.

[0086] FIG. **27**G illustrates the link with the deployment locking mechanism being rotated to the stowed configuration.

[0087] FIG. **27**H illustrates a section view of the link locked in the stowed configuration.

[0088] FIG. **28** illustrates an example stowage locking mechanism.

[0089] FIG. **29** illustrates a section view of the locking pin of the stowage locking mechanism in the locked position.

[0090] FIG. **30** illustrates an example emergency communication unit.

[0091] FIGS. **31-35** illustrate operation of an example handle device having a ratcheting mechanism.

[0092] FIGS. **36**A-B illustrate an example of the ratcheting mechanism of the device of FIG. **31**.

[0093] FIGS. **37**A and **37**B illustrate close-up views of the pawl, ratchet, and pawl trigger of the ratcheting mechanism of FIG. **31**.

[0094] FIG. **38** illustrates an exploded view of an example handle device.

[0095] FIG. **39** illustrates a top perspective view of an example of a bottom part of a handle housing.

[0096] FIG. **40** illustrates a bottom perspective view of the bottom part of FIG. **39**.

[0097] FIG. **41** illustrates a top plan view of the bottom part of FIG. **39**.

[0098] FIG. **42** illustrates a front plan view of the bottom part of FIG. **39**.

[0099] FIG. **42**A illustrates a section view taken along the **42**A-**42**A section line in FIG. **41**.

[0100] FIG. **43** illustrates an end elevation view of the bottom part of FIG. **39**.

[0101] FIG. **44** illustrates a bottom plan view of the bottom part of FIG. **39**.

[0102] FIG. **45** illustrates a rear plan view of the bottom part of FIG. **39**.

[0103] FIG. **46** illustrates a bottom perspective view of an example of a top part of a handle housing.

[0104] FIG. **47** illustrates a top perspective view of the top part of FIG. **46**.

[0105] FIG. **48** illustrates a top plan view of the top part of FIG. **46**.

[0106] FIG. **48**A illustrates a section view taken along the **48**A-**48**A section line in FIG. **48**.

[0107] FIG. **49** illustrates a front plan view of the top part of FIG. **46**.

[0108] FIG. **49**A illustrates a section view taken along the **49**A-**49**A section line in FIG. **49**.

[0109] FIG. **50** illustrates an end elevation view of the top part of FIG. **46**.

[0110] FIG. **51** illustrates a bottom plan view of the top part of FIG. **46**.

[0111] FIG. **52** illustrates a rear plan view of the top part of FIG. **46**.

[0112] FIGS. **53**A and **53**B illustrate perspective views of an example of a button.

[0113] FIG. **54** illustrates a front plan view of the button of FIG. **53**A.

[0114] FIG. **54**A illustrates a section view taken along the **54**A-**54**A section line in FIG. **54**.

[0115] FIG. **54**B illustrates a section view taken along the **54**B-**54**B section line in FIG. **54**.

[0116] FIG. **54**C illustrates a close-up view of a portion of FIG. **54**B.

[0117] FIG. **55** illustrates a side elevation view of the button of FIG. **53**A.

[0118] FIG. **56** illustrates a rear plan view of the button of FIG. **53**A.

[0119] FIG. **57** illustrates a top perspective view of an example of a linkage.

[0120] FIG. **58** illustrates a side elevation view of the linkage of FIG. **57**.

[0121] FIG. **59** illustrates an end elevation view of the linkage of FIG. **57**.

[0122] FIG. **60** illustrates a top plan view of the linkage of FIG. **57**.

[0123] FIG. **61** illustrates a rear plan view of the linkage of FIG. **57**.

[0124] FIG. **62** illustrates a perspective view of an example of a handle.

[0125] FIG. **63** illustrates a top plan view of the handle of FIG. **62**.

[0126] FIG. **64** illustrates a front plan view of the handle of FIG. **62**.

[0127] FIG. **64**A illustrates a section view taken along the **64**A-**64**A section line in FIG. **64**.

[0128] FIG. **64**B illustrates a section view taken along the **64**B-**64**B section line in FIG. **64**.

[0129] FIGS. **65** and **66** illustrate perspective views of an example of a pawl.
[0130] FIG. **67** illustrates a top plan view of the pawl of FIG. **65**.
[0131] FIG. **68** illustrates a first side plan view of the pawl of FIG. **65**.
[0132] FIG. **69** illustrates a front plan view of the pawl of FIG. **65**.
[0133] FIG. **70** illustrates a second side plan view of the pawl of FIG. **65**.
[0134] FIG. **71** illustrates a rear plan view of the pawl of FIG. **65**.
[0135] FIG. **72** illustrates a front perspective view of an example of a border plate.
[0136] FIG. **73** illustrates a front plan view of the border plate of FIG. **72**.
[0137] FIG. **73**A illustrates a section view taken along the **73**A-**73**A section line in FIG. **73**.
[0138] FIG. **73**B illustrates a close-up view of a section of FIG. **73**A.
[0139] FIG. **74** illustrates a top plan view of the border plate of FIG. **73**.
[0140] FIG. **75** illustrates a side elevation view of the border plate of FIG. **73**.
[0141] FIG. **76** illustrates a front perspective view of an example of a pawl trigger.
[0142] FIG. **77** illustrates a rear perspective view of the pawl trigger of FIG. **76**.
[0143] FIG. **78** illustrates a front plan view of the pawl trigger of FIG. **76**.
[0144] FIG. **79** illustrates a top plan view of the pawl trigger of FIG. **76**.
[0145] FIG. **80** illustrates a perspective view of an example of a ratchet.
[0146] FIG. **81** illustrates a top plan view of the ratchet of FIG. **80**.
[0147] FIG. **82** illustrates a front plan view of the ratchet of FIG. **80**.
[0148] FIG. **82**A illustrates a close-up view of a section of FIG. **82**.
[0149] FIG. **83** illustrates a front perspective view of an example of a bottom part of a handle arm.
[0150] FIG. **84** illustrates a rear perspective view of the bottom part of FIG. **83**.
[0151] FIG. **85** illustrates a top plan view of the bottom part of FIG. **83**.
[0152] FIG. **86** illustrates a bottom plan view of the bottom part of FIG. **83**.
[0153] FIG. **87** illustrates a side view of the bottom part of FIG. **83**.
[0154] FIG. **88** illustrates an end view of the bottom part of FIG. **83**.
[0155] FIG. **89** illustrates a front perspective view of an example of a top part of a handle arm.
[0156] FIG. **90** illustrates a rear perspective view of the top part of FIG. **89**.
[0157] FIG. **91** illustrates a top plan view of the top part of FIG. **89**.
[0158] FIG. **92** illustrates a bottom plan view of the top part of FIG. **89**.
[0159] FIG. **93** illustrates a side view of the top part of FIG. **89**.
[0160] FIG. **94** illustrates a first end view of the top part of FIG. **89**.
[0161] FIG. **95** illustrates a second end view of the top part of FIG. **89**.
[0162] FIG. **96** illustrates a perspective view of a handle extension member.
[0163] FIG. **97** illustrates a front perspective view of an example of an insert for a handle arm.
[0164] FIG. **98** illustrates a rear perspective view of the insert of FIG. **97**.
[0165] FIG. **99** illustrates a front plan view of the insert of FIG. **97**.
[0166] FIG. **100** illustrates a side elevation view of the insert of FIG. **99**.

DETAILED DESCRIPTION

[0167] Although certain embodiments and examples are described below, this disclosure extends beyond the specifically disclosed embodiments and/or uses and obvious modifications and equivalents thereof. Thus, it is intended that the scope of this disclosure should not be limited by any particular embodiments described below.

[0168] FIG. **1** illustrates a handle device **100**. The handle device **100** is coupled to and/or within the wall of a building, including bathing, environment, which can include showers, baths, bathtubs, etc. In some variants, the handle device **100**, and/other devices herein, can be used in other environments, such as cooking environments. The handle device **100** has an emergency contact device **110**. The emergency contact device **110** can be integrated with the handle device **100**, as illustrated, or a separate feature installed in the building, including bathing, environment. The emergency contact device **110** can have an electronic package **112** that can be easily removable to facilitate the changing of batteries. The emergency contact device **110** can have an emergency contact button **114**. The emergency contact device **110** can be pushed to alert a contact that the user is in need of assistance, such as emergency response services. The emergency contact button **114** can require that the button be held for a duration of time, such as several seconds, before an emergency contact is notified. The emergency contact button **114** can turn on a light, such as an LED, and/or make a sound, such as beep, when an alert has been sent to an emergency contact. The emergency contact device **110** can be an internet-of-things device. The emergency contact device **110** be connected to a network via Wi-Fi, Bluetooth, Ethernet, 3G, 4G, and/or other wireless or wired communication methods. In some embodiments, the emergency contact device **110** can have a speaker and microphone, allowing a user to speak to emergency contacts. The emergency contact device **110** can include an integrated two way voice over internet protocol (VOIP) emergency call button.

[0169] The handle **102** of the handle device **100** can be positioned in a stored or deployed position, providing improved aesthetics. The handle device **100** can have a spring (e.g., a continuous force spring and/or constant force spring) that deploys the handle **102** via a push to release latch. For example, a user can push against the handle **102** while the handle **102** is stored within a recess of the wall such that the latch releases and the spring is free to deploy the handle **102** away from the wall (e.g., place the handle device **100** in the deployed position). The handle **102** can be pushed back and retained in a recess of the wall of the bathing environment (e.g., placed in the stored positon). In some embodiments, the handle device **100** can have a hidden lock release that can be pushed, allowing the handle **102** to be collapsed by pushing the handle **102** back into the recess of the wall of the bathing environment to reload the springs. The hidden lock release can be on inner portion of the handle **102**. In some embodiments, the handle **102** is pushed back into the recess, against the force of the spring, until a latch retains the handle **102**. The handle **102**, as explained above, can be deployed from the stored position by again pushing on the handle **102**.

[0170] FIG. **2** schematically illustrates a sectional view of the handle device **100** with the handle **102** in the stored or stowed position. The handle **102** is positioned in the recess **120** of the wall of the environment. A frame or housing **126** (e.g., which can be thin) is positioned within the recess **120** to receive the handle **102**. The frame **126** can have adjustable positioning elements **124** (e.g., flanges) that allow for adjustable depth mounting of the frame **126** within a recess **120** of a wall to accommodate for varying thicknesses of different wall covers **122** (e.g., tiles, stone, etc.). As shown, the handle **102** can have a curved profile, including a slight curve, which can enable a user to conveniently push against the handle **102** for deployment.

[0171] FIG. **3**A illustrates a handle device **200** in the deployed position, and FIG. **3**B illustrates the handle device **200** in the stored position. The handle device **200** is vertically oriented. The handle device **200** can be elongate (e.g., long). The handle device **200** can have multiple rungs **230** to provide additional climbing assistance to a user. In the stowed position, the front surface of the handle device can be substantially flush with the surrounding wall.

[0172] FIG. **4**A illustrates the handle device **300** in the stowed position, and FIG. **4**B illustrates the handle device **300** in the deployed position. The handle device **300** can be frameless. The handle device **300** can have a handle cover **302** that overlaps a seam, but creates a lip on the handle **304**.

[0173] FIG. **5** illustrates a handle device **400** that is hinged one side, such that the handle **402** rotates between a deployed and stored position, rotating a free end towards and away from the wall of the building, including bathing, environment. The handle device **400** has an emergency contact device **410** that includes an emergency contact button on a free end of the handle, such that the button is exposed when the handle **402** is deployed. The button can be an integrated two way voice over internet protocol (VOIP) emergency call button.

[0174] FIGS. **6**A and **6**B illustrate an example handle device **500** that is hinged on two ends. FIG. **6**A illustrates the handle device **500** in a stowed position. FIG. **6**B illustrates the handle device **500** in a deployed position. The cover **502** of the handle device **500** can be pushed when in the stored position to allow the cover **502** to flip down and provide access to the handle **504**. In some embodiments, the hinges **506** can be goose neck hinges. The handle device **500** can have an emergency contact device **510**. The emergency contact device **510** can include an emergency contact button that is exposed when the cover **502** is flipped down and the handle **504** is deployed. The button can be an integrated two way voice over internet protocol (VOIP) emergency call button. In some embodiments, the emergency contact device **510** includes a keypad to place emergency calls.

[0175] FIGS. **7**A and **7**B illustrate an example handle device **600** in the stored and deployed positions, respectively. The handle device **600** can have a handle **602** and back frame **604** that slide straight out together in the direction of arrow **601** when deployed to expose an emergency contact device **610**. The emergency contact device **610** can include an emergency contact button that is exposed when the handle **602** is deployed. The button can be an integrated two way voice over internet protocol (VOIP) emergency call button.

[0176] FIGS. **8**A and **8**B illustrate an example handle device **700** in the stored and deployed positions, respectively. The handle device **700** can have a hidden release button that flips the door **702** (e.g., 180 degrees) in the direction of arrow **701** to provide access to the handle **704**. The handle device **700** can be closed by manually disengaging an open lock and pushing the door **702** closed to reset a release spring.

[0177] FIG. **9** illustrates an example handle device **800**. The handle device **800** can include a pocketed frame **802**. The handle device **800** can include a front plate or recessed plate **804** as part of the handle **808** that can accept various custom cut wall finish materials **806** (e.g., tile, stone, etc.). This can enable the front plate **804** of the handle device **800** to have the same appearance as that of the surrounding wall of the building, including bathing, environment, such that the handle device **800** is substantially concealed when in the stored position.

[0178] FIG. **10** illustrates example universal handle devices **900** mounted in various positions on the contoured walls **930** of a free standing tub **912**. The universal handle devices **900** can be mounted on surfaces with different contours, enabling handles **902** of varying shapes and configurations to be coupled to one or both sides of the contoured walls **930** of free standing tubs of different configurations. In some embodiments, the universal handle devices **900** are mounted to walls of a building, including bathing, environment.

[0179] FIG. **11** illustrates an enlarged view of the universal handle device **900** mounted on the contoured walls **930** of a free standing tub. The universal handle device **900** can mount to contoured walls **930** of a free standing tub while enabling the handle **902** to extend away at different angles, depending on the desired configuration.

[0180] FIG. **12** illustrates an exploded view of the universal handle device **900**. The universal handle device **900** can include an adhesive element **950** (e.g., tape, very high bond (VHB) tape) that enables coupling to the contoured walls **930** of a free standing tub. The universal handle device **900** can include a base plate **952**, angle lock **954**, top cover **956**, angle lock top **958**, bolt **960**, elbow start **962**, and/or other features. The angle lock **954**, and/or other features (such as the angle lock top **958**), can engage with the bolt **960** to orient the adhesive element **950** to engage the surface of a contoured wall **930** while allowing the elbow start **962** to extend away at a desired orientation. The elbow start **962** can connect to handles of various configurations. FIG. **13** schematically illustrates the universal handle device **900** at different orientations.

[0181] FIG. **14**A-**21**C illustrate a handle device **1000**, also referred to as a grab bar device, and components thereof. FIG. **14**A illustrates the handle device **1000** disposed within a wall **1003**. The wall **1003** may be in a building environment, including bathing (e.g., shower, bath, etc.), cooking (e.g., kitchen, etc.), or other environment. The handle device **1000** can be disposed at various orientations, such as vertically, horizontally, etc. In some variants, the handle device

1000 can be disposed with a handle **1002** thereof being perpendicular, parallel, or otherwise angled relative to a floor of the environment.

[0182] The handle device **1000** can be suspended between two supports **1010**, **1012** of the wall **1003**. In some variants, the handle device **1000** can be supported by a single support, such as one of support **1010** or support **1012**. The two supports **1010**, **1012** can be wall studs, such as a two inch by four inch wall studs.

[0183] The handle device **1000** can be disposed between a front wall **1006** and rear wall **1008** of the wall **1003**. The front wall **1006** and rear wall **1008** can be drywall, also referred to as plasterboard, sheet rock, wall board, etc. The handle device **1000** can at least partially extend through the front wall **1006** such that a user may contact the handle **1002** of the handle device **1000**. An outer layer **1004**, such as tile, can overlay the front wall **1006** and be disposed around the handle **1002**.

[0184] With the handle device **1000** in the stowed configuration, as illustrated in FIG. **14**A, an exposed panel **1001**, also referred to as exposed surface or front panel, of the handle **1002** can be substantially flush with the outer layer **1004**, which can be aesthetically pleasing and/or at least partially disguise or conceal the presence of the handle device **1000** to an observer. For example, when the handle device **1000** is in the stowed configuration, the wall **1003** can have a substantially continuous appearance that appears to be free of handles. In some variants, the exposed panel **1001** can have a similar appearance as the surrounding outer layer **1004**, further disguising or concealing the handle device **1000** from view when stowed. In some variants, the exposed panel **1001** can incorporate the same material as the outer layer **1004**.

[0185] To deploy the handle **1002**, the user can push the handle **1002** (i.e., contacting the exposed panel **1001**) in the direction of arrow **1014** (e.g., toward the wall **1003**, rear wall **1008**, etc.), which can release the handle **1002** to deploy from within a cavity, such as within a housing, of the handle device **1000**, as illustrated in FIG. **14**B. The handle **1002** can automatically deploy after the user pushes in the direction of arrow **1014**, which can be facilitated by one or more springs (e.g., constant or continuous force springs). In the deployed configuration, the handle **1002** can be spaced away from the wall **1003**, enabling the user to be able to grasp the handle **1002**. In the deployed configuration, the handle **1002** can be grasped by the user for increased stability when navigating an environment, such as a slippery bathing or cooking environment. The handle device **1000** may be returned to the stowed configuration, illustrated in FIG. **14**A, by pushing the handle **1002** back into the cavity, e.g., housing, of the handle device **1000** (e.g., toward the wall **1003**, rear wall **1008**, etc.). In some variants, the user must over disengage a locking mechanism and/or overcome the biasing force of one or more springs to move the handle **1002** into the stowed configuration. The handle device **1000** can retain the handle **1002** in the stowed configuration until the user once again pushes the handle **1002** in the direction of arrow **1014**, as described in reference to FIG. **14**A.

[0186] As illustrated in FIG. **15**A, the handle device **1000** can be supported between the supports **1010**, **1012**. Brackets **1018**, **1020**, also referred to as mounts, can couple the handle device **1000** to the supports **1010**, **1012**, which can be via bolts, screws, latches, clips, fasteners, and/or other suitable devices. The brackets **1018**, **1020** can be C-shaped to receive the handle device **1000** for coupling. The handle device **1000** can be coupled to the brackets **1018**, **1020** with one or more bolts **1022**, also referred to as screws. In some variants, the handle device **1000** can be coupled to the brackets **1018**, **1020** with latches, clips, fasteners, and/or other suitable devices. In some variants, only one bracket is used to couple the handle device **1000** to a support. In some variants, the handle device **1000** can be installed in the wall of an environment, e.g., a bathing environment, with construction of the wall **1003** and/or other structure or object. In some variants, the handle device **100** can be retrofit within the wall **1003** and/or other structure or object. For example, a hole can be cut in the front wall **1006** of the wall **1003**, enabling the handle device **1000** to be inserted through the hole and screwed, bolted, fastened, and/or otherwise coupled to one or both of the supports **1010**, **1012**.

[0187] FIG. **15**B illustrates an exploded view of various components of the handle device **1000**. The handle device **1000** can include a handle housing **1026**, also referred to as an enclosure or shell. The housing **1026** can define a cavity **1016**, as shown in FIG. **15**C, to house a handle assembly **1024**. The housing **1026** can include an upper housing portion **1028** and lower housing portion **1030** that can be coupled together to form the housing **1026** and cavity **1016**. The reference to upper and lower can be used solely to facilitate description and should not be considered limiting. In some variants, the brackets **1018**, **1020** can be used to couple the upper housing portion **1028** and lower housing portion **1030** together. The housing **1026** can be various shapes, such as generally a rectangular prism. The housing **1026**, i.e., the upper housing portion **1028** and lower housing portion **1030**, can include one or more flanges **1034**, also referred to as ribs. The one or more flanges **1034** can provide rigidity to the housing **1026**.

[0188] The handle device **1000** can include a collar **1032**, also referred to as a border member. The collar **1032** can define an opening **1033** that provides access into the cavity **1016**. The handle **1002** can extend or retract through the opening **1033** when being placed in the deployed or stowed configurations. The collar **1032** can couple to the housing **1026**. The collar **1032** can couple to the upper housing portion **1028** and lower housing portion **1030**. The collar **1032** can be placed over a lip **1036** of the upper housing portion **1028** and lip **1038** of the lower housing portion **1030**. The collar **1032** can include one or more clips **1042**, also referred to as fasteners, hooks, or hooked tabs, to couple the collar **1032** to the housing **1026**. In some variants, the clips **1042** can extend through one or more openings **1040** disposed on the upper housing portion **1028** and/or lower housing portion **1030** to facilitate coupling. In some variants, the clips **1042** can couple to the upper housing portion **1028** and/or lower housing portion **1030** via a snap fit, press fit, and/or other suitable technique.

[0189] The handle device **1000** can include a handle assembly **1024**. The handle assembly **1024** can translate within the housing **1026** to deploy or stow the handle **1002**. A portion of the handle assembly **1024** can be disposed within the housing **1026** in the deployed configuration while the handle **1002** can extend outside the housing **1026** to be grasped by the user. The handle assembly **1024**, and handle device **1000**, can include various features to facilitate automatic deployment and stowage of the handle **1002**, which can include gears, spring, joints, pivoting members, etc.—as described herein.

[0190] FIG. **15**C illustrates an exploded view of the handle assembly **1024** with other components of the handle device **1000**. The handle assembly **1024**, as described herein, can have a handle **1002** (also referred to as a grab bar or bar). The handle **1002** can include a handle frame **1054** (also referred to as a handle support structure). The handle frame **1054** can include a front portion **1120**. The front portion **1120** can be substantially flat and/or include features that are substantially flat on a front side. The front portion **1120** can couple to a front panel support **1064**, which can be via bolts, screws, fasteners, adhesive, and the like. The front panel support **1064** can receive, which can include couple to, the exposed panel **1001**. As described above, the exposed panel **1001** can be similar in appearance and/or material to the surrounding outer layer **1004**. In some variants, different exposed panels **1001** can be received within a recess of the front panel support **1064** to match the surrounding outer layer **1004**. In some variants, the exposed panel **1001** can be retained, which can include being coupled to, the front panel support **1064** via an adhesive, bonding agent, clips, screws, fasteners, and/or other suitable techniques.

[0191] The handle frame **1054** can include a curved surface **1096**, which can be on an opposing surface of the handle frame **1054** relative to the front portion **1120**. The curved surface **1096** can improve user comfort when grasping the handle **1002**. The handle **1002** can include other ergonomic contours to improve user comfort.

[0192] The handle frame **1054** can include arms **1122**, **1124**. The arms **1122**, **1124** can extend in a direction away from the front portion **1120**. The arms **1122**, **1124** can space the handle **1002** away from the housing **1026** in the deployed configuration such that the user can grasp the handle **1002**. The arms **1122**, **1124** can include one or more tabs **1076** that can be inserted into one or more holes **1078** of a support panel **1050**, which can be a press fit, interference fit, etc., as described in more detail elsewhere herein for coupling. The arms **1122**, **1124** can respectively include cavities **1126** to at least partially receive lock mechanisms **1060**, **1062** in conjunction with lock housing **1056** and/or lock housing **1058**, described below. The arms **1122**, **1124** can enclose the lock mechanisms **1060**, **1062**, respectively, in the lock housings **1056**, **1058**.

[0193] The handle **1002** can include a lock housing **1056** and/or lock housing **1058**. In some variants, the handle **1002** only includes one lock housing. The lock housing **1056** and/or lock housing **1058** can be coupled to the handle frame **1054**, which can include the arms **1122**, **1124** of the handle frame **1054**. The lock housings **1056**, **1058** can define sides of the handle **1002**. The lock housings **1056**, **1058** can respectively receive a lock mechanism **1060** and lock mechanism **1062**, which can be in conjunction with the cavities **1126** of the arms **1122**, **1124**. The locking mechanisms **1060**, **1062** can selectively retain the handle **1002** in the deployed position, as described herein. The lock mechanisms **1060**, **1062** can automatically retain the handle **1002** in the deployed position and be selectively actuated to enable the handle **1002** to be placed in the stowed position, as described herein.

[0194] The handle assembly **1024** can include a support panel **1050**. The support panel **1050**, also referred to as the support structure, back member, or back wall, can couple to the handle frame **1054** and/or lock housings **1056**, **1058**. For example, the handle frame **1054** can include one or more tabs **1076** that can be inserted into one or more holes **1078** of the support panel **1050**, which can be a press fit, interference fit, etc. The support panel **1050** can be coupled to the lock housings **1056**, **1058** via bolts, screws, fasteners, or the like. This can enable translation of the handle **1002** and support panel **1050** together.

[0195] The support panel **1050** can support a gear rod **1048**, also referred to as a shaft or axle, which can assist the handle **1002** in proper deployment (e.g., deploy substantially straight out of the housing **1026** and/or smooth deployment as well a straight and smooth movement into the stowed position), as described herein. The gear rod **1048** can help the handle **1002** to smoothly deploy from the housing **1026**, which can include deploying substantially straight. The gear rod **1048** can include gears that can engage with features of the upper housing portion **1028**, such as gear racks, to facilitate smooth, straight, and/or even deployment or stowage, as described herein.

[0196] The support panel **1050** can include a lock spring that enables the handle **1002** to be stowed upon the user pushing the handle **1002** into the housing **1026** from the deployed position and free to deploy upon pushing the handle **1002** into the housing **1026** from the stowed position, as described herein. The lock spring can include a lock spring arm **1066** and/or lock spring arm **1068**, also referred to as spring arms. In some variants, the lock spring arm **1066** and lock spring arm **1068** are joined or separate. The lock spring arm **1066** and lock spring arm **1068** can engage with features of the upper housing portion **1028**, e.g., guide channels, to facilitate the push-to-lock and push-to-release functions described herein.

[0197] A back panel **1052** can be coupled to the support panel **1050**. The panel **1052** can be exposed when the handle device **1000** is in the deployed configuration. In some variants, the panel **1052** can have an appearance similar to the exposed panel **1001** or outer layer **1004**. The panel **1052** can protect the support panel **1050**. In some variants, the panel **1052** can be flush with the surrounding outer layer **1004** when the handle device **1000** is in the deployed configuration.

[0198] The handle assembly **1024** can include links, also referred to as scissor links, members, supports, struts, etc., that can expand or collapse upon deployment or stowage of the handle **1002**. The links can include a first link **1044**, second link **1046**, first link **1045**, and second link **1047**. The first link **1044** and first link **1045** can be rotatably coupled to the housing **1026**, such as the upper housing portion **1028** and the lower housing portion **1030**. The second link **1046** can be rotatably coupled to the first link **1044** and the lock housing **1056**. The second link **1047** can be rotatably coupled to the first link **1045** and the lock housing **1058**. As described herein, movement of the handle **1002** can correspond to the collapsing and expansion of the first link **1044**, second link **1046**, first link **1045**, and second link **1047**. The links can increase structural stability of the handle assembly **1024** with deployment and stowage.

[0199] FIG. **15**D illustrates an assembled handle assembly **1024**. The support panel **1050** can support the gear rod **1048** via one or more support flanges **1051** connected to, integrated, or formed with the support panel **1050**; the support flanges can also be referred to as walls or structures. The gear rod **1048** can include one or more gears, such as the gears **1070**, **1072**. In some variants, the one or more support flanges **1051** can maintain the position of the gear rod **1048** and/or gears **1070**, **1072**. In some variants, the gear rod **1048**

is rotatably fixed and the gears **1070**, **1072** rotate independently thereon (e.g., the gears **1070**, **1072** rotate about the gear rod **1048** with the gear rod **1048** being a central axis of rotation as the gear rod **1048** remains relatively fixed while the gears **1070**, **1072** rotate). In some variants, the gear rod **1048** rotates with the gears **1070**, **1072** such that the rotation of the gears **1070**, **1072** is the same. In some variants, the gears **1070**, **1072** can be formed with and/or be integral with the gear rod **1048** (e.g., the gears **1070**, **1072** and gear rod **1048** can be formed from a monolithic piece of material). In some variants, the gears **1070**, **1072** can be fixed to, connect with, mate with, and/or engage with the gear rod **1048**. The gears **1070**, **1072** can be slid and/or positioned onto the gear rod **1048**. The gear rod **1048** can have one or more engagement features such flat, concave, and/or convex surfaces with the gears **1070**, **1072** having corresponding engagement features, (e.g., corresponding surfaces) such that when the gears **1070**, **1072** are positioned onto the gear rod **1048**, the gears **1070**, **1072** are fixed relative to the gear rod **1048** (e.g., the gears **1070**, **1072** are radially fixed to fixedly rotate with the gear rod **1048**). In some variants, the gears **1070**, **1072** and the gear rod **1048** can have corresponding protrusion and detents that engage one another when the gears **1070**, **1072** are positioned onto the gear rod **1048**.

[0200] The gears **1070**, **1072** can be fixed onto the gear rod **1048** via one or more washers, nuts, and/or crimpers **1073**. The crimpers **1073** can be slid and/or positioned onto the gear rod **1048**. The crimpers **1073** can be crimped and/or deformed onto the gear rod **1048** to fix the gears **1070**, **1072** onto the gear rod **1048**. The crimpers **1073** and the gear rod **1048** can have corresponding engagement features (e.g., flat surfaces and/or protrusion and detents) to axially and/or radially fix the crimpers **1073** relative to the gear rod **1048**. The gears **1070**, **1072** can be positioned against a relatively thicker or larger radius central portion of the gear rod **1048** against which the crimpers **1073** press and/or positioned the gears **1070**, **1072** such that the gears **1070**, **1072** are axially fixed onto the gear rod **1048** between the central portion of the gear rod **1048** and the crimpers **1073** while being radially fixed onto gear rod **1048** via the engagement features as discussed herein.

[0201] The gears **1070**, **1072** can engage with one or more gear racks **1080**, **1082** (also referred to as gear tracks), as shown in FIG. **16**A, disposed in the upper housing portion **1028**, as described herein. The gears **1070**, **1072** can rotate from engagement with the one or more gear racks **1080**, **1082**, respectively, as the handle assembly **1024** is translated relative to the upper housing portion **1028** during deployment or stowage of the handle **1002**.

[0202] The engagement between the gears **1070**, **1072** and gear racks **1080**, **1082** can assist in straight and/or smooth deployment of the handle **1002**. For example, if the user applies a force on the handle **1002** that is not perpendicular to the longitudinal length of the handle **1002** and/or the constant force springs **1108**, **1110** (described in more detail herein) or applies a force that is not centered on the handle **1002**, the user may apply unequal biasing forces on the handle assembly **1024**. In stowing the handle **1002**, the user can push on the handle **1002** as discussed herein. In some instances, the user may apply a force on the handle **1002** that acts on the handle **1002** to deviate the handle **1002** from translating directly and/or straightly into the housing **1026**, e.g., pushing on a portion of the handle **1002** proximate to one of the lateral sides of the housing **1026** relative to the other later side of the housing **1026**, which may cause forces at least partially in the direction of one of the lateral sides of the housing **1026**. The application of such a force, without the engagement between the gears **1070**, **1072** and gear racks **1080**, **1082**, can cause the handle assembly **1024** to become askew or deviate from its path of travel between the deployed and stowed configuration as discussed herein, which may cause the handle assembly **1024** to push against one of the surfaces of the housing **1026** and/or other surfaces handle device **1000** as discussed herein, which can result in increased friction and slower stowage, binding, or even break components of the handle assembly **1024**.

[0203] Similarly, the biasing forces of the constant force springs **1108**, **1110** may be unequal in some instances. The engagement between the gears **1070**, **1072** and gear racks **1080**, **1082** can prevent and/or at least reduce twisting or rotation of the handle assembly **1024** due to imbalanced biasing forces of the constant force springs **1108**, **1110**—helping the handle **1002** to deploy substantially straight out of the housing **1026**.

[0204] The engagement between the gears **1070**, **1072** and gear racks **1080**, **1082** (discussed in further detail herein) can reduce such negative effects by controlling movement of the handle assembly **1024** straight in and out of the housing **1026**. In some variants, the gears **1070**, **1072** can be fixedly coupled to the gear rod **1048** such that the gears **1070**, **1072** rotate in unison (e.g., at the same rate of rotation, together, at the same speed, etc.) to reduce and/or eliminate movement of the handle assembly **1024** in other directions other than translating straight in or out of the housing **1026** that may otherwise occur with the user pushing on the handle **1002** and/or unequal biasing forces from the constant force springs **1108**, **1110**. For example, as one of the gears **1070** rotates and linearly travels/translates along one of the corresponding gear rack **1080**, the other gear **1072** will rotate at the same rate via the fixed connection to the gear rod **1048**. The other gear **1072** will then travel/translate along the other corresponding gear rack **1082** substantially the same distance or extent to cause the other corresponding gear rack **1082** to linearly translate or move along the direction of movement at the same rate as the corresponding gear rack **1080** to cause the handle assembly **1024** to linearly translate in a straight and/or non-skewed direction relative to the housing **1026** (e.g., sides and/or surfaces of the handle assembly **1024** remain at substantially same distances relative to corresponding sides and/or surfaces of the housing **1026** that are moving relative to the each parallel to the direction of travel of the handle assembly **1024** between the stowed and deployed configurations) for smooth and straight movement as discussed herein.

[0205] The support panel **1050** can include lock spring arms **1066**, **1068**. The lock spring arms **1066**, **1068** can respectively engage with guide channels **1084**, **1086**, as shown in FIGS. **16**A and **18**A-**18**F-**2**. The lock spring arms **1066**, **1068** can move within the guide channels **1084**, **1086**, also referred to as guide grooves, as the handle **1002** is moved between the stowed and deployed configurations. The lock spring arms **1066**, **1068** can enable the handle **1002** to be retained in the stowed configuration from the deployed configuration by pushing the handle **1002** into the housing **1026**. The lock spring arms **1066**, **1068** can enable the handle **1002** to move to the deployed configuration from the stowed configuration by the user pushing the handle **1002** into the housing **1026** and releasing. Further details regarding the lock spring arms **1066**, **1068** are described in reference to FIGS. and **18**A-**18**F-**2**.

[0206] FIG. **15**D illustrates the first link **1044**, second link **1046**, first link **1045**, and second link **1047** in an expanded configuration. As illustrated, the first link **1044** and first link **1045** can be coupled to a pin **1074** and pin **1075**, respectively. The pin **1074** and pin **1075** can engage with the housing **1026** such that the first link **1044**, second link **1046**, first link **1045**, and second link **1047** move between an expanded and collapsed configuration upon movement of the handle **1002**. Specifically, the pin **1074** can rotatably engage with retainer **1088** of the upper housing portion **1028**, illustrated in FIG. **16**A, and the retainer **1092** of the lower housing portion **1030**, illustrated in FIG. **19** (e.g., the pin **1074** can be inserted into the retainer **1088** and retainer **1092**). The pin **1075** can rotatably engage with the retainer **1090** of the upper housing portion **1028**, illustrated in FIG. **16**A, and the retainer **1094** of the lower housing portion **1030**, illustrated in FIG. **19** (e.g., the pin **1075** can be inserted into the retainer **1090** and retainer **1094**).

[0207] FIG. **16**A illustrates the upper housing portion **1028**. As described herein, the upper housing portion **1028** can include the guide channels **1084**, **1086**. The guide channel **1084** and guide channel **1086** can be in mirrored configurations relative to each other. As described, the ends of the lock spring arm **1066** and lock spring arm **1068** can, respectively, move within the guide channels **1084**, **1086**. The guide channels **1084**, **1086** can be formed within the upper housing portion **1028**. The guide channels **1084**, **1086** can also be referred to as grooves. The guide channels **1084**, **1086** can include one or more contours to control the stowage and deployment of the handle **1002**, as described in reference to FIGS. **18**A-**18**F-**2**.

[0208] The upper housing portion **1028** can include a gear rack **1080** and a gear rack **1082**. The gear racks **1080**, **1082** can engage with the gears **1070**, **1072** during translation of the handle **1002**, which can assist in smooth and/or straight deployment and stowage. The gear racks **1080**, **1082** can be formed in the upper housing portion **1028**. The teeth of the gear racks **1080**, **1082** can correspond to (e.g., mesh with) teeth of the gears **1070**, **1072**. In some variants, a single gear rack and gear is included in the handled device **1000**. In some variants, the gear racks **1080**, **1082** and/or guide channels **1084**, **1086** can be disposed on the upper housing portion **1028** to avoid and/or reduce moisture (e.g., liquid from a bathing or cooking environment) gathering therein. The gear racks **1080**, **1082** and/or guide channels **1084**, **1086** can be disposed on the upper housing portion **1028** to avoid and/or reduce issues with mold or mildew that may otherwise develop if positioned on the lower housing portion **1030**.

[0209] The upper housing portion **1028** can include retainers **1088**, **1090**. As described, the retainers **1088**, **1090** can receive the pins **1074**, **1075**—rotatably coupling the first link **1044** and first link **1045** to the upper housing portion **1028**. The retainers **1088**, **1090** can be annular structures protruding from the upper housing portion **1028**. The retainers **1088**, **1090** can be formed in the upper housing portion **1028**.

[0210] The upper housing portion **1028** can include a recessed opening **1104** and/or recessed opening **1106**. The recessed openings **1104**, **1106** can interface with constant force springs **1108**, **1110** (also referred to as continuous force springs or springs), illustrated in FIGS. **16**B and **16**C, that are configured to bias the handle device **1000** to the deployed configuration. The constant force springs **1108**, **1110** can apply a substantially constant force biasing the handle **1002** to the deployed position. As illustrated in FIG. **16**C, the constant force spring **1108** can include a clip **1112** that enables the constant force spring **1108** to clip onto the upper housing portion **1028** via the recessed opening **1104**, as illustrated in FIG. **16**B. Returning to FIG. **16**C, the constant force spring **1108** can include a coil **1114** that uncoils as the handle device **1000** is moved to the stowed configuration and recoils as the handle device **1000** is moved to the deployed configuration. For example, the coil **1114** can be moved by engagement with the support panel **1050**, causing uncoiling and coiling. The constant force spring **1110** can be the same or similar to the constant force spring **1108**. In some variants, the handle device **1000** can include a single spring (e.g., constant force spring) that biases the handle device **1000** toward the deployed configuration. The single spring can be centrally positioned on the upper housing portion **1028** (e.g., positioned equidistantly between lateral sides of the upper housing portion **1028**). The central position of the single spring can apply a centrally located biasing force, which can enable the handle **1002** to smoothly (e.g., avoid binding) and straightly move to the deployed configuration.

[0211] FIGS. **17**A-**17**C illustrate the handle device **1000** in various configurations during use. For descriptive and illustrative purposes, the lower housing portion **1030** has been removed in FIGS. **17**A-**17**C.

[0212] FIG. **17**A illustrates the handle device **1000** in the deployed configuration with the handle **1002** extending outside of the housing **1026**. The coils **1114** of the constant force springs **1108**, **1110** can contact the support panel **1050** to apply a force, such as a constant force, in the direction of arrow **1017** which can bias the handle device **1000** toward the deployed configuration. The first link **1044**, second link **1046**, first link **1045**, and second link **1047** can be in an extended position to provide structural support to the handle **1002**.

[0213] To stow the handle **1002**, the user can actuate the lock mechanisms **1060**, **1062**, described in detail in reference to FIGS. **20**A-**21**C, freeing movement of the handle **1002** in the direction of arrow **101**, and apply a force, e.g., push force, to the handle **1002** in the direction of arrow **1014**, which can include overcoming the force applied by the constant force springs **1108**, **1110** in the direction of arrow **1017**. The application of the force in the direction of arrow **1014** can cause movement of the handle **1002** and handle assembly **1024** in the direction of arrow **1014**.

[0214] FIG. **17**B illustrates the handle device **1000** in a transitional configuration between the deployed configuration, illustrated in FIG. **17**A, and the stowed configuration, illustrated in FIG. **17**C. As illustrated in FIG. **17**B, the gears **1070**, **1072** can be meshed with the gear racks **1080**, **1082** during translation of the handle **1002** and handle assembly **1024** as shown in FIG. **17**D, which can assist in providing smooth and/or straight deployment and stowage of the handle **1002** as described herein. The engagement between the gears **1070**, **1072** and gear racks **1080**, **1082** can prevent or at least reduce twisting or rotation of the handle assembly **1024** that may otherwise occur if the user applies a force that deviates from directly into the housing **1026**, e.g., pushing the handle **1002** at least partially in the direction of one of the lateral sides of the housing **1026**. As described herein, the gears **1070**, **1072** can rotate in unison, which may be due to being fixedly coupled to the gear rod **1048**, to facilitate substantially straight deployment in and out of the housing **1026**.

[0215] Returning to FIG. **17**B, the ends of the lock spring arms **1066**, **1068** can move within the guide channels **1084**, **1086** during translation of the handle **1002** and handle assembly **1024**, as described in more detail in reference to FIGS. **18**A-**18**F-**2**. The first link **1045** and second link **1047** can begin to collapse as the handle **1002** is pushed in the direction of arrow **1014**. The first link **1044** and second link **1047** can begin to collapse as the handle **1002** is pushed in the direction of arrow **1014**. The support panel **1050** can translate with the handle **1002** as the handle **1002** is pushed in the direction of the arrow **1014**. The support panel **1050** can remain engaged with the coils **1114** of the constant force springs **1108**, **1110** during movement thereof while the clips **1112** of the constant force springs **1108**, **1110** remain coupled to the upper housing portion **1028** at the recessed openings **1104**, **1106**, which can cause the coils **1114** to uncoil to provide or maintain a force, such as a constant force, in the direction of arrow **1017**.

[0216] FIG. **17**C illustrates the handle device **1000** in the stowed configuration with the handle **1002** disposed inside the housing **1026**. The support panel **1050**, which can move with the handle **1002**, can be disposed proximate the rear wall of the upper housing portion **1028** moving the coils **1114** of the constant force springs **1108**, **1110** therewith, which can result in further uncoiling of the constant force springs **1108**, **1110** as shown by the uncoiled portions **1115** of the constant force springs **1108**, **1110**. The constant force springs **1108**, **1110** can continue to apply a biasing force on the handle **1002** and handle assembly **1024** in the direction of arrow **1017** but the engagement of the ends of the lock spring arms **1066**, **1068** with the guide channels **1084**, **1086** can maintain the handle device **1000** in the stowed configuration, as described in more detail in reference to FIGS. **18**A-**18**F-**2**. As illustrated, the gears **1070**, **1072** can remain meshed with the gear racks **1080**, **1082** in the stowed configuration. The first link **1045** and second link **1047** can be collapsed (e.g., pivoted to positions adjacent and/or parallel to each other) with the handle **1002** stowed. The first link **1044** and second link **1047** can be collapsed (e.g., pivoted to positions adjacent and/or parallel to each other) with the handle **1002** stowed.

[0217] The user can deploy the handle **1002** from the stowed configuration by, once again, applying a force, e.g., a push force, in the direction of arrow **1014** that facilitates movement of the ends of the lock spring arms **1066**, **1068** within the guide channels **1084**, **1086** to allow the biasing force of the constant force springs **1108**, **1110** in the direction of arrow **1017** to deploy the handle **1002**, as described in more detail in reference to FIGS. **18**A-**18**F-**2**. As described herein, the engagement between the gears **1070**, **1072** and gear racks **1080**, **1082** can prevent and/or at least reduce twisting or rotation of the handle assembly **1024** that may occur from the constant force springs **1108**, **1110** applying unequal biasing forces on the handle assembly **1024**, which can facilitate smooth and/or straight deployment. This can, in some variants, be due to the gears **1070**, **1072** rotating in unison, as described herein.

[0218] FIGS. **18**A-**18**F-**2** depict the movement of the ends of the lock spring arms **1066**, **1068** within the guide channels **1084**, **1086**. FIG. **18**A illustrates the ends of the lock spring arms **1066**, **1068** within the guide channels **1084**, **1086** with the handle device **1000** in the stowed configuration. The ends of the lock spring arms **1066**, **1068** can be retained within bends **1100**, **1102**, also referred to as contours or turns, of the guide channels **1084**, **1086**. The lock spring arms **1066**, **1068** can be biased outward such that the lock spring arm **1066** and lock spring arm **1068** are biased away from each other. Specifically, the lock spring arm **1066** can be biased in the direction of arrow **1098** and the lock spring arm **1068** can be biased in the direction of arrow **1099**.

[0219] To place the handle device **1000** in the deployed configuration, the user can push the handle **1002** inward (e.g., into the housing **1026**) in the direction of arrow **1014**. The pushing force of the user, in combination with the bias of the lock spring arms **1066**, **1068** can place the ends of the lock spring arms **1066**, **1068** in the configuration shown in FIG. **18**B. As illustrated in FIGS. **18**C-**1** and **18**C-**2**, the pushing force of the user can move the ends of the lock spring arms **1066**, **1068** into a position such that the bias of the lock spring arms **1066**, **1068** can move the ends thereof outward (i.e., away from each other) to the positions shown in FIGS. **18**B, **18**C-**1**, and **18**C-**2** because outward movement of the ends of the lock spring arms **1066**, **1068** is not obstructed by the guide channels **1084**, **1086**. The dashed lines in FIGS. **18**C-**1** and **18**C-**2** can indicate the travel path of the ends of the lock spring arms **1066**, **1068** between the stowed configuration in FIG. **18**A and the transitional configuration in FIG. **18**B. As described herein, the handle device **1000** can include one or more springs, e.g., constant force springs **1108**, **1110**, that can bias the handle device **1000** toward the deployed configuration. Specifically, the one or more springs **1108**, **1110** can apply a biasing force in the direction of arrow **1017** to move the handle device **1000** to the deployed configuration with the handle **1002** extending outside the housing **1026**. With the ends of the lock spring arms **1066**, **1068** in the bends **1100**, **1102**, the handle device **1000** can be maintained in the stowed configuration despite the handle device **1000** being biased by the springs **1108**, **1110** to the deployed configuration. However, the user can overcome the biasing force of the one or more springs in the direction of arrow **1017** to enable the outward bias of the lock spring arms **1066**, **1068** to move the ends thereof to the transitional positions illustrated in FIGS. **18**B, **18**C-**1**, and **18**C-**2**.

[0220] With the ends of the lock spring arms **1066**, **1068** in the positions illustrated in FIGS. **18**B, **18**C-**1**, and **18**C-**2**, the handle device **1000** can be placed in the deployed configuration by the one or more constant force springs **1108**, **1110** applying a force in the direction of arrow **1017** because the contours of the guide channels **1084**, **1086** do not obstruct such movement. Accordingly, the ends of the lock spring arms **1066**, **1068** can move to the positions within the guide channels **1084**, **1086** illustrated in FIG. **18**D which correspond to the handle device **1000** being in the deployed configuration. The guide channels **1084**, **1086** can each include a stepped portion **1085**, also referred to as an indented portion, depressed portion, lower-elevation portion, recessed portion, or slot, that is depressed or recessed relative to the adjacent portion of the guide channels **1084**, **1086** such that the ends of the lock spring arms **1066**, **1068** drop into the stepped portions **1085** upon movement to the positions shown in FIG. **18**D. The outward biasing force of the lock spring arms **1066**, **1068** can push the ends of the lock spring arms **1066**, **1068** into the side walls of the stepped portions **1085**. The side walls of the stepped portions **1085** can prevent movement of the ends of the lock spring arms **1066**, **1068** back toward the positions illustrated in FIG. **18**B or, stated differently, out of the stepped portions **1085**.

[0221] To place the handle device **1000** in the stowed configuration from the deployed configuration illustrated in FIG. **18**D, the user can push the handle **1002** in the direction of arrow **1014**, overcoming the force applied by the one or more constant force springs **1108**, **1110** in the direction of arrow **1017**, to move the ends of the lock spring arms **1066**, **1068** to the transitional positions illustrated in FIG. **18**E. The stepped portions **1085** can include a gradual incline in elevation between the location of the ends of the lock spring arms illustrated in FIG. **18**D and the location of the ends of the lock spring arms **1066**, **1068** illustrated in FIG. **18**E. The stepped portions **1085** can end at the location of the ends of the lock spring arms **1066**, **1068** illustrated in FIG. **18**E such that the outward biasing forces of the lock spring arms **1066**, **1068** can move the ends of the lock spring arms to the bends **1100**, **1102**. Outside the recesses of the stepped portions **1085**, the outward biasing force of the lock spring arms **1066**, **1068** can then be free to move the ends of lock spring arms **1066**, **1068** outward in the direction of arrows **1098**, **1099** such that the ends of lock spring arms **1066**, **1068** are positioned in the bends **1100**, **1102** as illustrated in FIG. **18**A, which corresponds to the handle device **1000** being in the stowed configuration. With the ends of lock spring arms **1066**, **1068** positioned in the bends **1100**, **1102**, the handle device **1000** is retained in the stowed configuration despite the one or more constant force springs **1108**, **1110** applying a force in the direction of arrow **1017**. FIGS. **18**F-**1** and **18**F-**2** depict, via dashed lines, the movement of the ends of the lock spring arms **1066**, **1068** within the guide channels **1084**, **1086** from the positions illustrated in FIG. **18**E to the those illustrated in FIGS. **18**A, **18**F-**1**, and **18**F-**2**, which correlate to the handle device **1000** being in the stowed configuration with the handle **1002** retained within the housing **1026**.

[0222] FIG. **19** illustrates the lower housing portion **1030**. As described elsewhere herein, the lower housing portion **1030** can include retainers **1092**, **1094**. The retainers **1092**, **1094** can receive the pins **1074**, **1075**—rotatably coupling the first link **1044** and first link **1045** to the lower housing portion **1030**. The retainers **1092**, **1094** can be annular structures protruding from the lower housing portion **1030**. The retainers **1092**, **1094** can be formed in the lower housing portion **1030**.

[0223] The lower housing portion **1030** can include protrusions **1116**, **1118**, also referred to as elongate protrusions, protuberances, or elongate protuberances. The protrusions **1116**, **1118** can maneuver the lock mechanisms **1060**, **1062**, as described in reference to FIGS. **20**A-**21**C, to maintain the handle **1002** in the deployed configuration and/or enable the handle **1002** to be stowed. The protrusions **1116**, **1118** can be formed in the lower housing portion **1030**. The protrusions **1116**, **1118** can have rounded ends.

[0224] FIG. **20**A illustrates a bottom perspective view of the handle device **1000**. As described elsewhere herein, the handle device **1000** can include lock mechanisms **1060**, **1062** disposed in the lock housings **1056**, **1058**. In some variants, the arms **1122**, **1124** of the handle frame **1054** can enclose the lock mechanisms **1060**, **1062** within the lock housings **1056**, **1058**. The lock housings **1056**, **1058** can respectively include openings **1128**, **1130** through which the lock mechanisms **1060**, **1062** may extend to enable the user to be able to access the lock mechanisms **1060**, **1062**. In some variants, the arms **1122**, **1124** can include gaps **1132**, which can enable the user to be able to access the lock mechanisms **1060**, **1062**. The lock mechanisms **1060**, **1062** can retain the handle **1002** in the deployed position until manipulated. To stow the handle **1002**, the user can manipulate the lock mechanisms **1060**, **1062**, which can include pushing them inward, to allow the handle **1002** to be pushed into the housing **1026**. In some variants, only one lock mechanism is used. In some variants, the handle **1002** cannot be pushed into the stowed position without the user manipulating the lock mechanisms **1060**, **1062**, which can prevent the user's hand from being pinched between the handle **1002** and the wall **1003** and/or prevent unintentional movement of the handle **1002**.

[0225] FIG. **20**B illustrates a bottom perspective view of the handle assembly **1024** decoupled from the housing **1026**. As illustrated, the lock housing **1056** can include a groove **1134**. In some variants, the lock housing **1058** can include a groove **1134** that is the same or similar to the groove **1134**. The groove **1134** can enable the lock housing **1056** to pass over the protrusion **1118** as the handle device **1000** is moved between the stowed and deployed configurations.

[0226] FIG. **20**C illustrates a view of the lock housing **1058** with the arm **1122** decoupled therefrom. The lock housing **1058** can include a cavity **1136** that can receive the lock mechanism **1060** therein. The lock housing **1058** can include a support **1138**, also referred to as a strut, bridge, or tab, that can retain the lock mechanism **1060** within the cavity **1136**. The lock mechanism **1060** when positioned within the cavity **1136** can have an engagement portion or body **1140**, also referred to as an engagement cylinder, cylinder, or button, that can extend through the opening **1128** enabling the user to contact the engagement portion **1140**. The engagement portion **1140** can be a cylinder, prism, or other structure. The lock mechanism **1060** can include an arm **1142**. The arm **1142** can contact the support **1138** when the lock mechanism **1060** is disposed within the cavity **1136** to suspend the lock mechanism **1060** therein. The arm **1142** can include one or more curves that position the arm **1142** around the support **1138**. The arm **1142** can be disposed and/or extend into the groove **1134** of the lock housing **1058** in the deployed configuration. In use, the user can apply a force, e.g., push, the engagement portion **1140** in the direction of the arrow **1144**, e.g., upward, raising the arm **1142** out of the groove **1134** such that the lock housing **1056** can slide over the protrusion **1118** as the handle **1002** is pushed into the housing **1026**.

[0227] FIGS. **21**A-**21**C illustrate various sectional views of the lock housing **1056**, lock mechanism **1060**, protrusion **1118**, and other components of the handle device **1000** in the deployed, transitional, and stowed configurations. FIG. **21**A illustrates the handle device **1000** in the deployed configuration. As illustrated, the lock mechanism **1060** can be disposed within the cavity **1136** of the lock housing **1056**. The lock mechanism **1060** can be supported by the support **1138** within the cavity **1136**. The arm **1142** of the lock mechanism **1060** can be disposed in the groove **1134** of the lock housing **1056**, which can prevent the protrusion **1118**, also referred to as the elongate protrusion, from entering the groove **1134** and, as a result, the lock housing **1056** from being translated in the direction of arrow **1014**. As in FIG.

20C, the engagement portion **1140** of the lock mechanism **1060** is extending through the opening **1128** of the lock housing **1056** such that the user can contact the engagement portion **1140**.

[0228] To begin stowage of the handle **1002**, the user can apply a force to the engagement portion **1140** of the lock mechanism **1060** in the direction of arrow **1144**, moving the lock mechanism **1060** in the direction of arrow **1144** such that the arm **1142** is moved out of the groove **1134**. With the arm **1142** out of the groove **1134**, the user can apply a force to the handle **1002** in the direction of arrow **1014**, pushing the handle **1002** into the housing **1026**, as the lock housing **1056** is moved over the protrusion **1118** such that the protrusion **1118** is disposed in the groove **1134**, as illustrated in FIG. **21**B.

[0229] As the handle **1002** is pushed into the housing **1026** in the direction of arrow **1014**, the arm **1142** can slide over the protrusion **1118** and, once the engagement portion **1140** passes over protrusion **1118**, the engagement portion **1140** can slide over the protrusion **1118** as well, as illustrated in FIG. **21**C. The handle **1002** can be pushed in the direction of arrow **1014** until the handle **1002** is placed in the stowed configuration as illustrated in FIG. **21**C with the lock mechanism **1060** disposed on the protrusion **1118**. The handle **1002** can be deployed, as described elsewhere herein, by the user, once again, pushing the **10002** in the direction of arrow **1014**. The handle **1002** can be deployed via the biasing force of the constant force springs **1108**, **1110** moving the handle **1002** to the transition configuration in FIG. **21**B with the lock mechanism **1060** sliding over the protrusion **1118** and then to the deployed configuration in FIG. **21**A. Gravity can position the engagement portion **1140** in the opening **1128** and/or arm **1142** in the groove **1134**, locking the handle **1002** in the deployed configuration. In some variants, a spring can bias the lock mechanism **1060** toward a direction opposite the arrow **1144**. For example, a spring can be place in the cavity **1136** to force the engagement portion **1140** in a direction opposite the arrow **1144**. In some variants, the handle frame **1054**, e.g., arm **1122**, can include tabs **1146** that extend into the cavity **1136** to prevent excessive movement of the lock mechanism **1060** in a direction opposite the arrow **1144**, as illustrated in FIG. **21**A.

[0230] FIG. **22**A-**29** illustrate views of a handle device **2000** and various components thereof. The handle device **2000** can be incorporated in the same environment described elsewhere herein, which can include within a wall **1003**. The handle device **2000** can include a swing action for the handle **1002** for deployment/stowage. The handle device **2000** may, in some variants, be suitable for narrow installation parameters, such as the annular pace of a tub wall, on a shelf surrounding a tub, on a door, on a bedframe, and/or other locations. The handle device **2000** can be relatively thin compared to other variations and involve fewer components to reduce cost.

[0231] FIG. **22**A illustrates the handle device **2000** in a stowed configuration. The handle device **2000** can be suspended between supports **1010**, **1012**. Specifically, the handle device **2000** can include brackets **2004**, **2005**, also referred to as mounts, that can be respectively coupled, via bolts, screws, fasteners, or the like, to the supports **1010**, **1012**. In the stowed configuration, a handle **1002** of the handle device **2000** can be housed within a housing **2002** while an exposed panel **1001** remains accessible to a user's touch. In the stowed configuration, the exposed panel **1001** can be flush with a surrounding wall and/or outer layer **1004**, as described elsewhere herein to provide an aesthetically pleasing appearance. The exposed panel **1001** can be similar in appearance or even the same material as a surrounding wall and/or outer layer **1004** to assist in concealing the presence of the handle device **2000** to an observer. The exposed panel **1001** can have an hole **2026** through which a button **2008** can extend.

[0232] To deploy the handle **1002** from within the housing **2002**, the user can press the button **2008** releasing the handle **1002** to rotate out from within a cavity **2018** of the housing **2002**, as illustrated in FIG. **22**B. FIG. **22**B illustrates the handle device **2000** in a transition configuration between the stowed configuration, illustrated in FIG. **22**A, and a deployed configuration, illustrated in FIG. **22**C. The handle device **2000** can include one or more springs that can facilitate automatic deployment upon release via pressing the button **2008**. As illustrated in FIG. **22**B, the handle **1002** can be rotatably coupled to a first link **2016** and second link **2017**, also referred to as members, supports, struts, etc., that are each rotatably coupled to the housing **2002**. In some variants, a pin can be respectively inserted through a joint connecting the handle **1002** to the first link **2016**, handle **1002** to the second link **2017**, first link **2016** to the housing **2002**, and/or second link **2017** to the housing **2002**. For example, a pin, or the like, can be inserted through a hole **2006** of the housing **2002** to rotatably couple the first link **2016** to the housing **2002** and/or a pin, or the like, can be inserted through a hole **2007** of the housing **2002** to rotatably couple the second link **2017** to the housing **2002**. As illustrated, the first link **2016** and second link **2017** can pivot relative to the housing **2002** and out from the cavity **2018** of the housing **2002** to begin deployment.

[0233] The rotation of the handle **1002** out from the housing **2002** can expose an emergency communication unit **2010**. The emergency communication unit **2010** can be used to contact an emergency contact in the event of an emergency, such as a fall of the user. The emergency communication unit **2010** can be activated to initiate an emergency response protocol and/or commanded to contact the emergency contact via manipulation of a user interface **2012**, such as a button or switch. For example, the user may press or otherwise interact with the user interface **2012**, such as a button, to initiate communication with an emergency contact, which can include initiating a call with an emergency service. The user may then communicate with (e.g., speak to and hear) the emergency contact via a speaker and/or microphone **2014**. In some variants, a wearable device (e.g., pendant, bracelet, watch, etc.) can interact with the emergency communication unit **2010** to enable the user to contact an emergency contact, such as an emergency service, without interaction with the user interface **2012**. The wearable device can be worn by the user such that the user can communicate with an emergency contact in the event of a fall or slip resulting in the user being out of reach of the emergency communication unit **2010**. In some variants, the wearable device can detect if the user has fallen and automatically contact an emergency contact and/or begin an emergency response protocol. The wearable device can be an internet-of-things device (i.e., IoT device). The wearable device be connected to the emergency communication unit **2010** and/or a network via Wi-Fi, Bluetooth, Ethernet, 3G, 4G, 5G, and/or other wireless or wired communication methods. In some embodiments, the wearable device can have a speaker and microphone, allowing a user to speak to emergency contacts. The wearable device can include an integrated two way voice over internet protocol (VOIP) emergency call button.

[0234] FIG. **22**C illustrates the handle device **2000** in the deployed configuration. In the deployed configuration, the handle **1002** can be spaced away from the housing **2002** and/or a surrounding wall, enabling the user to comfortably grasp the handle **1002** for support. The first link **2016** and second link **2017** can be arranged such that longitudinal lengths thereof are perpendicular to a surface, e.g., a wall, in which the handle device **2000** is incorporated. The first link **2016** and second link **2017** can be rotated to extend straight, e.g., perpendicularly relative to a longitudinal length of the housing **2002**, in the deployed configuration. In the deployed configuration, the handle **1002** can be releasably locked to reduce the likelihood of unintentional stowage of the handle **1002**.

[0235] As illustrated in FIG. **22**D, the first link **2016** can include a deployment locking mechanism **2038** that locks the handle **1002**, which can include the first link **2016**, in the deployed configuration. To stow the handle **1002**, the user can push the deployment locking mechanism **2038** in the direction of arrow **2108**, unlocking the handle **1002** and/or first link **2016**, and apply a force to the handle **1002** in the direction of arrow **2110** (e.g., inward toward the housing **2002**). The handle **1002** can then be rotated in the direction of arrow **2110** toward placement in the cavity **2018** of the housing **2002**. The first link **2016** and second link **2017** can be rotated in the direction of arrow **2110** toward respective stowage positions within a first recess **2020** and a second recess **2021**, illustrated in FIG. **22**C. The placement of the handle **1002** within the cavity **2018** can lock the handle **1002** within the cavity **2018** via at least the mechanism(s) described herein.

[0236] FIG. **23** illustrates an exploded view of the handle device **2000**. As illustrated, the handle **1002** can include a handle frame **2048**, also referred to as a handle structure or support. The handle frame **2048** can include a recess **2028**, also referred to as a receiving space, that can receive stowage locking mechanism **2022** that can lock and release the handle **1002** from the stowage position in the cavity **2018**.

[0237] The stowage locking mechanism **2022** can include a button **2008** that can be manipulated by the user. The button **2008** can extend through a hole **2026** of the exposed panel **1001** that can be received within a recess **2040** of the handle frame **2048**. The stowage locking mechanism **2022** can include a locking pin **2024**, also referred to as an arm, that can be moved via manipulation of the button **2008** to lock and unlock the handle **1002** from the stowed position. The stowage locking mechanism **2022** can be biased by a spring **2088** that pushes the button **2008** through the hole **2026** of the exposed panel **1001** and the locking pin **2024** in a position that releasably locks the handle **1002** in the stowed configuration.

[0238] The handle **1002** can include an exposed panel **1001** as described elsewhere herein that can be exposed when the handle **1002** is in the stowed configuration. The exposed panel **1001** can be coupled to the handle frame **2048** within a recess **2040** thereof via a variety of techniques, which can include adhesive, bonding, screws, bolts, fasteners, and the like.

[0239] The handle frame **2048** can include a curved surface **1096** that can increase user comfort when grasping the handle **1002**. The handle frame **2048** can include a first tab **2030** and/or a second tab **2031**. The first tab **2030** can rotatably couple to the first link **2016** such that the first link **2016** can pivot. Specifically, a first end **2032** of the first link **2016** can rotatably couple to the first tab **2030**, which can be via a pin. The second tab **2031** can rotatably couple to the second link **2017** such that the second link **2017** can pivot. Specifically, a first end **2032** of the second link **2017** can rotatably couple to the second tab **2031**, which can be via a pin.

[0240] The first link **2016** and/or second link **2017** can each include a second end **2034** that can rotatably couple to a support housing **2044** of the housing **2002**. Specifically, the second end **2034** of the first link **2016** can couple to a first interface **2046** of the support housing **2044**, which can be via insertion of a pin through the hole **2006**, and the second end **2034** of the second link **2017** can couple to a second interface **2047** of the support housing **2044**, which can be via insertion of a pin through the hole **2007**. The first link **2016** can include a torsion spring **2042** that can bias the first link **2016** toward the deployed configuration. The second link **2017** can include a torsion spring **2043** that can bias the second link **2017** toward the deployed configuration. Specifically, the second ends **2034** of each of the first link **2016** and second link **2017** can include a cylindrical structure **2074**. The torsion spring **2042** can be positioned around the cylindrical structure **2074** of the first link **2016** and the torsion spring **2043** can be positioned around the cylindrical structure **2074** of the second link **2017**. The torsion spring **2042** can engage with features of the support housing **2044**, e.g., the surrounding walls of the first recess **2020**, to bias the first link **2016** toward the deployed configuration. The torsion spring **2043** can engage with features of the support housing **2044**, e.g., the surrounding walls of the **20241**, to bias the second link **2017** toward the deployed configuration. In some variants, only one of the torsion spring **2042** or torsion spring **2043** is included in the handle device **2000**.

[0241] The handle device **2000** can include a deployment locking mechanism **2038**. The deployment locking mechanism **2038** can releasably lock the handle device **2000** in the deployed configuration. Specifically, the deployment locking mechanism **2038** can be inserted into a cavity **2036** of the first link **2016**. The deployment locking mechanism **2038** can engage with features of the first tab **2030** to prevent the first link **2016** from pivoting with respect to the first tab **2030**. The deployment locking mechanism **2038** can be disengaged from one or more features of the first tab **2030** to allow the handle **1002** to be stowed via application of a force, as described in reference to FIG. **22**D. In some variants, the deployment locking mechanism **2038** can be inserted into a cavity **2036** of the second link **2017** and releasably engage with features of the second tab **2031**.

[0242] The support housing **2044** can define the first recess **2020** and second recess **2021**. The support housing **2044** can house the emergency communication unit **2010**. The support housing **2044** can be positioned within the housing **2002** such that stowage of the handle **1002** within the cavity **2018** places the exposed panel **1001** substantially flush with a surrounding wall or surface.

[0243] FIGS. **24**A-**24**C illustrate the handle frame **2048**. As illustrated in FIG. **24**A, the handle frame **2048** can include a recess **2028** that can receive the stowage locking mechanism **2022**, such that the stowage locking mechanism **2022** is disposed between the handle frame **2048** and the exposed panel **1001**. The handle frame **2048** can include a fulcrum **2050** at which the stowage locking mechanism **2022** pivots such that pushing of the button **2008** causes insertion and removal (e.g., forward and backward) movement of the locking pin **2024** within a hole **2052**. The hole **2052** can extend through the handle frame **2048** to engage with the first link **2016** as described herein to lock the first link **2016** and/or handle **1002** in the stowed configuration.

[0244] As illustrated in FIGS. **24**B and **24**C, the handle frame **2048** can include a first tab **2030** and/or second tab **2031**. The first tab **2030** can couple with the first link **2016**. The second tab **2031** can couple with the second link **2017**. In some variants, the first tab **2030** and second tab **2031** can be the same. The first tab **2030** can include a slot **2054** that can receive a tab of the deployment locking mechanism **2038**. The tab of the deployment locking mechanism **2038** can be positioned within the slot **2054** to lock relative rotation between the first tab **2030** and the first link **2016** such that the first link **2016** and/or handle **1002** are locked in the deployed configuration. A force can be applied to the deployment locking mechanism **2038**, such as an upward force, such that the tab of the deployment locking mechanism **2038** is moved out of the slot **2054** and is free to move through an upper recess **2056**, also referred to as a cutout, gap, channel, etc., as the first link **2016** pivots relative to the first tab **2030** to place the handle **1002** in the stowed configuration. The tab of the deployment locking mechanism **2038** can slide across a shelf **2076** of the first tab **2030** during movement between the deployed and stowed configurations. The first tab **2030** can include a lower recess **2058**, also referred to as a cutout, gap, channel, etc. The lower recess **2058** can provide space for the first link **2016** to rotate relative to the first tab **2030**. The shelf **2076** can separate the upper recess **2056** from the lower recess **2058**. As described elsewhere herein, the locking pin **2024** can extend through the hole **2052** to engage with a portion of the first link **2016** that is positioned within the lower recess **2058** to lock the handle **1002** in the stowed configuration. The second tab **2031** can be the same as the second tab **2031**. In some variants, the second tab **2031** does not include the lower recess **2058** and hole **2052**.

[0245] FIGS. **25**A-**25**C illustrate the first link **2016**. The first link **2016** can include a first end **2032** that can rotatably couple to the first tab **2030**. The first end **2032** can include a receiving region **2064**, also referred to as a gap, that can receive the first tab **2030**. The first end **2032** can include a flange **2060**, also referred to as a tab. The flange **2060** can be positioned within the receiving region **2064**. The flange **2060** can include a hole **2062** that can be positioned coaxially with the hole **2052** such that the locking pin **2024** can extend into the hole **2062** to lock the first link **2016** in the deployed configuration, which also place the handle **1002** in the deployed configuration. As described herein, the button **2008** of the stowage locking mechanism **2022** can be pushed, causing the locking pin **2024** to be removed from the hole **2062** such that the first link **2016** can be rotated toward the stowed configuration by the user pushing on the handle **1002**.

[0246] The first link **2016** can include a cavity **2036** to receive the deployment locking mechanism **2038**. The first link **2016** can include a slot **2066**, which can extend into the cavity **2036**. The tab of the deployment locking mechanism **2038** can be moved within the slot **2066** to position the tab inside the slot **2054** of the first tab **2030**, locking the first link **2016** in the deployed configuration, or moved out of the slot **2054** of the first tab **2030**, allowing the tab to be rotated through the upper recess **2056** such that the first link **2016** can be placed in the stowed configuration. The cavity **2036** can include a recess **2037** that can receive an orienting protrusion of the deployment locking mechanism **2038**.

[0247] The first link **2016** can include a second end **2034**, which can be opposite the first end **2032**. The second end **2034** can be rotatably coupled to the first interface **2046** of the support housing **2044**. The second end **2034** can include a tab **2035**. The tab **2035** can be positioned within a gap **2112** of the first interface **2046** to facilitate coupling, which can be via a pin. The second end **2034** can include a cylindrical structure **2074**. As described elsewhere herein, a torsion spring **2042** can be positioned around the cylindrical structure **2074** to secure the torsion spring **2042** in place and position the torsion spring **2042** to engage features of the second end **2034** and first recess **2020** to bias the first link **2016** toward the deployed configuration.

[0248] The second link **2017** can be the same or similar to the first link **2016**. The second link **2017** can include a second end **2034** that can be rotatably coupled to the second interface **2047**. The second end **2034** can include a tab **2035**. The tab **2035** can be positioned within a gap **2114** of the second interface **2047** to facilitate coupling, which can be via a pin. The second end **2034** can include a cylindrical structure **2074**. As described elsewhere herein, a torsion spring **2043** can be positioned around the cylindrical structure **2074** to secure the torsion spring **2042** in place and position the torsion spring **2042** to engage features of the second end **2034** and first recess **2020** to bias the first link **2016** toward the deployed configuration.

[0249] FIG. **26** illustrates a deployment locking mechanism **2038**. The deployment locking mechanism **2038** can be a cylinder, prism, and/or other shape. The deployment locking mechanism **2038** can include a cavity **2078**. The cavity **2078** can receive a spring **2068** to bias the deployment locking mechanism **2038** in a direction, which can provide for automatic locking of the handle **1002** in the stowed configuration. The deployment locking mechanism **2038** can include an orientation protrusion **2070**, also referred to as a protrusion, protuberance, elongate guide, etc., which can move within the recess **2037** of the cavity **2036** of the first link **2016**. The orientation protrusion **2070** can prevent excessive forces from being applied on the tab **2072**. The deployment locking mechanism **2038** can include a tab **2072**, also referred to as a flange, which can extend from the cavity **2036** of the first link via the slot **2066**, such that the tab **2072** can be moved in and out of the slot **2054** of the first tab **2030**.

[0250] FIGS. **27**A-**27**B illustrate the deployment locking mechanism **2038** within the cavity **2036** of the first link **2016**. As illustrated, the deployment locking mechanism **2038** extends out of the cavity **2036** to be accessible by the user, such that the user can apply a force thereto. The tab **2072** of the deployment locking mechanism **2038** is disposed in the slot **2066** of the first link **2016**. In use, the user can apply a force in the direction of arrow **2108**, which can move the deployment locking mechanism **2038** in the direction of arrow **2108** within the cavity **2078** and the tab **2072** in the direction of arrow **2108** within the slot **2066** of the first link **2016**. The spring **2068** can engage a surface of the first link **2016** to apply a biasing force in the direction of arrow **2109**, which can bias the tab **2072** in the direction of arrow **2109** such that the deployment locking mechanism **2038** is biased toward the position illustrated in FIGS. **27**A and **27**B. The user may overcome the biasing force of the spring **2068** in the direction of arrow **2109** to move the deployment locking mechanism **2038** in the direction of arrow **2108**.

[0251] FIGS. **27**C and **27**D illustrate the first link **2016** with the deployment locking mechanism **2038** disposed within the cavity **2036** and rotatably coupled to the first tab **2030** of the handle frame **2048** in a locked deployed configuration. As illustrated, the first tab **2030** is disposed within the receiving region **2064**, also referred to as a gap, of the first end **2032** of the first link **2016**. As described elsewhere herein, a pin can be inserted through a hole **2082**, extending through the first end **2032** and first tab **2030**, to rotatably couple the first link **2016** to the first tab **2030**. As illustrated in FIG. **27**C, the tab **2072** of the deployment locking mechanism **2038** is positioned within the slot **2066** of the first tab **2030**, rotatably locking the first link **2016** to the first tab **2030** such that the handle **1002** is locked in the deployed configuration. The second end **2034** can also include a hole **2080**, which can extend through the cylindrical structure **2074**. The hole **2080** cab be coaxially aligned with the hole **2006** in the housing **2002** and hole in the first interface **2046** such that a pin can be inserted therethrough to rotatably couple the first link **2016** to the support housing **2044**. The second link **2017** can be coupled to the support housing **2044** in a similar manner. FIG. **27**D illustrates a section view showing the position of the tab **2072** of the deployment locking mechanism **2038** within the slot **2066** of the first tab **2030**.

[0252] FIGS. **27**E and **27**F illustrate side section views of the deployment locking mechanism **2038** disposed within the cavity **203** and the first link **2016** rotatably coupled to the first tab **2030** of the handle frame **2048**. FIG. **27**E illustrates the deployment locking mechanism **2038** in a locked position. The deployment locking mechanism **2038** is biased by the spring **2068** toward the position wherein the tab **2072** is disposed within the slot **2066** of the first tab **2030**. FIG. **27**F illustrates the deployment locking mechanism **2038** in an unlocked positon. To move the deployment locking mechanism **2038** from the locked position in FIG. **27**E to the unlocked position in FIG. **27**F, the user can apply a force to the deployment locking mechanism **2038** in the direction of arrow **2108**, which can move the deployment locking mechanism **2038** and tab **2072** in the direction of arrow **2108** such that the tab **2072** is moved out of the slot **2066** and is free to move through the upper recess **2056** of the first tab **2030**. As described elsewhere herein, the spring **2068** can apply a biasing force in the direction of arrow **2109**. Accordingly, the user can apply a force in the direction of arrow **2108** that overcomes the biasing force in the direction of arrow **2109** to move the deployment locking mechanism **2038** to enable the handle **1002** to be rotated from the deployed position to the stowed position.

[0253] FIG. **27**G illustrates the first link **2016** in a transition configuration, pivoting relative to the first tab **2030** toward the stowed position. As described elsewhere herein, the user can apply a force in the direction of arrow **2110** after applying a force to the deployment locking mechanism **2038** to rotate the handle **1002** toward the stowed configuration. The tab **2072** can move through the upper recess **2056** and over the shelf **2076** as the first link **2016** is rotated to the stowed configuration. The user can cease applying a force to the deployment locking mechanism **2038** after the tab **2072** is rotated over the shelf **2076**. The shelf **2076** can support the tab **2072** in the unlocked position out of the slot **2066**, as illustrated in FIG. **27**H, despite the biasing force of the spring **2068**. FIG. **27**H illustrates a section view of the first link **2016** in the stowed configuration. As illustrated, the tab **2072** can contact the shelf **2076**. The spring **2068** can continue to apply a force in the direction of arrow **2109** but the contact between the tab **2072** and the shelf **2076** can maintain the deployment locking mechanism **2038** in the position illustrated in FIG. **27**H. As illustrated in FIG. **27**H, the locking pin **2024** can be positioned within the hole **2062** of the flange **2060** of the first link **2016** in the stowed configuration, which can releasably lock the handle **1002** in the stowed configuration.

[0254] FIG. **28** illustrates the stowage locking mechanism **2022**, also referred to as the lever, lever lock, actuator lock, actuator lever, stowage lock, stowage locking lever. The stowage locking mechanism **2022**, as described elsewhere herein, can include a button **2008** that can be actuated, e.g., pressed, pushed, etc., by a user to move the locking pin **2024** in and out of the hole **2062** of the flange **2060** of the first link **2016**. The stowage locking mechanism **2022** can include a retaining corner **2086**. The stowage locking mechanism **2022** can include a stepped portion **2084**.

[0255] FIG. **29** illustrates a section view of the stowage locking mechanism **2022** with the locking pin **2024** positioned within the hole **2062** of the flange **2060** of the first link **2016**, locking the handle **1002** in the stowed configuration. The stowage locking mechanism **2022** can be positioned within the recess **2028** of the handle frame **2048**, which can be between the handle frame **2048** and the exposed panel **1001**. The button **2008** can be biased by a spring **2088** through a hole **2026** of the exposed panel **1001** such that the button **2008** is accessible to the user. The retaining corner **2086** can interface with a fulcrum **2050**, also referred to as a corner, point, or edge, such that the biasing force of the spring **2088** on the button **2008** in the direction of arrow **2116** can bias the locking pin **2024** in the direction of arrow **2122** which can place the locking pin **2024** in the hole **2062** of the flange **2060** of the first link **2016** in the stowed configuration. The biasing force of the spring **2088** in the direction of arrow **2116** can automatically place the locking pin **2024** in the hole **2026** of the flange **2060** as the user pushes the handle **1002** into the stowed configuration in the housing **1026** such that the handle **1002** is automatically locked in the stowed configuration upon placement therein.

[0256] To deploy the handle **1002**, the user can overcome the biasing force of the spring **2088** by applying a force to the button **2008** in the direction of arrow **2118** such that the locking pin **2024** moves in the direction of arrow **2120**, which can be facilitated by the fulcrum **2050** acting as a pivot point. The retaining corner **2086** can interface with the fulcrum **2050** to maintain the position of the stowage locking mechanism **2022** and/or facilitate smooth pivoting. When the user applies a force to the button **2008** in the direction of arrow **2118**, the locking pin **2024** can move in the direction of arrow **2120** such that the locking pin **2024** is removed from the hole **2062** of the flange **2060** of the first link **2016**—allowing the torsion spring **2042** and/or torsion spring **2043** to automatically deploy the handle **1002** to the deployed configuration. The locking pin **2024** can move within the hole **2052**, as described herein.

[0257] FIG. **30** schematically illustrates an emergency communication unit **2010**. The emergency communication unit **2010** can be incorporated in any of the handle devices disclosed herein. The emergency communication unit **2010** can include a transceiver **2124** that can communicate with a remote electronic device **2090** to facilitate communication between a user and an emergency contact. The emergency communication unit **2010** can include a power supply interface **2092** that facilitates connection with a power supply to power the emergency communication unit **2010**. In some variants, the emergency communication unit **2010** can include a battery **2104** that can power the emergency communication unit **2010** when a power supply is not available or active. The emergency communication unit **2010** can include a speaker **2094** that can emit sound enabling an emergency contact to audibly communicate with the user and/or emit an alarm. The emergency communication unit **2010** can include a microphone **2096** that can enable the user to speak to an emergency contract. The emergency communication unit **2010** can, in some variants, include a light source **2106** that can be activated in an emergency to provide light. The emergency communication unit **2010** can include a processor **2098** that can execute instructions **2102** stored on a memory system **2100** to perform the operations, tasks, and methods described herein. The incorporation of an emergency communication unit **2010** can enable a user to reach an emergency contact in an emergency when a phone or other communication device may not be accessible.

[0258] Referring to FIGS. **31-100**, deployable handle device **3000** is illustrated. Examples of deployable handle device **3000** have a handle **3012** configured to be grasped by a user, the handle **3012** configured to move between a stowed configuration (FIG. **31**) and a deployed configuration (FIG. **33**), wherein in the stowed configuration, handle **3012** is configured to be substantially flush with a surrounding wall (not pictured), and in the deployed configuration, handle **3012** is configured to protrude from the surrounding wall for the user to grasp handle **3012**. Handle device **3000** has a handle housing **3016** configured to be positioned in the surrounding wall with the handle **3012** at least partially positioned in the handle housing (FIGS. **31-35**).

[0259] Handle device **3000** has handle arm **3018** pivotably connected to handle housing **3016** at first end **3020**A and pivotably connected to handle **3012** at a second, opposing end **3020**B. Handle device **3000** has pawl **3022** connected to handle arm **3018** such that pawl **3022** moves with handle arm **3018**, such as when handle arm **3018** pivots. Handle arm **3018** and pawl **3022** are configured to move in the first direction **3028** as handle **3012** moves from the stowed configuration to the deployed configuration. Handle arm **3018** and pawl **3022** are configured to move in the second direction **3030** as handle **3012** moves from the deployed configuration to the stowed configuration.

[0260] Pawl **3022** is configured to move independently from handle arm **3018** between a first pawl position and a second pawl position (FIGS. **37**A and **37**B). The first pawl position may be defined as a position in which pawl **3022** is engaged with ratchet **3024**, such that the pawl **3022** cannot move in the second direction **3030** (FIGS. **36**A, **36**B, and **37**B). The second pawl position may be defined as any position in which the pawl **3022** is disengaged from ratchet **3024**, such that the pawl **3022** can move in the first and second directions **3028**, **3030**.

[0261] Ratchet **3024** is mounted within housing **3016**. Ratchet **3024** defines teeth **3026** biased to, when pawl **3022** is in the first pawl position, allow movement of pawl **3022** in a first direction and restrict movement of pawl **3022** in a second direction **3030**. Handle device **3000** has linkage **3032** movably connected to handle housing **3016**, linkage **3032** being configured to move from a first linkage position to a second linkage position (FIGS. **36**A and **36**B). Linkage **3032** is configured to move pawl **3022** from the first pawl position to the second pawl position as linkage **3032** moves from the first linkage position to the second linkage position. Actuator **3034** is connected to linkage **3032**, actuator **3034** being configured to move linkage **3032** from the first linkage position to the second linkage position when actuated.

[0262] FIGS. **31-36** illustrate operation of a handle device **3000**, also referred to as a grab bar device, and components thereof. Handle device **3000** may be operated by pressing actuator button **3010** to deploy the handle or bar **3012** from a stowed position, as referred to as stowed configuration herein, to a first position (FIGS. **31-32**). Handle **3012** may be pushed to the first position by a spring, such as a torsion spring **3036**, biased to push handle **3012** from the stowed configuration in a deployment direction **3038**. In the first position, a ratcheting mechanism **3014** may prevent the bar from returning to the stowed position (FIGS. **36**A-B). The handle may then be manually pulled in the direction **3038** to a second position, also referred to as a deployed configuration herein (FIG. **33**). Ratcheting mechanism **3014** may prevent the handle **3012** from moving to the first position or stowed position while pawl **3022** is engaged with ratchet **3024** in the first pawl position. Once the handle **3012** has been pushed at least partially from the housing **3016**, the user may release the button **3010**.

[0263] Once in the deployed configuration, a user may press and hold actuator button **3010** to allow handle **3012** to move in the stowing direction **3040** towards the first position or stowed position (FIG. **35**). Pressing actuator button **3010** may release a pawl **3022** from engagement with teeth **3026** of ratchet **3024** in ratcheting mechanism **3014** to permit movement of the handle **3012** between the stowed configuration, first position and/or deployed configuration. A user may hold the button **3010** down to keep pawl **3022** disengaged and permit the movement of handle **3012** in the stowed direction **3040**.

[0264] Operation of handle device **3000** may require a user to use two hands. For example, a user may press button **3010** with one hand and move the handle **3012** between stowed configuration, first position and/or deployed configuration with the other hand. If button **3010** is released during operation of the device **3000**, the handle **3012** may be unable to move in the deployed direction and/or the stowed direction **3040**. In some embodiments, releasing button **3010** prevents the handle from moving in the stowed direction **3040**. Locking the handle **3012** may prevent a user from pinching their hand between the wall and handle **3012** when moving the handle to the stowed position. Once the handle **3012** has been returned to the stowed configuration, the user may release button **3010** to lock or retain the handle **3010** within handle housing **3016**. Although three positions are depicted (stowed, first and deployed positions), the handle device **3000** may have more than three positions. The number of positions/configurations may generally correspond with the number of teeth **3026** on ratchet **3024**.

[0265] Referring to FIGS. **36**A, **36**B, **37**A, and **37**B, an example of ratcheting mechanism **3014** is depicted. Ratcheting mechanism **3014** may comprise pawl **3022**, ratchet **3024**, actuator **3034**, and linkage **3032**. FIGS. **36**A, **36**B, **37**A, and **37**B depict the pawl **3022** in the first pawl position and linkage **3032** in the first linkage position. As shown in FIG. **36**B and **37**B, pawl **3022** is engaged with ratchet **3024** in the first pawl position, such that movement of the pawl **3022** and handle arm **3018** in second direction **3030** is hindered or substantially prevented by edge **3025** of teeth **3026**. While edge **3025** of teeth **3026** may hinder movement of an engaged (ie. in the first pawl position) pawl **3022** in the second direction, ramp **3027** of teeth **3026** may allow movement of an engaged pawl **3022** in the first direction **3028**.

[0266] As shown in FIGS. **36**A, **36**B, **37**A, and **37**B, pawl **3022** may be disengaged (ie. moved to the second pawl position) by actuating linkage **3032**. Linkage **3032** may be moved from the first linkage position (as depicted in FIGS. **36**A, **36**B, **37**A, and **37**B) to a second linkage position by actuator **3034**. For example, a user pushing button **3010** and actuator **3034** in direction **3042** will actuate linkage **3032** to move from the first linkage position to the second linkage position. Movement of linkage **3032** from the first linkage position to the second linkage position may occur in direction **3044**. Linkage **3032** may be actuated by actuator **3034** via cam **3046**, such as a third linear cam. Linear cam **3046** may be comprised of a cam surface **3048** defined by the actuator **3034** and a third follower **3050** disposed on linkage **3032**. Cam **3046** may be configured such that linear movement of the actuator, for example in direction **3042**, results in linear movement of linkage **3032**, for example in direction **3044**.

[0267] Linkage **3032** and pawl **3022** may be configured such that movement of linkage **3032** to the second linkage position moves pawl **3022** to the second pawl position. For example, linkage **3032** and pawl **3022** may be connected through a first cam **3052**, such as a first linear cam, comprised of cam surface **3054** and first follower **3056**. Cam surface **3054** may be defined by pawl trigger **3058** connected to linkage **3032**, and follower **3056** may be formed by a pawl arm **3060**. Cam **3052** may drive follower **3056** and pawl **3022** in direction **3** (FIG. **37**A) and move pawl **3022** from the first pawl position (pictured) to the second pawl position. Moving pawl **3022** into the second pawl position (not pictured) in direction **3** (FIG. **37**A) and direction **4** (FIG. **37**B) results in pawl **3022** no longer abutting or otherwise engaging with teeth **3026** of ratchet **3024**. As will be apparent from the disclosure, continuous force applied to button **3010** will keep linkage **3032** in the second linkage position, and therefore pawl **3022** in the second pawl position, permitting movement of pawl **3022** and handle arm **3018** in the second direction **3030** towards the stowed configuration.

[0268] Referring to FIG. **37**A, movement of pawl trigger **3058** in direction **2** may be driven by second cam **3062**, such as a second linear cam. Cam **3062** may comprise cam surface **3064** defined by pawl trigger **3058** and a second follower **3066** formed or mounted on housing **3016**, such as support **3067**. Cam **3062** may be configured such that linear movement of linkage **3032** in direction **1** drives pawl trigger **3058** in linear direction **2**. Movement of pawl trigger **3058** in direction **2** may drive pawl arm **3060** and pawl **3022** in direction **3**, thus moving pawl **3022** into the second pawl position and out of abutment or engagement with teeth **3026** of ratchet **3024**. Pawl trigger **3058** may be pivotably connected to linkage **3032**.

[0269] Referring to FIG. **38**, spring **3036**, such as a torsion spring, may be used to bias handle **3012** and handle arm **3018** toward the deployed configuration. Spring **3036** may be mounted to handle housing **3016**. For example, spring **3036** may be mounted to support **3068**. Spring adjustment carriage **3070** may be used to adjust the strength of spring **3036**. A bolt of spring adjustment carriage **3070** may be tightened or loosened to adjust the strength of spring **3036**. Spring **3072** may be used to bias or push linkage **3032** to the first linkage position. For example, spring **3072** may be housed in spring housing **3074** (FIG. **41**) and may be compressed as linkage **3032** is actuated to the second linkage position by actuator **3034**. Once button **3010** is released by the user, the compressed spring **3072** may push the linkage **3032** to the first linkage position as the spring relaxes. For example, spring **3072** may push on tab **3084** of linkage **3032**.

[0270] Referring to FIGS. **39-45**, an example of bottom part **3016**A of handle housing **3016** is depicted. Bottom part **3016**A may be shaped or configured to mate with top part **3016**B. Handle housing **3016** may define an inner cavity in which the ratcheting mechanism **3014** is disposed. Bottom part **3016**A may comprise one or more inner walls **3076**. Inner walls **3076** may be shaped or configured to mate with corresponding inner walls **3086** of top part **3016**B. Inner walls **3076** may provide structural support and abutments for parts of handle device **3000**. For example, wall **3076**A may provide an abutment for handle arm **3018** when the handle **3012** is in the deployed position. Bottom part **3016**A may also comprise mounts for components of the ratcheting mechanism **3014**. For example, one or more supports **3078** may mate with corresponding supports or recesses, such as recess **3079** of top part **3016**B to align the top part **3016**B and bottom part **3016**A in use. Bottom part **3016**A may also comprise support **3067** that acts as follower **3066** in cam **3062** (FIG. **37**). Support **3068** may also be used to mount one or more of spring **3036**, ratchet **3024**, and handle arm **3018**. Support **3068** (with an optional fastener) may act as a pivot point for a first end **3018**A of handle arm **3018**. Bottom part **3016**A may also comprise support **3080** as a guide for linkage **3032**. For example, an upper part **3080**A of support **3080** may be at least partially inserted into aperture **3082** of linkage **3032** and have a smaller diameter to allow linear movement of linkage **3032**. Bottom part **3016**B may also comprise spring housing **3074** for housing spring **3072**.

[0271] Referring to FIGS. **46-52**, an example of top part **3016**B of handle housing **3016** is depicted. Top part **3016**B may be shaped or configured to mate with bottom part **3016**A. Top part **3016**B may comprise one or more inner walls **3086**. Inner walls **3086** may be shaped or configured to mate with corresponding inner walls **3076** of bottom part **3016**A. Inner walls **3086** may provide structural support and abutments for parts of handle device **3000**. Top part **3016**B may supports **3079** for mating with supports of the bottom part **3016**A. Once corresponding supports of the bottom part **3016**A and top part **3016**B have been mated, one or more fasteners may be threaded through said supports to secure the bottom part **3016**A and top part **3016**B to one another.

[0272] Referring to FIGS. **53**A-**56**, actuator button **3010** may have a suitable shape and configuration, such as an "L" shape in cross-section. Other shapes may be used. Actuator button **3010** may have one or more lights to facilitate location and operation in dim or low light. Button **3010** may have a member **3088** shaped and configured for a user to grip or push. Button **3010** may also define a recess **3090** configured for receiving part of actuator **3034**. Button **3010** may also comprise one or more gripping members, such as ribs **3092** that grip or provide an interference fit when mating with actuator **3034**. For example, ribs **3092** may be crush ribs.

[0273] Referring to FIGS. **57**-**61**, an example of linkage **3032** is depicted. Linkage **3032** is movably connected to handle housing **3016**. For example, linkage **3032** defines apertures **3082** which are configured to receive part of housing **3016**, such as supports **3080**. Linkage **3032** may be slidably mounted to supports **3080** such that linkage **3032** can move linearly between the first linkage position and the second linkage position. Linkage **3032** may be configured to move pawl **3022** from the first pawl position to the second pawl position as linkage **3032** moves from the first linkage position to the second linkage position. For example, linkage **3032** may be pivotably mounted to pawl trigger **3058**, such as by a fastener or rod inserted through aperture **3096** of pawl trigger **3058** and aperture **3098** of linkage **3032**. In some embodiments, the linkage comprises a rod that is inserted into apertures **3096** and **3086** to form the pivotable connection. Linkage **3032** may comprise tab **3084** which provides an abutment for spring **3072** to push linkage **3032** back into the first linkage position from the second linkage position. Linkage **3032** may comprise a locking tab **3094** configured to engage with a ledge **3101** of complementary recess **3100** of handle arm **3018**, for example top part **3104** and/or insert **3108**, when handle **3012** is in the stowed configuration and linkage **3032** is in the first linkage position. When a user moves actuator **3034** and moves linkage **3032** from the first linkage position to the second linkage position, locking tab **3094** may disengage from recess **3100** and spring **3036** may move handle arm **3018** and handle **3012** toward the deployed configuration, such as the first position (FIG. **32**). Linkage **3032** may also comprise member **3102** configured to engage with complementary recess **3107** of handle arm **3018**, for example top part **3104** or recess **3122** and tooth **3123** of handle arm insert **3108**. Member **3102** and tooth **3123** may be configured to engage and lock to one another when the handle **3012** is in the deployed configuration. Linkage **3032** may define aperture **3099** for mounting a follower **3050**.

[0274] Referring to FIGS. **62**-**64**B, an example of handle **3012** is depicted. Handle **3012** may be configured to be grasped by a user. Handle **3012** may define a recess **3110** shaped to receive an insert (not pictured). The insert may be a front plate, such as front plate **804**, or a panel, such as panel **1001**. The insert may be a solid (or otherwise single piece) insert, such as a piece of foam or decorative material, or it may be formed by a moldable/formable material or process within recess **3110**. For example, an insert comprising tiles and adhesive may be installed into recess **3110**. In cases where the insert is one or more continuous pieces, the insert may be removable from its recess. The insert may be shaped to be received within recess **3110**. The insert may be composed of a suitable material, such as glass, tile, metal, plastic, wood, or a combination thereof. The insert may have a suitable colour, such as a colour with a high visibility. The insert may be retained within recess **3110** by suitable means, such as adhesive, welding, fasteners, interference fit, and others. For example, one or more surfaces may define one or more grooves, such as grooves **3112**, for receiving adhesive for securing the insert to the handle **3012**.

[0275] Referring to FIGS. **65**-**71**, an example of pawl **3022** is depicted. Pawl **3022** may be shaped to ride along ramp **3027** of ratchet **3024** as pawl **3022**, in the first pawl position, moves in first direction **3028**. For example, pawl **3022** may define a ramp **3114** that is shaped to ride complementarily shaped ramp **3027**. Pawl **3022** may also define riser **3116** that abuts against edge **3025** of teeth **3026** when the pawl is in the first pawl position. Pawl **3022** may comprise pawl arm **3060** with follower **3056**.

[0276] Referring to FIGS. **72**-**75**, an example of border plate **3118** is depicted. Border plate **3118** may connect and mount to handle housing **3016**, such as via clips **3120** which may provide an interference fit against one or more parts of housing **3016**. Border plate **3118** may be mounted to a front face of housing **3016**.

[0277] Referring to FIGS. **76**-**79**, an example of pawl trigger **3058** is depicted. Pawl trigger **3058** may drive pawl **3022** from the first pawl position to the second pawl position via cam **3052**. Pawl trigger **3058** may define cam surface **3054** configured to drive follower **3056** and pawl **3022** in a linear direction, such as towards handle **3012**. Cam surface **3054** may be a ramp. Pawl trigger **3058** may define cam surface **3064**, which may be part of cam **3062**. Pawl trigger **3058** may also define aperture **3096** which may form a pivotable connection with linkage **3032** along with a fastener or rod.

[0278] Referring to FIGS. **80**-**82**A, an example of ratchet **3024** is depicted. Ratchet **3024** may be mounted within handle housing **3016**, for example via fasteners through one or more apertures **3124**. Ratchet **3024** defines teeth **3026** biased to, when pawl **3022** is in the first pawl position, allow movement of pawl **3022** in first direction **3028** and restrict movement of pawl **3022** in second direction **3030**. Each tooth of teeth **3026** may define edge **3025** and ramp **3026**. Ratchet **3024** may define at least 3 teeth, providing at least three positions for handle **3012**. Ratchet **3024** may comprise more than 3 teeth, which would provide more positions for handle **3012** between stowed and deployed configurations. Ratchet **3024** may be mounted to handle housing **3016** by mating or threading support **3068** through aperture **3126**.

[0279] Referring to FIG. **38**, handle arm **3018** may comprise top part **3104**, handle arm insert **3108**, and bottom part **3106**. Top part **3104**, bottom part **3106** and insert **3108** may cooperate to form an enclosure for pawl **3022** such that the pawl **3022** moves with the movable handle arm **3018**. For example, as handle **3012** moves between the stowed and deployed configurations, handle arm **3018** swings on a pivot point within handle housing **3016**, and pawl **3022** swings with handle arm **3018**. The enclosure of top part **3104**, bottom part **3106** and insert **3108** may also allow pawl **3022** to move independently of handle arm **3018**, such as linear movement between the first and second pawl positions.

[0280] Handle arm **3018** may be pivotably connected to handle housing **3016** at first end **3020**A and pivotably connected to the handle at a second, opposing end **3020**B. Handle arm **3018** may have two pivot points for pivot connections at first end **3020**A and second end **3020**B. First end **3020**A may be configured to cooperate with supports of the top and bottom parts **3016**B, **3016**A of housing **3016**, such as support **3068**, to form a pivot connection. Second end **3020**B may be configured to cooperate with handle extension member **3136** and form a pivot connection.

[0281] Referring to FIGS. **83-88**, an example of bottom part **3106** of handle arm **3018** is depicted. Bottom part **3106** may be shaped to cooperate with complimentary shaped top part **3104**. Bottom part **3106** may define apertures **3128** shaped to receive one or more fasteners to secure bottom part **3106** to top part **3104**. Bottom part **3106** may define a pawl aperture **3130** shaped to receive pawl **3022**. Pawl aperture may be sized to allow slidable movement of pawl **3022** within pawl aperture **3130**. Bottom part **3106** may define apertures **3134** for receiving a fastener or pin as a pivot point. Bottom part **3106** may be composed of a suitable material, such as molded plastic or metal.

[0282] Referring to FIGS. **89-95**, an example of top part **3104** of handle arm **3018** is depicted. Top part **3104** may be shaped to cooperate with complimentary shaped bottom part **3106**. Top part **3104** may define apertures **3136** shaped to receive one or more fasteners to secure top part **3104** to bottom part **3106**. Top part **3104** may define apertures **3138** for receiving a fastener or pin as a pivot point. Top part **3104** may define recess **3100** and ledge **3101** configured to engage with locking tab **3094** of linkage **3032** when the linkage arm is in the first linkage position and the handle **3012** is in the stowed configuration. Top part **3104** may define channel **3132** configured to receive part of pawl **3022**, such as pawl arm **3060**. Recess **3107** may be shaped to receive part of pawl **3022**, such as pawl arm **3060**. Top part **3104** may be composed of a suitable material, such as molded plastic or metal.

[0283] Referring to FIG. **96**, an example of handle extension member **3136** is depicted. Handle extension member **3136** may act to provide a pivoting connection with handle arm **3018**. For example, first end **3136**A of handle extension member **3136** may be shaped to receive an end of handle arm **3018**. A fastener or pin may be inserted through apertures **3140** of handle extension member **3136**, apertures **3138** of top part **3104**, and apertures **3134** of bottom part **3106** to form a pivoting connection.

[0284] Referring to FIG. **97-100**, an example of handle arm insert **3108** is depicted. Insert **3108** may provide rigidity and/or strength to the handle arm **3018**. Handle arm insert **3108** act to increase the load bearing capabilities of handle device **3000**, for example the handle device may meet or exceed the 300 lbs ADA weight withstand requirements. Handle arm insert **3108** may be at least partially composed of metal. Handle arm insert **3108** may have recess **3122** and tooth **3123** configured to engage with complementary member **3102** of linkage **3032**. Member **3102** and tooth **3123** may be configured to engage and lock to one another when the handle **3012** is in the deployed configuration. Handle arm insert **3108** may also comprise recess **3100** and shelf **3101**, both configured to receive and/or engage with locking tab **3094** of linkage **3032**. Recess **3100** and shelf **3101** may engage with locking tab **3094** when handle **3012** is in the stowed configuration and linkage **3032** is in the first linkage position. Movement of linkage **3032** from the first linkage position to the second linkage position may disengage locking tab **3094** from recess **3100**, unlocking handle **3012** from linkage **3032**, and spring **3036** may be free to push handle **3012** toward the deployed configuration. Spring **3036** may push handle **3012** and handle arm **3018** may swing until pawl **3022** engages with the first tooth of ratchet **3024**.

[0285] Handle arm insert **3108** may have additional locking means disposed at or near tooth **3123** that engages a part of linkage **3032** and prevents movement of handle arm **3018** when the handle **3012** is in the deployed configuration and linkage **3032** is in the first linkage position. Movement of the of the linkage **3032** to the second linkage position may disengage the locking means. Such locking means may be used to provide additional locking and support to the handle **3012**.

[0286] Handle device **3000** may have two handle arms **3018**, each having their own respective pawl **3022**, each connected to handle **3012** and linkage **3032**, and each connected to their respective ratchet **3024**. It will be understood that more than two handle arms **3018** may be used with a similar configuration as depicted herein.

[0287] The handles and handle devices described herein can be made of a variety of materials, which can include metal (e.g., steel, aluminum), metal alloys, wood, polymers (e.g., plastic), glass (which could include metal reinforcing features, such as a center steel rod), and/or others. In some variants, the handles, housing, collar, and/or other features may include lights, such as an LED light channel, to help a user locate the handle and/or handle device. The handles and handle devices can be scaled up/down or made more or less robust depending on application, such as for use in cabinetry, heavy duty use, and/or for hanging objects.

[0288] In some variants, the handles and handle devices described herein can be incorporated into a drawer or other cabinetry fronts. For example, the handle device can form a front of the drawer. The handle device can include a mechanism that locks the drawer shut such that use of the handle device does not open the drawer. For example, the handle device can include deployable rods that can extend into surrounding features of the drawer, e.g., supporting structures below, above, or otherwise around. The deployable rods can be retracted to allow the drawer to be opened. The handle device can incorporate a second handle (e.g., smaller handle) coupled to a front of the handle of the handle device that can be used to open and close the drawer when the drawer is not locked shut by the deployable rods. Such an arrangement can be beneficial to persons in a seated position needing assistance to assume a standing position.

[0289] Conditional language, such as "can," "could," "might," or "may," unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include or do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments.

[0290] Conjunctive language, such as the phrase "at least one of X, Y, and Z," unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require the presence of at least one of X, at least one of Y, and at least one of Z.

[0291] Terms of orientation used herein, such as "top," "bottom," "horizontal," "vertical," "longitudinal," "lateral," and "end" are used in the context of the illustrated embodiment. However, the present disclosure should not be limited to the illustrated orientation. Indeed, other orientations are possible and are within the scope of this disclosure. Terms relating to circular shapes as used herein, such as diameter or radius, should be understood not to require perfect circular structures, but rather should be applied to any suitable structure with a cross-sectional region that can be measured from side-to-side. Terms relating to shapes generally, such as "circular" or "cylindrical" or "semi-circular" or "semi-cylindrical" or any related or similar terms, are not required to conform strictly to the mathematical definitions of circles or cylinders or other structures, but can encompass structures that are reasonably close approximations. The term "linear" is not required to confirm strictly to a mathematically straight line, but can encompass generally straight, or reasonably close approximations.

[0292] The terms "approximately," "about," and "substantially" as used herein represent an amount close to the stated amount that still performs a desired function or achieves a desired result. For example, in some embodiments, as the context may permit, the terms "approximately", "about", and "substantially" may refer to an amount that is within less than or equal to 10% of the stated amount. The term "generally" as used herein represents a value, amount, or characteristic that predominantly includes or tends toward a particular value, amount, or characteristic. As an example, in certain embodiments, as the context may permit, the term "generally parallel" can refer to something that departs from exactly parallel by less than or equal to 20 degrees.

[0293] Although the handle devices, handle assemblies, systems, and/or methods have been disclosed in the context of certain embodiments and examples, the scope of this disclosure extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the embodiments and certain modifications and equivalents thereof. Various features and aspects of the disclosed embodiments can be combined with or substituted for one another in order to form varying modes of the conveyor. The scope of this disclosure should not be limited by the particular disclosed embodiments described herein.

[0294] Certain features that are described in this disclosure in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations, one or more features from a claimed combination can, in some cases, be excised from the combination, and the combination may be claimed as any subcombination or variation of any subcombination.

[0295] Moreover, while operations may be depicted in the drawings or described in the specification in a particular order, such operations need not be performed in the particular order shown or in sequential order, and all operations need not be performed, to achieve the desirable results. Other operations that are not depicted or described can be incorporated in the example methods and processes. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the described operations. Further, the operations may be rearranged or reordered in other implementations. Also, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations. The described components and systems can generally be integrated together in a single product or packaged into multiple products. Additionally, other implementations are within the scope of this disclosure.

[0296] Some embodiments have been described in connection with the accompanying drawings. The figures are drawn to scale where appropriate, but such scale should not be interpreted as limiting, since dimensions and proportions other than what are shown are contemplated and are within the scope of the disclosed invention. Distances, angles, etc. are merely illustrative and do not necessarily bear an exact relationship to actual dimensions and layout of the devices illustrated. Components can be added, removed, and/or rearranged. Further, the disclosure herein of any particular feature, aspect, method, property, characteristic, quality, attribute, element, or the like in connection with various embodiments can be used in all other embodiments set forth herein. Additionally, any methods described herein may be practiced using any device suitable for performing the recited steps.

[0297] In summary, various embodiments and examples of handle devices, handle assemblies, systems, and/or methods have been disclosed. This disclosure expressly contemplates that various features and aspects of the disclosed embodiments can be combined with, or substituted for, one another. Accordingly, the scope of this disclosure should not be limited by the particular disclosed embodiments and examples described above, but should be determined only by a fair reading of the entire disclosure.

What is claimed is:

1. A deployable handle device comprising:

a handle configured to be grasped by a user, the handle configured to move between a stowed configuration and a deployed configuration, wherein in the stowed configuration, the handle is configured to be substantially flush with a surrounding wall, and in the deployed configuration, the handle is configured to protrude from the surrounding wall for the user to grasp the handle;

a handle housing configured to be positioned in the surrounding wall with the handle at least partially positioned in the handle housing;

a handle arm pivotably connected to the handle housing at a first end and pivotably connected to the handle at a second, opposing end;

a pawl connected to the handle arm such that the pawl moves with the handle arm, the pawl being configured to move independently from the handle arm between a first pawl position and a second pawl position;

a ratchet mounted within the housing, the ratchet defining teeth biased to, when the pawl is in the first pawl position, allow movement of the pawl in a first direction and restrict movement of the pawl in a second direction;

a linkage movably connected to the handle housing, the linkage being configured to move from a first linkage position to a second linkage position, the linkage being configured to move the pawl from the first pawl position to the second pawl position as the linkage moves from the first linkage position to the second linkage position;

an actuator connected to the linkage, the actuator being configured to move the linkage from the first linkage position to the second linkage position when actuated;

wherein the pawl is engaged with the ratchet in the first pawl position and the pawl is disengaged with the ratchet in the second position such that the pawl can move in the second direction; and

wherein the handle arm and pawl are configured to move in the first direction as the handle moves from the stowed configuration to the deployed configuration, and the handle arm and pawl are configured to move in the second direction as the handle moves from the deployed configuration to the stowed configuration.

2. The handle device of claim **1**, wherein the linkage comprises a first linear cam, the pawl comprises a respective first follower, and movement of the pawl from the first pawl position to the second pawl position is caused by movement of the first follower along the first linear cam.

3. The handle device of claim **2**, wherein the first linear cam is a cam surface of a pawl trigger connected to the linkage.

4. The handle device of claim **3**, wherein the pawl trigger defines a second linear cam, and the linkage comprises a respective second follower, and movement of the linkage from the first linkage position to the second linkage position actuates the pawl trigger to move the pawl from the first pawl position to the second pawl position.

5. The handle device of claim **1**, wherein the actuator defines a third linear cam, the linkage comprises a respective third follower, and movement of the third linear cam drives the third follower and the linkage from the first linkage position to the second linkage position.

6. The handle device of claim **1**, wherein the linkage comprising a locking clip configured to mate with a corresponding ledge of the handle arm when the handle is in the stowed configuration, and movement of the linkage from the first linkage position to the second linkage position disengages the locking clip from the ledge.

7. The handle device of claim **1**, further comprising a spring connected to the handle housing and configured to bias the handle toward the deployed configuration.

8. The handle device of claim **1**, further comprising a spring connected to the handle housing and configured to bias the linkage arm toward the first position.

9. The handle device of claim **1**, further comprising a button connected to the actuator such that force exerted on the button actuates the actuator.

10. The handle device of claim **1**, wherein the handle defines a recess configured for receiving an insert.

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